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Kim et al.

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(54) **ICE MAKER AND REFRIGERATOR**

(71) Applicant: **LG Electronics Inc.**, Seoul (KR)

(72) Inventors: **Yonghyun Kim**, Seoul (KR); **Jinil Hong**, Seoul (KR); **Hyunji Park**, Seoul (KR)

(73) Assignee: **LG Electronics Inc.**, Seoul (KR)

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(30) **Foreign Application Priority Data**

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Jul. 22, 2019 (KR) 10-2019-0088299

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F25C 1/24 (2018.01)
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CPC **F25C 5/08** (2013.01); **F25C 1/18** (2013.01); **F25C 1/24** (2013.01); **F25C 1/243** (2013.01);
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CPC F25C 1/06; F25C 1/18; F25C 1/24; F25C 5/08; F25C 5/22; F25C 5/02; F25C 5/04
See application file for complete search history.

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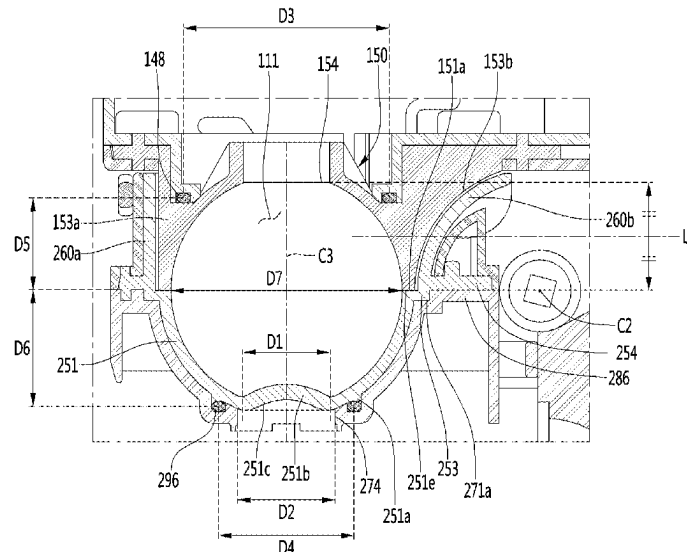
Primary Examiner — Jonathan Bradford

(74) *Attorney, Agent, or Firm* — Fish & Richardson P.C.

(57) **ABSTRACT**

An ice maker includes: an upper tray defining an upper chamber that is a portion of an ice chamber; a lower tray rotatable relative to the upper tray and defining a lower chamber that is another portion of the ice chamber disposed under the upper chamber; an upper heater disposed around the upper tray for providing heat to the upper chamber; and a lower heater disposed around the lower tray for providing heat to the lower chamber. In an ice making position, a distance from a horizontal central line passing a contact surface of the upper tray and the lower tray to the upper heater is shorter than a distance from the horizontal central line to the lower heater.

20 Claims, 33 Drawing Sheets



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F25C 1/243 (2018.01)
F25C 5/02 (2006.01)
F25C 5/04 (2006.01)
F25C 5/08 (2006.01)
F25C 5/20 (2018.01)

- (52) **U.S. Cl.**
CPC **F25C 5/02** (2013.01); **F25C 5/04**
(2013.01); **F25C 5/22** (2018.01)

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FIG. 1

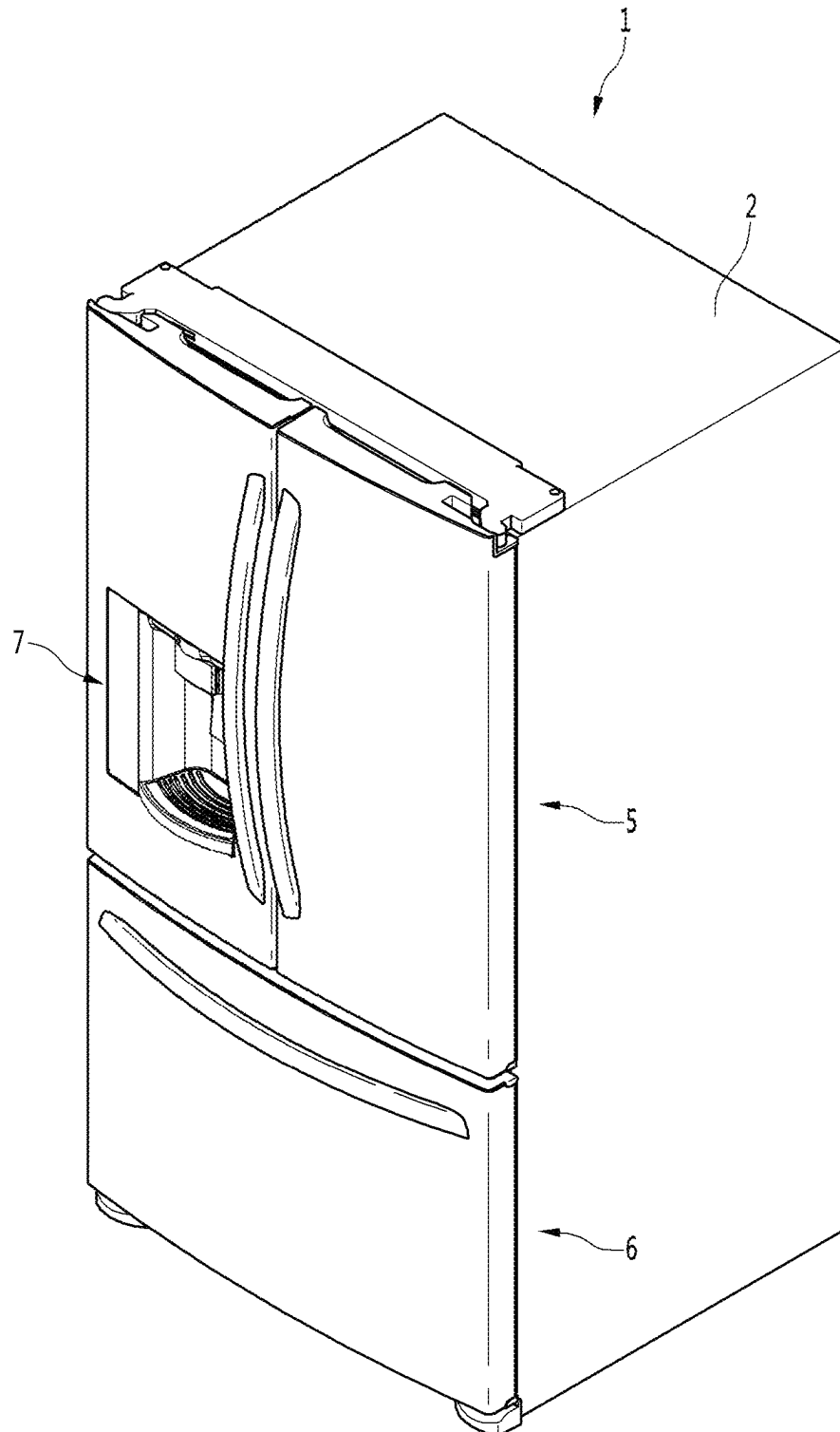


FIG. 2

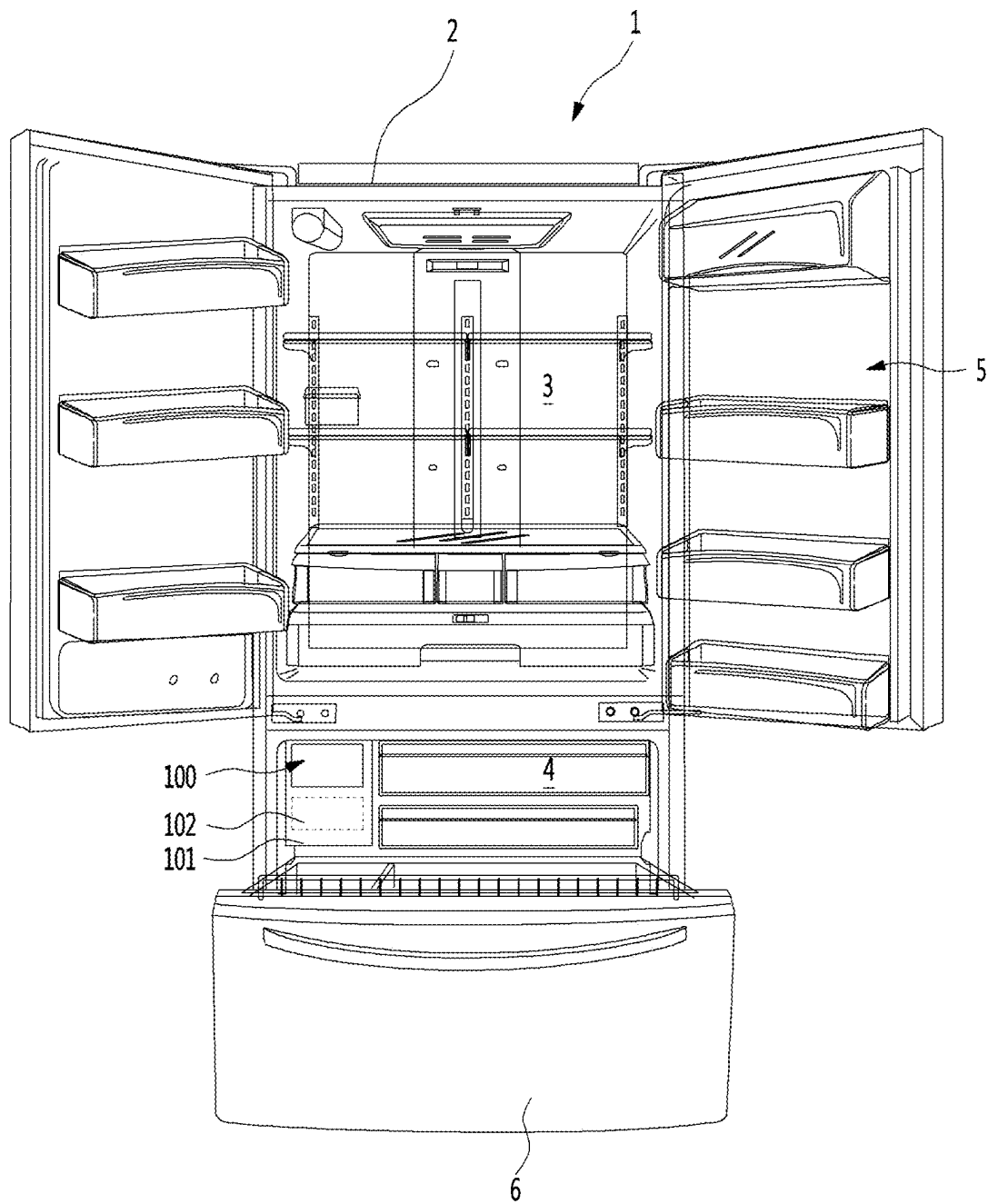


FIG. 3

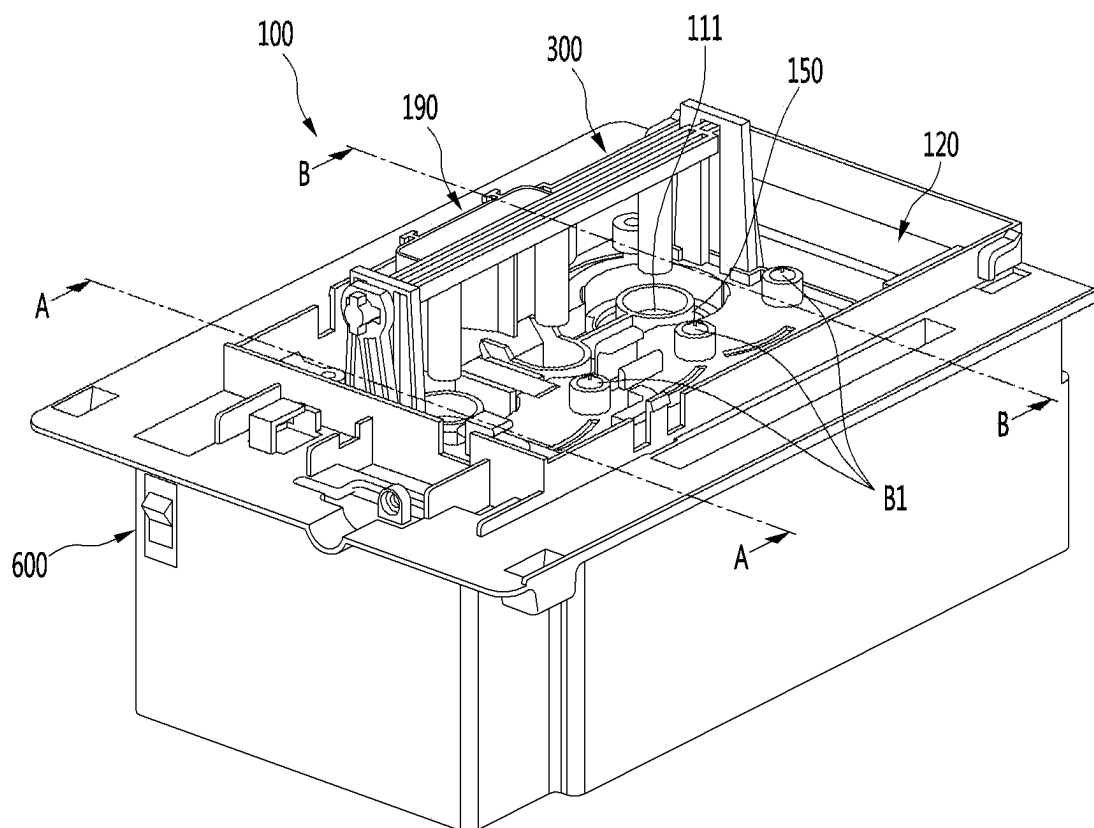


FIG. 4

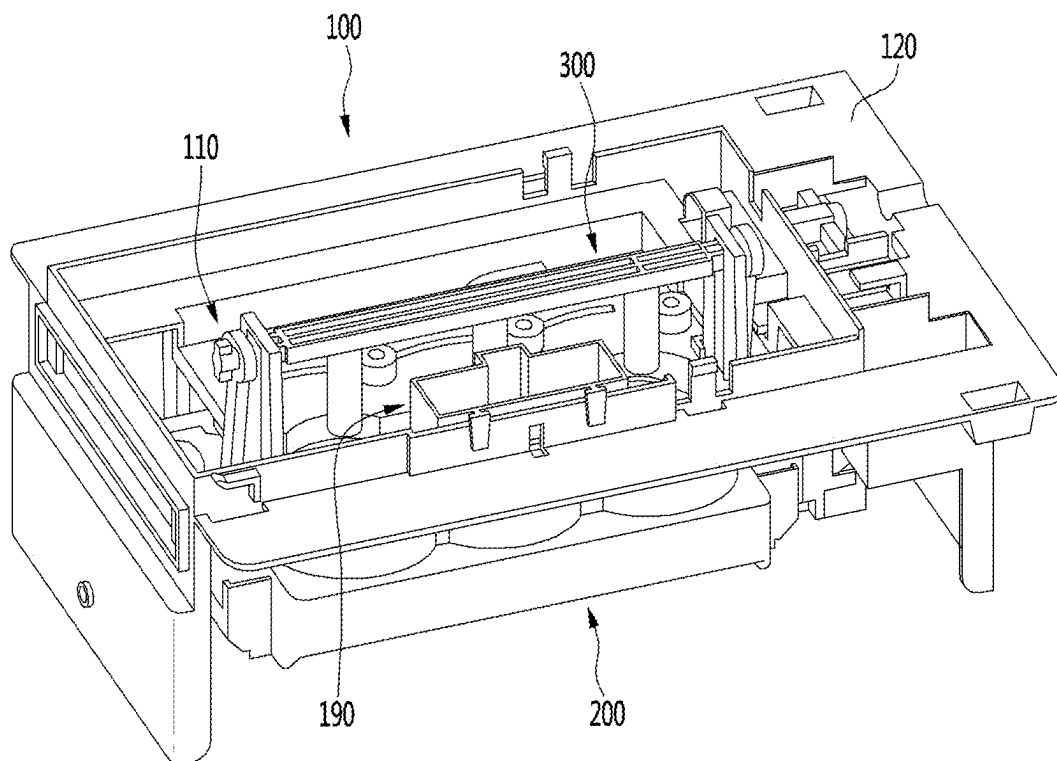


FIG. 5

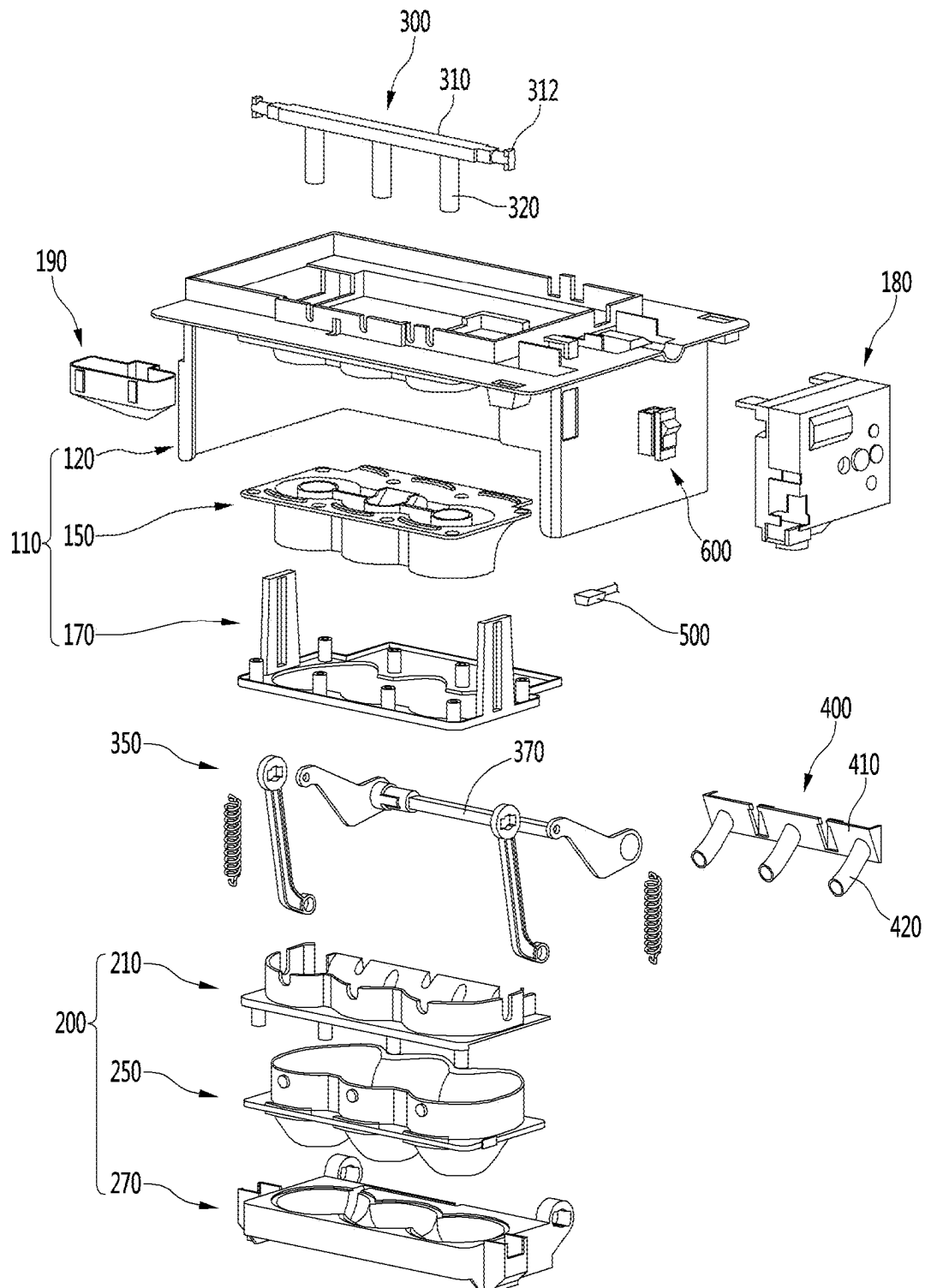


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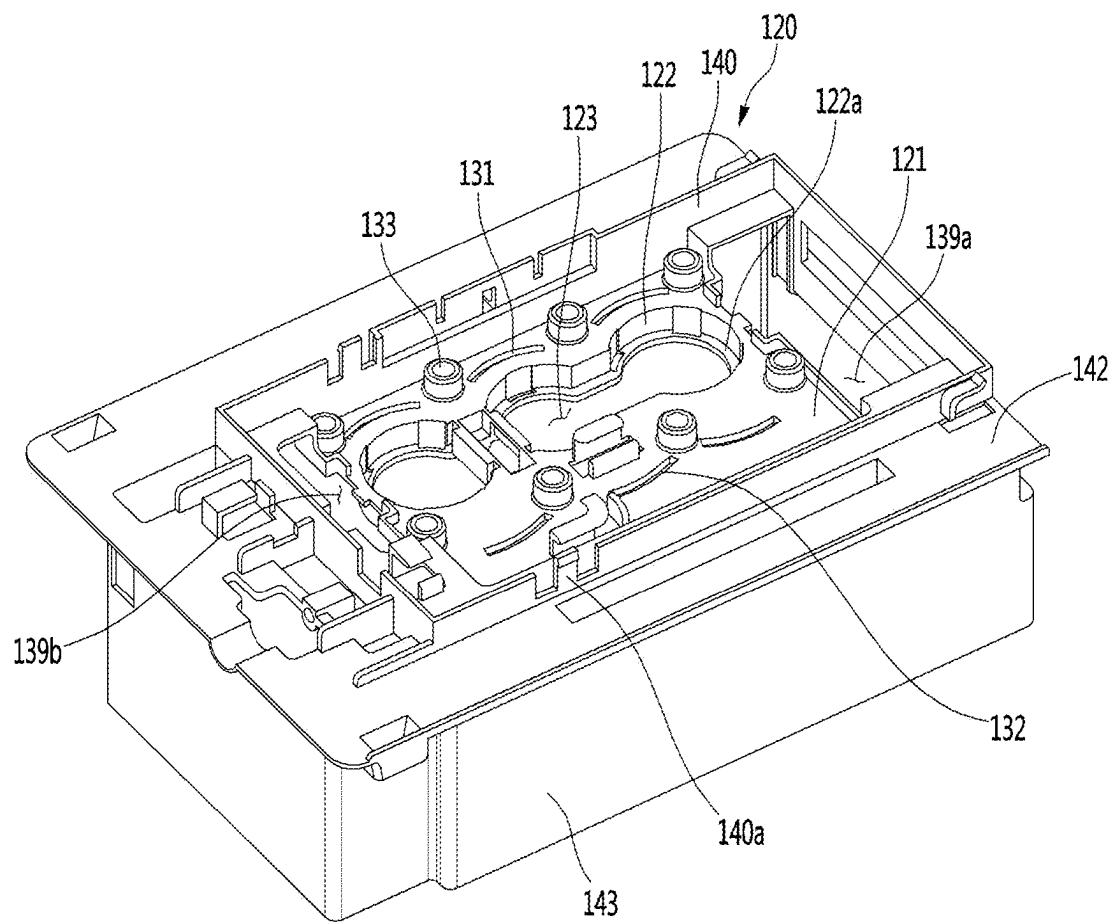


FIG. 7

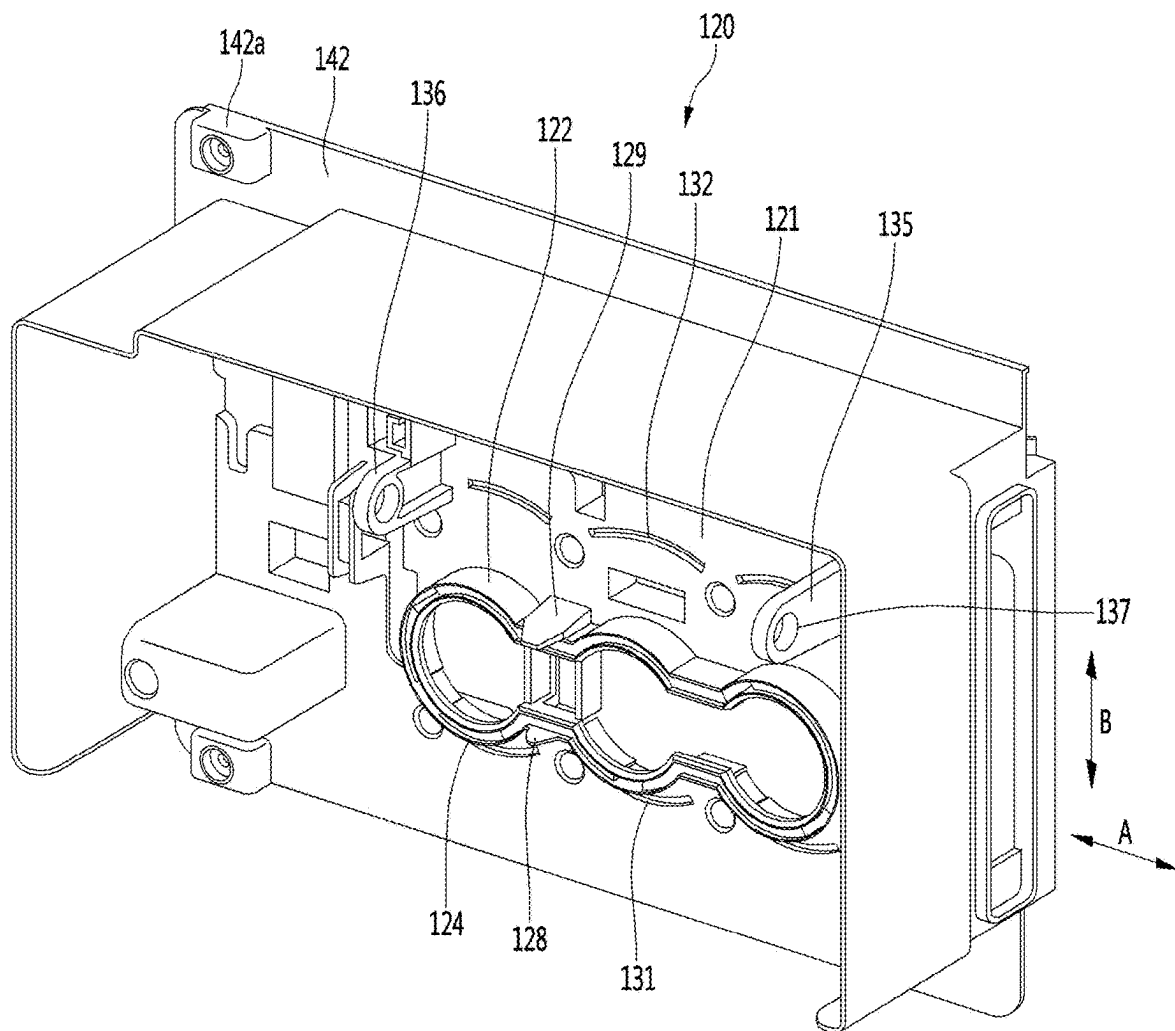


FIG. 8

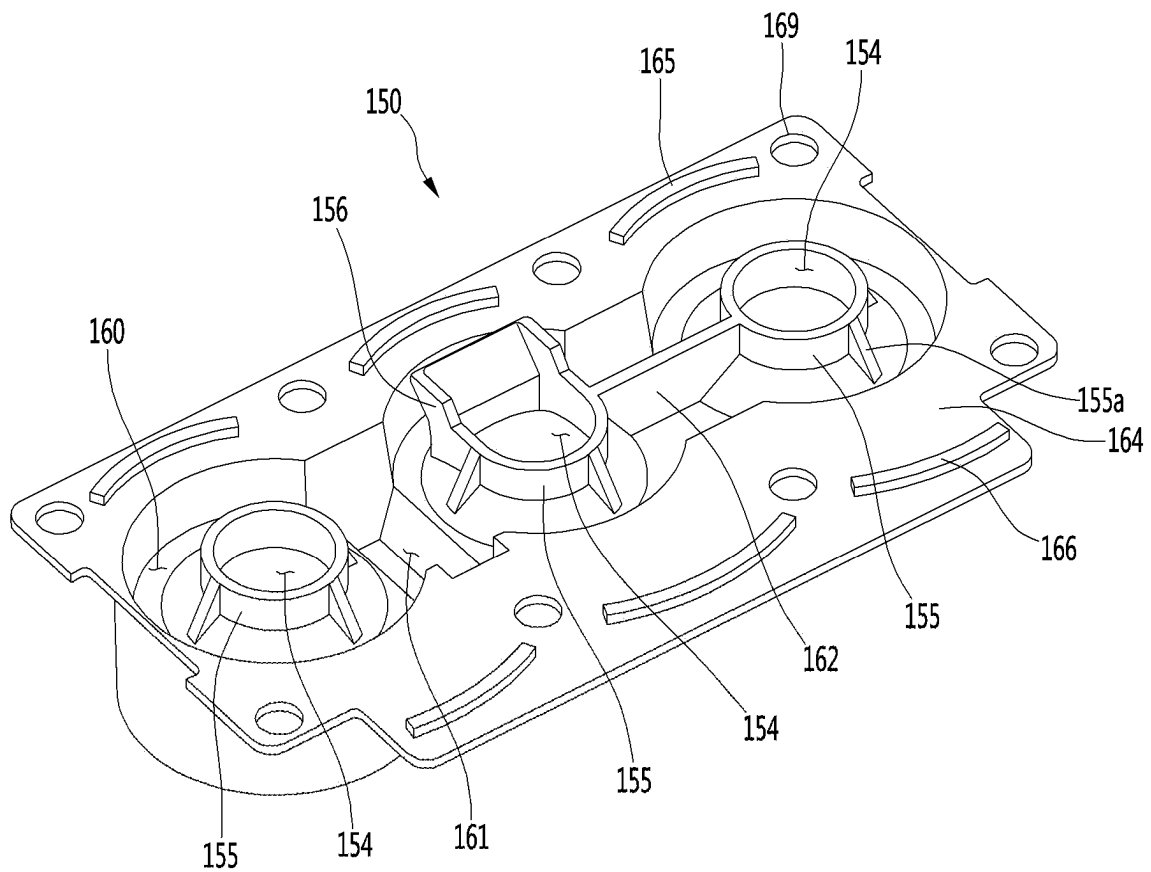


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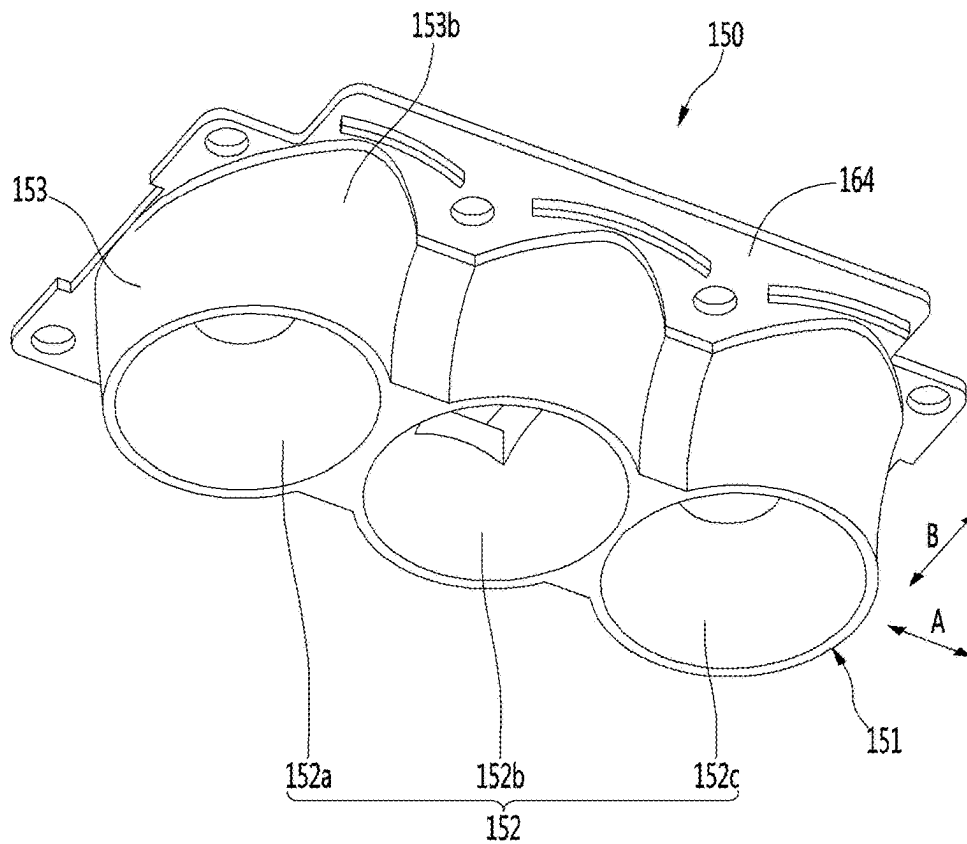


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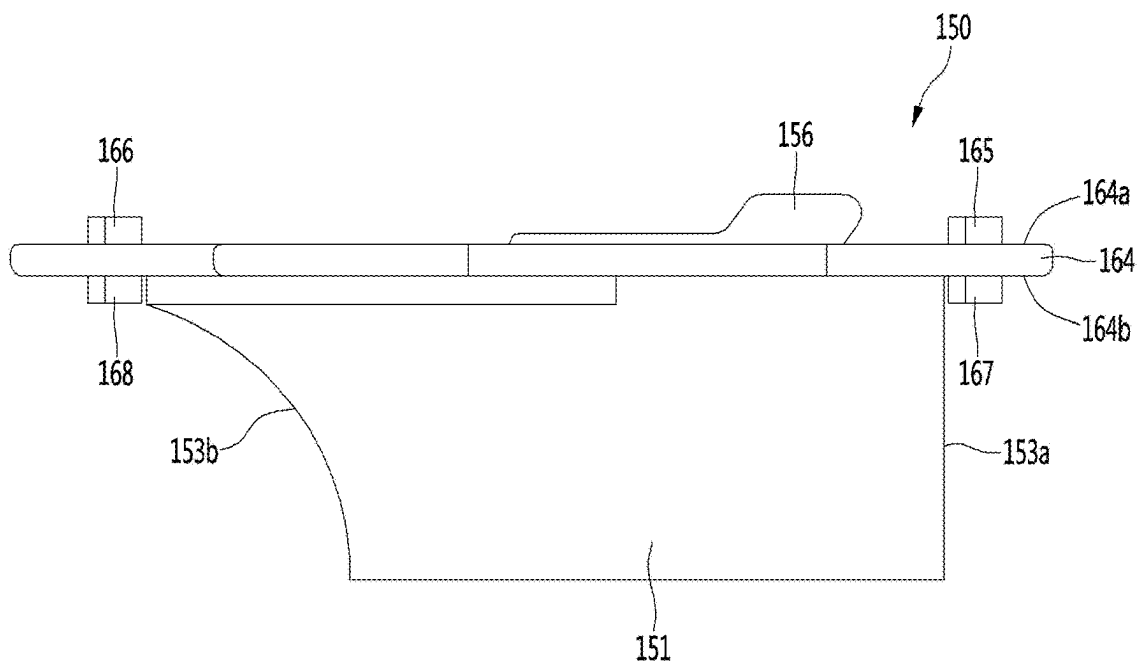


FIG. 11

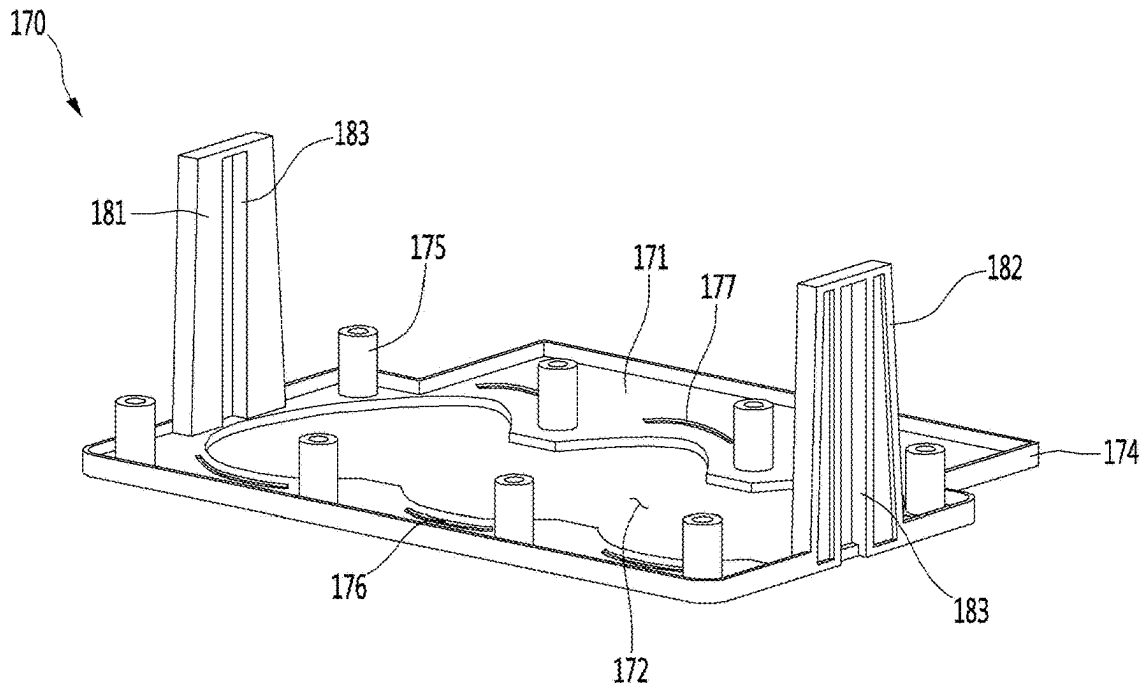


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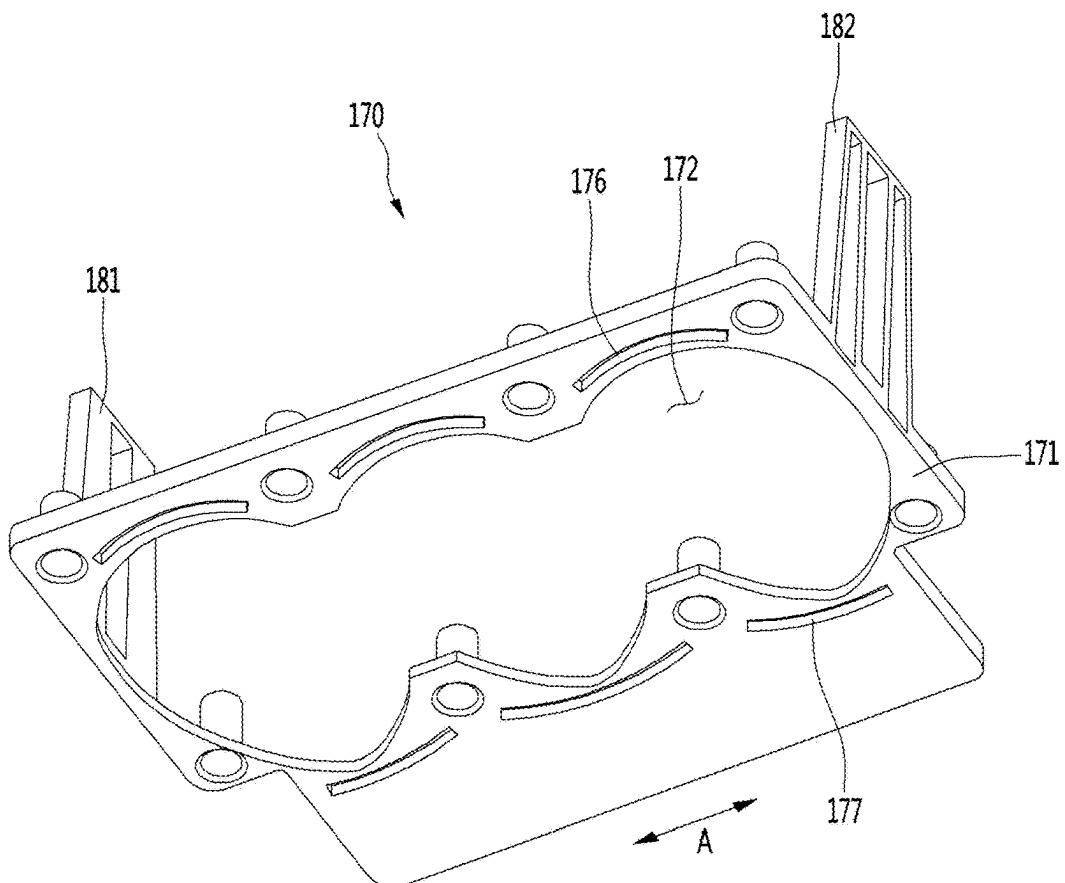


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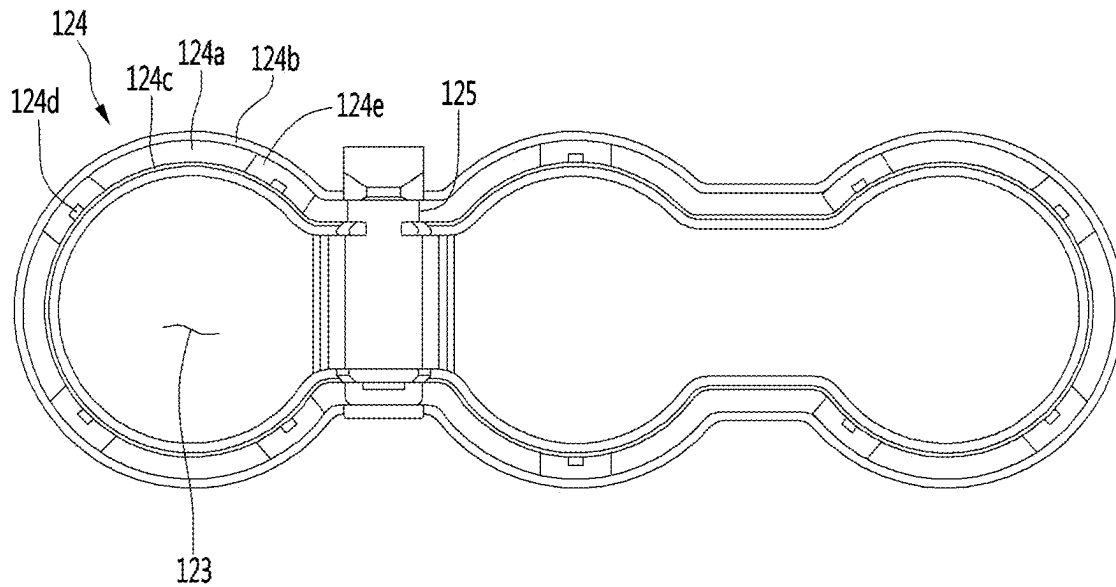


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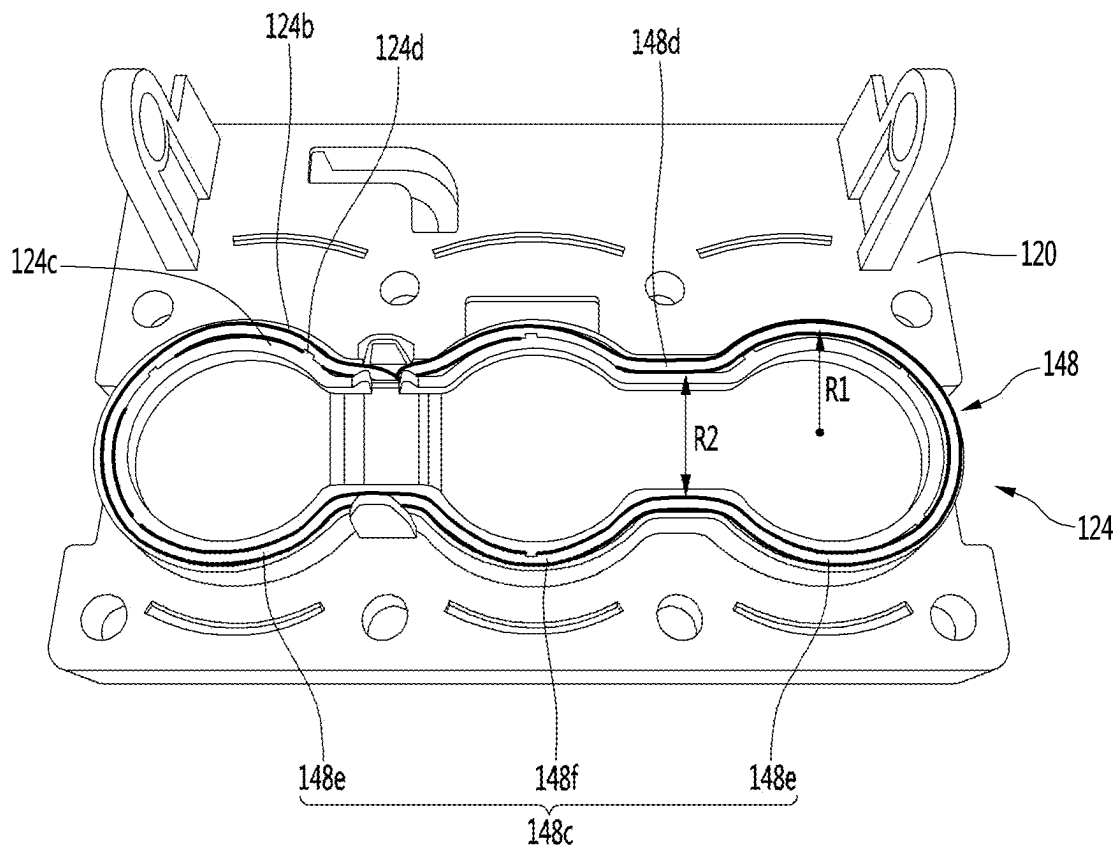


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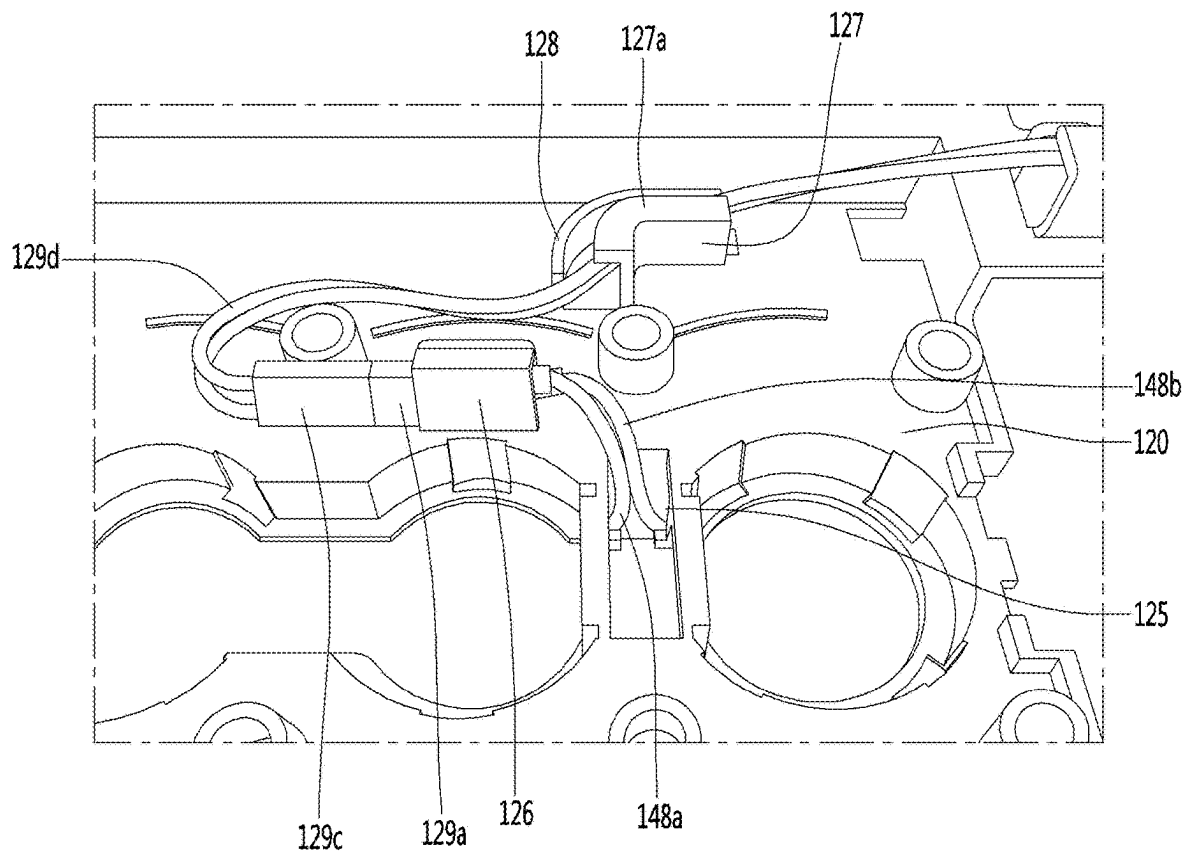


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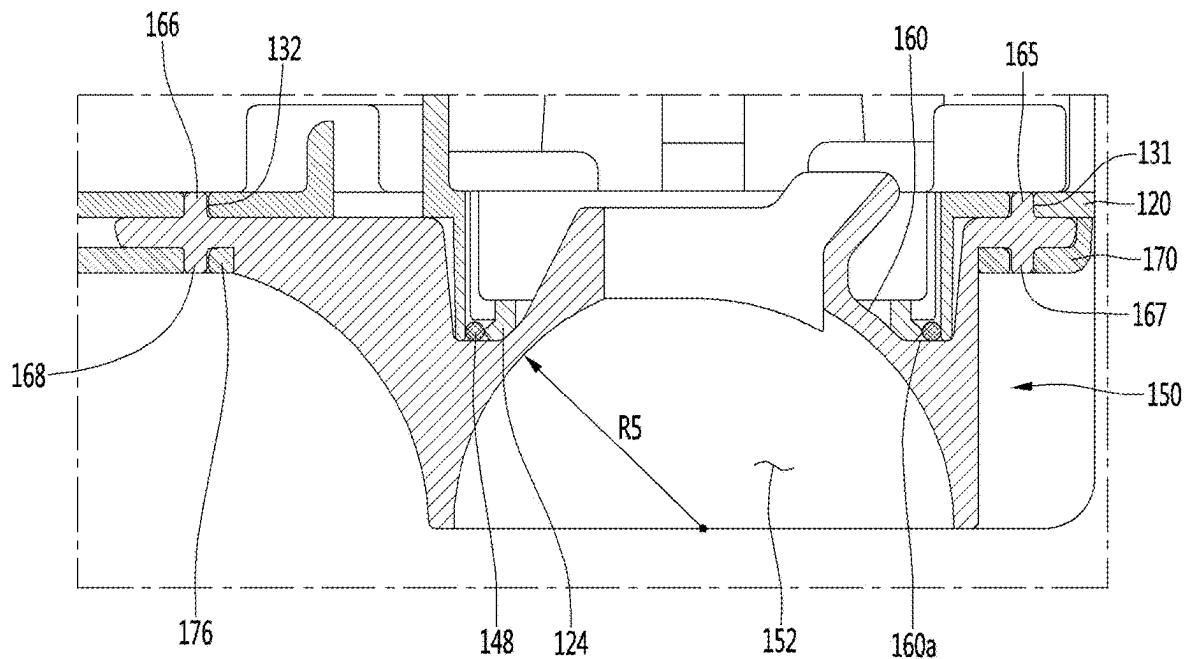


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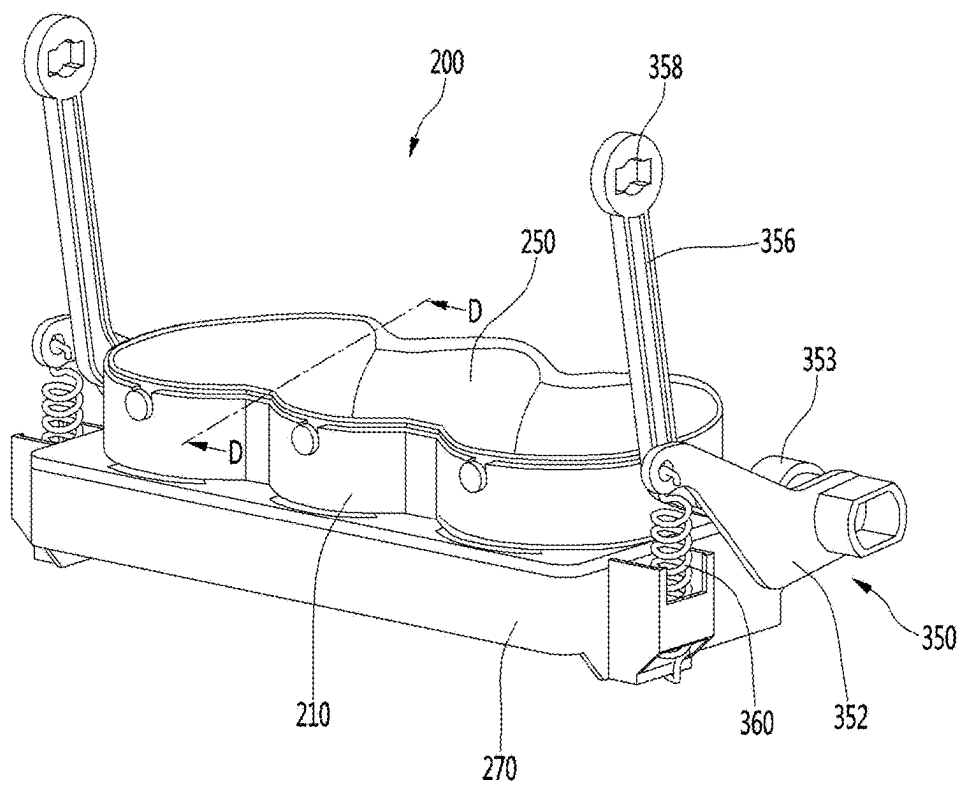


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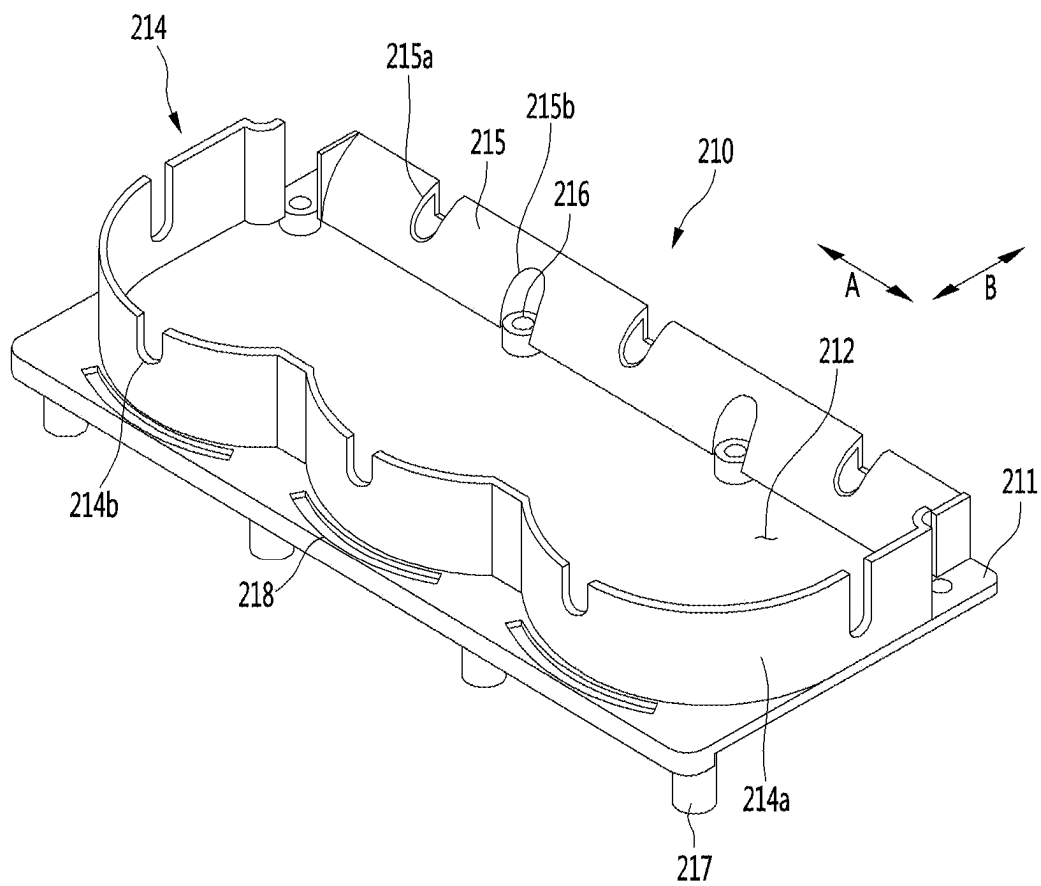


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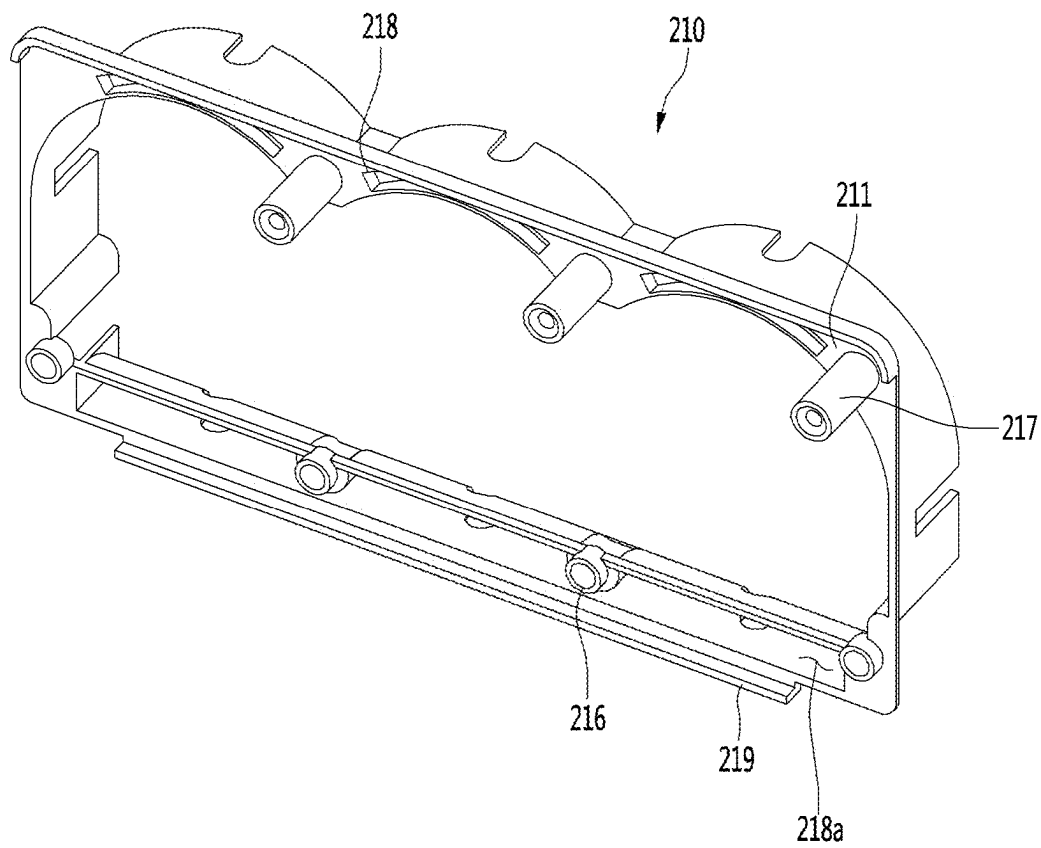


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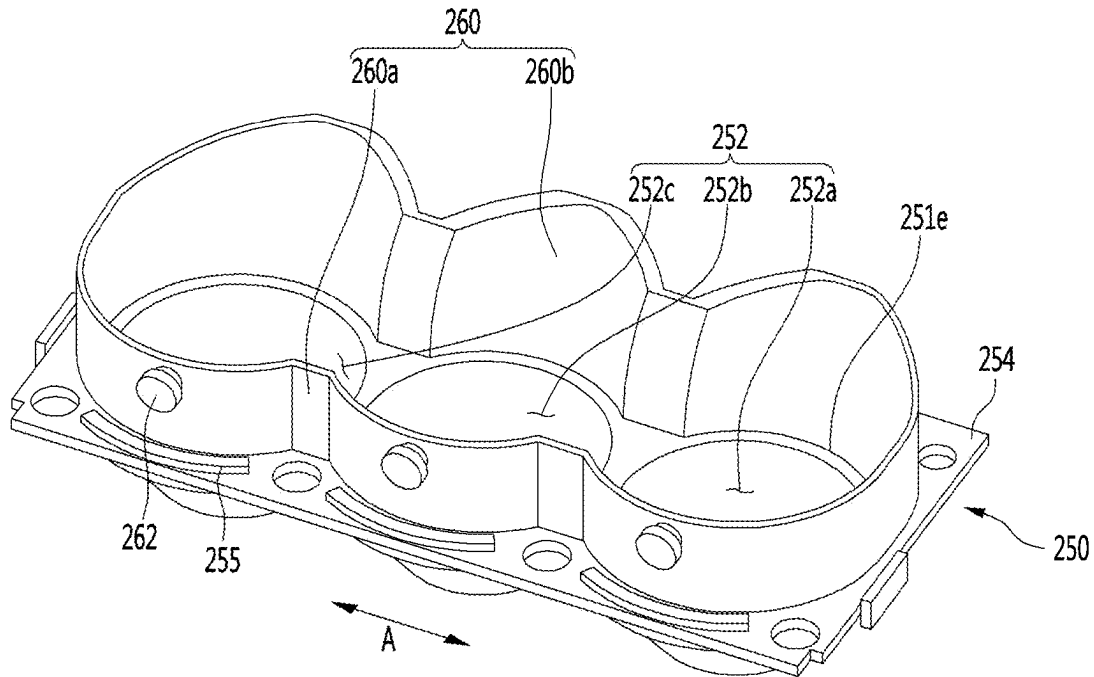


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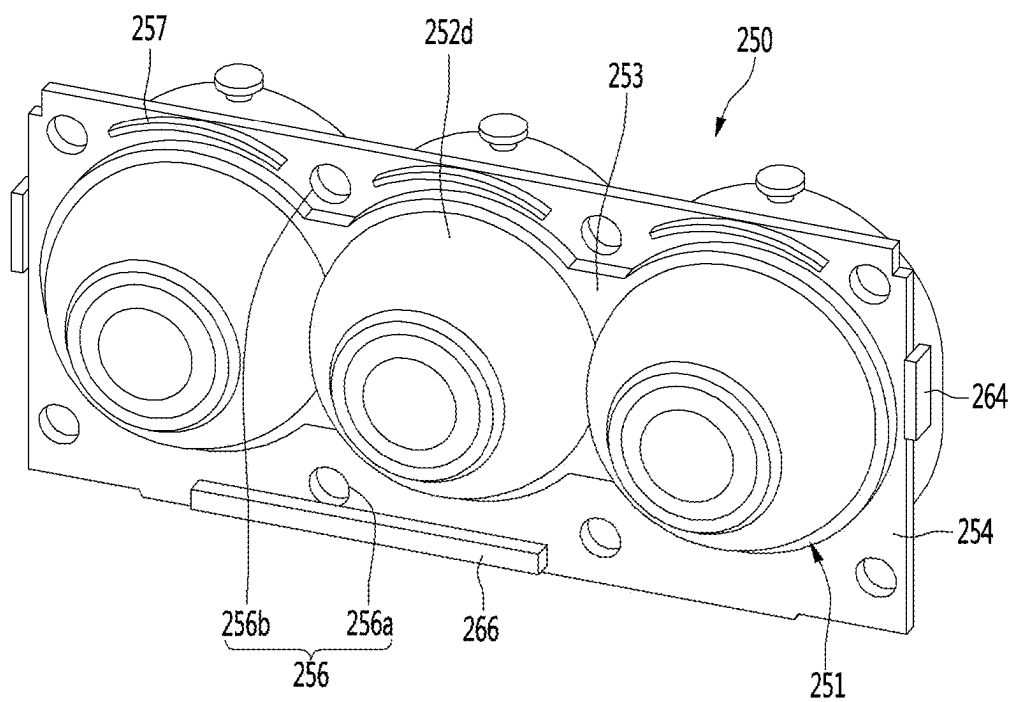


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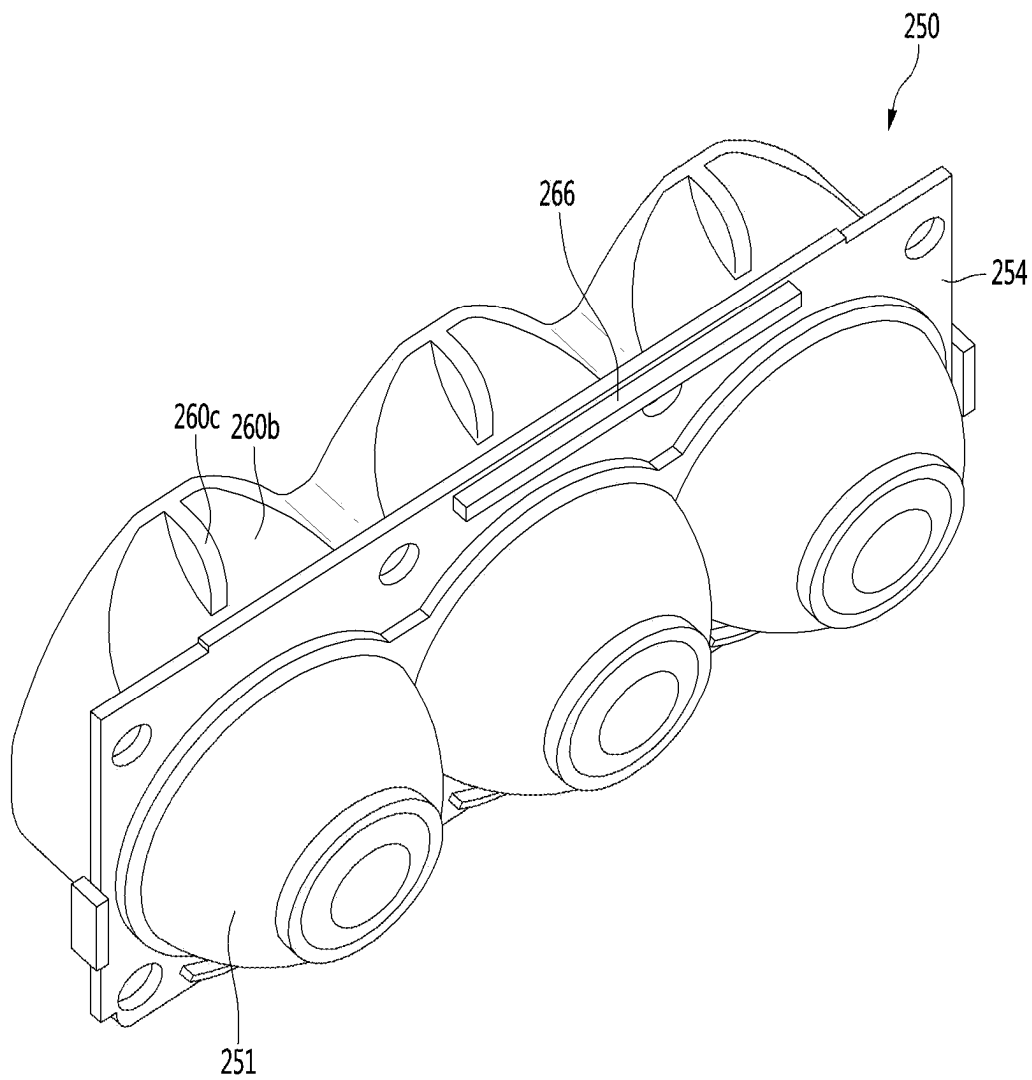


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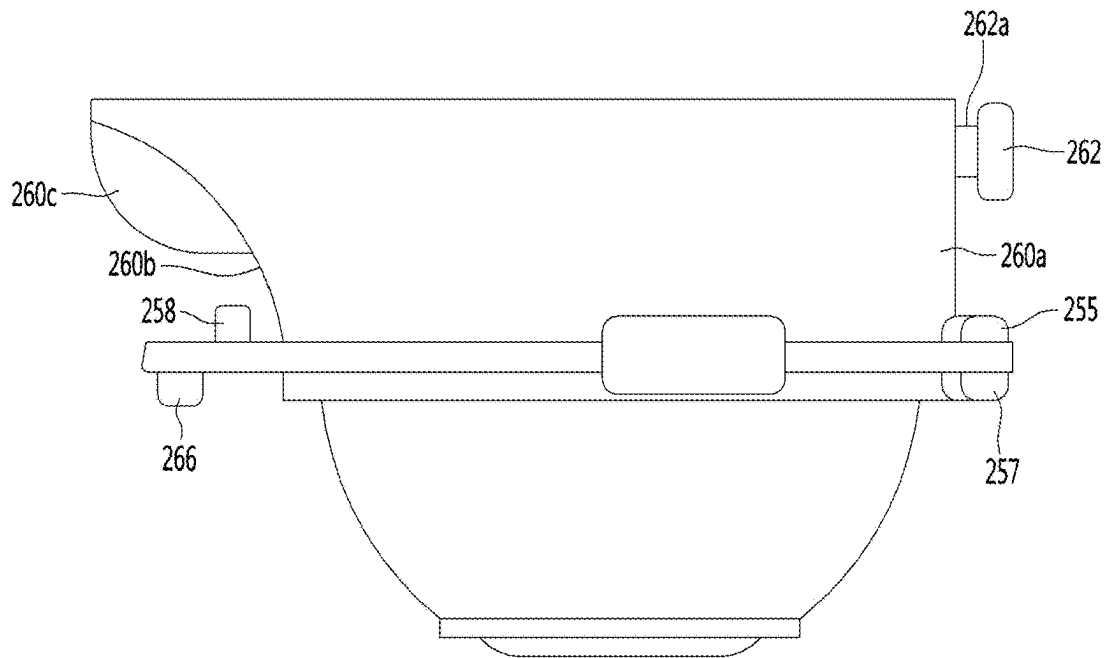


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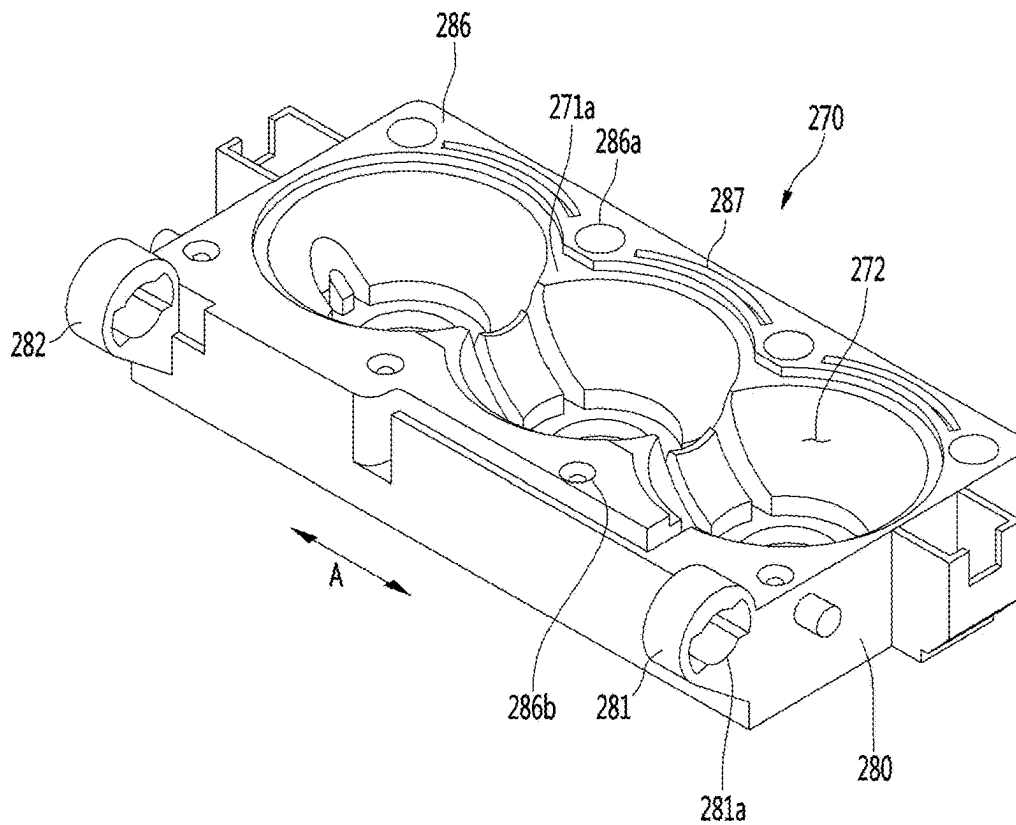


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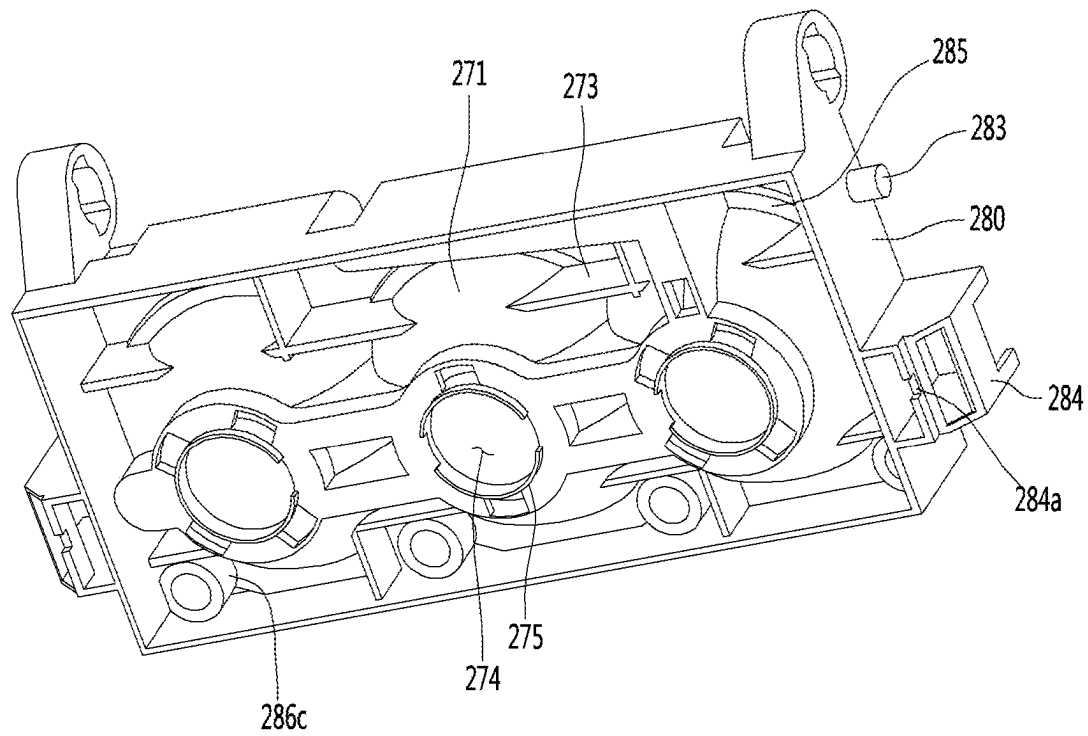


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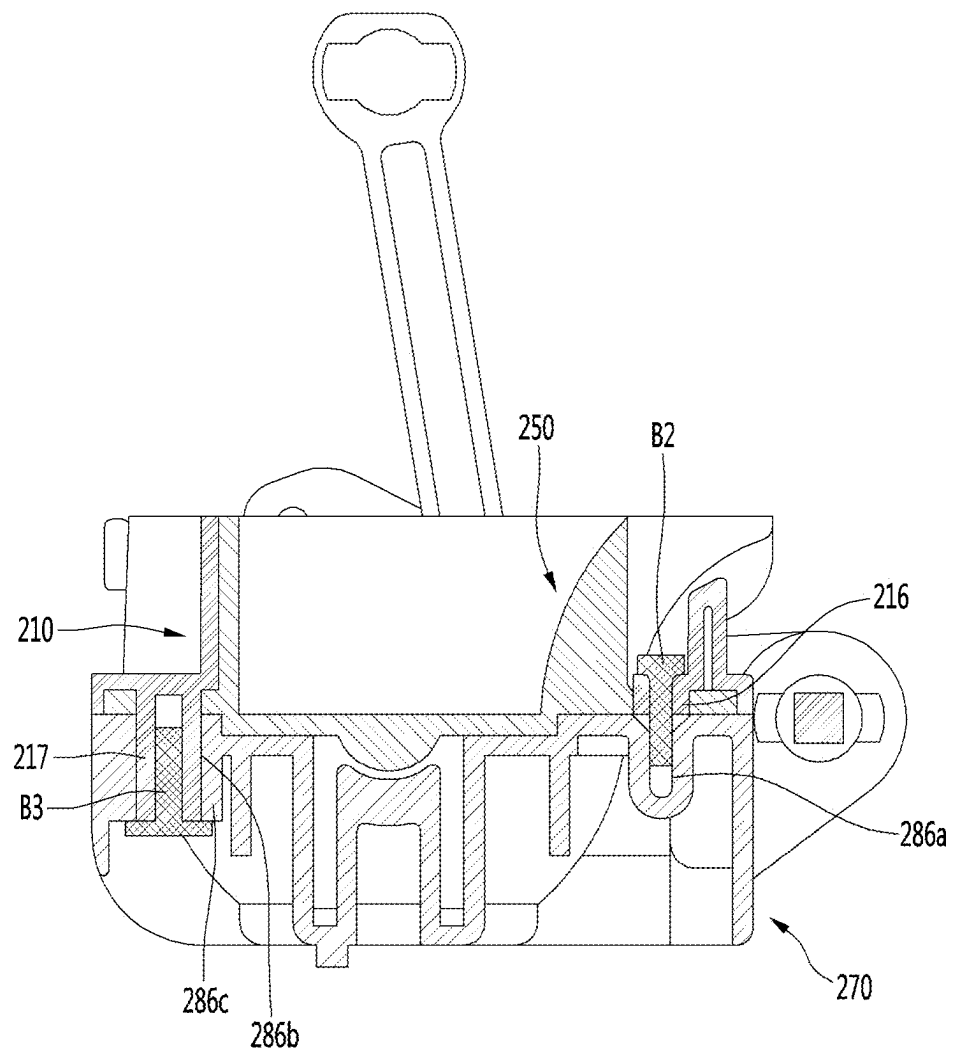


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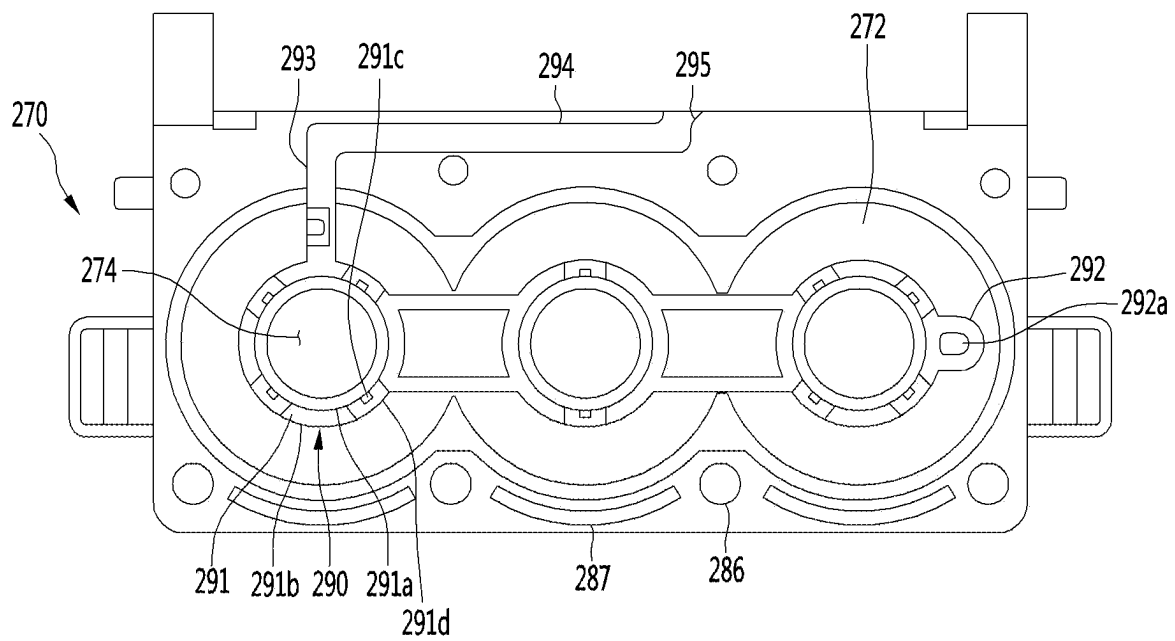


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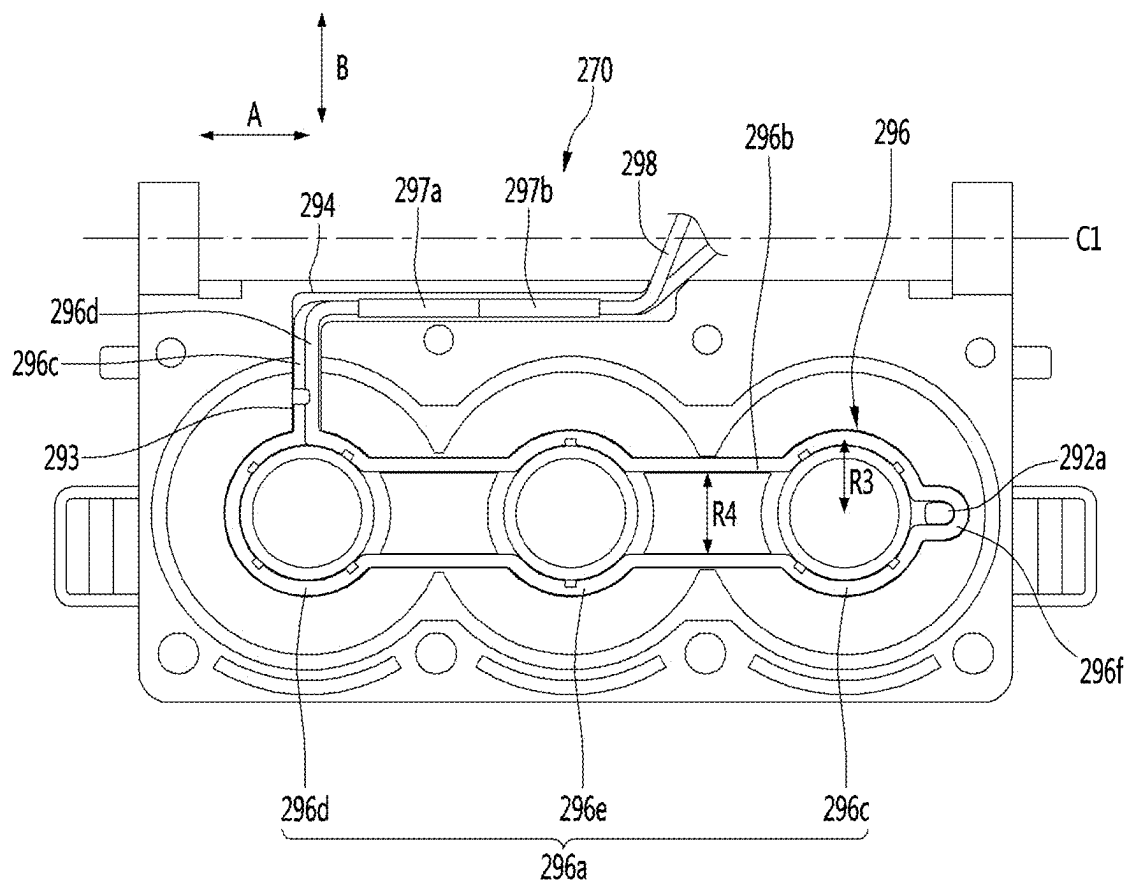


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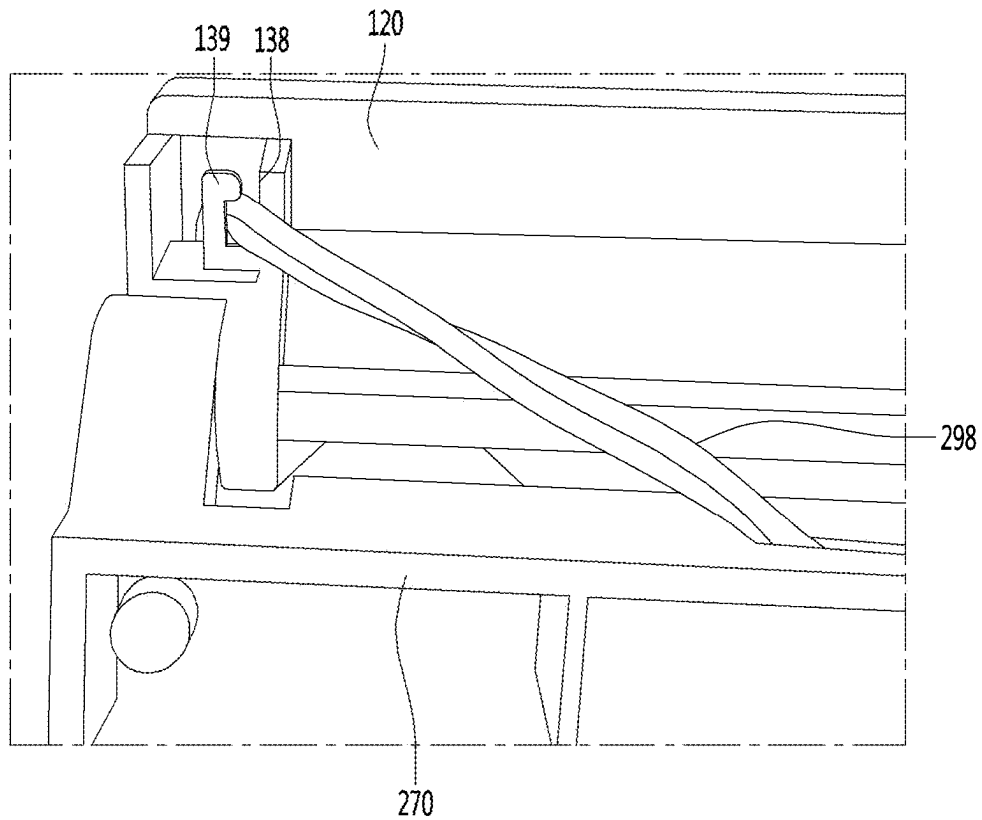


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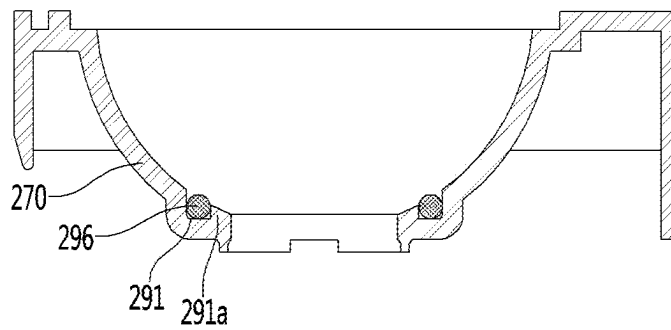


FIG. 31

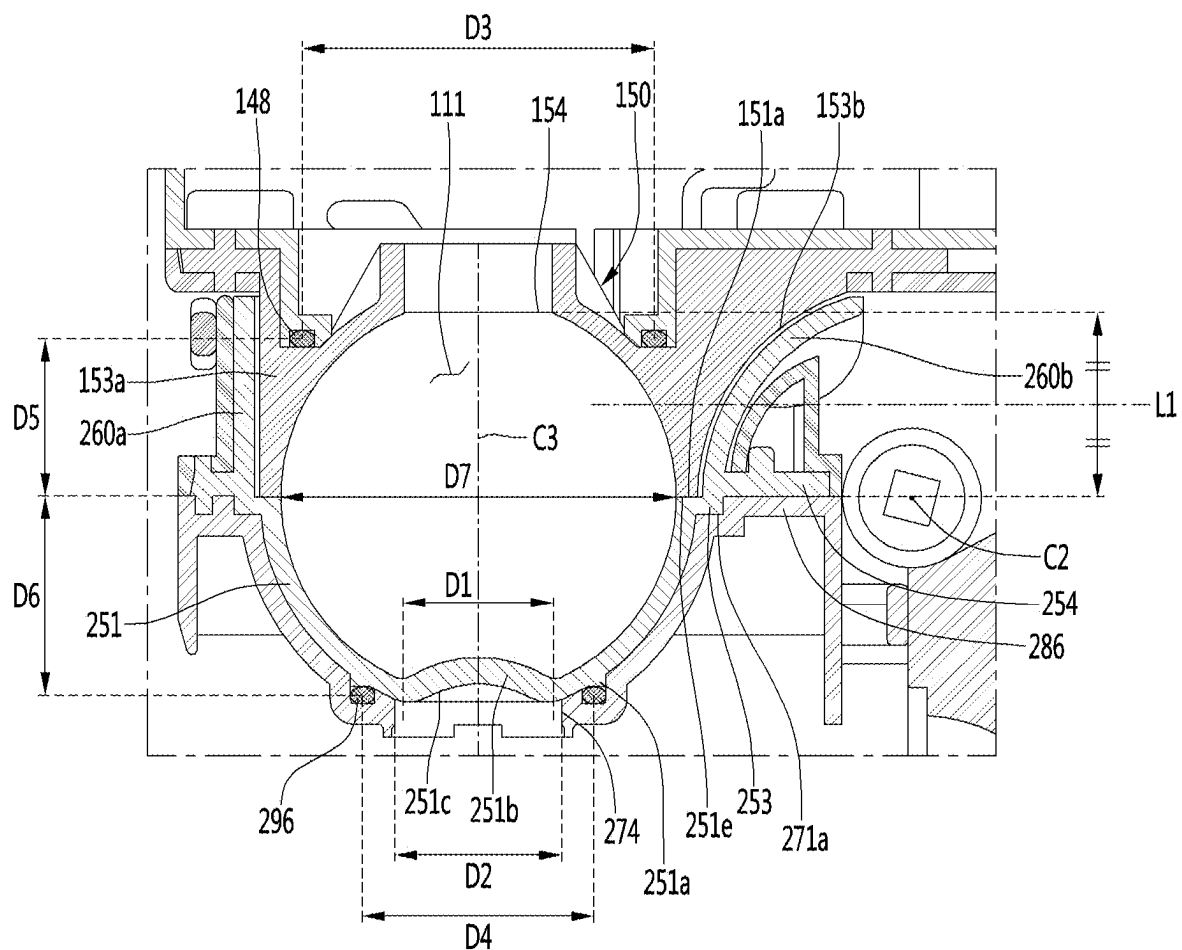


FIG. 32

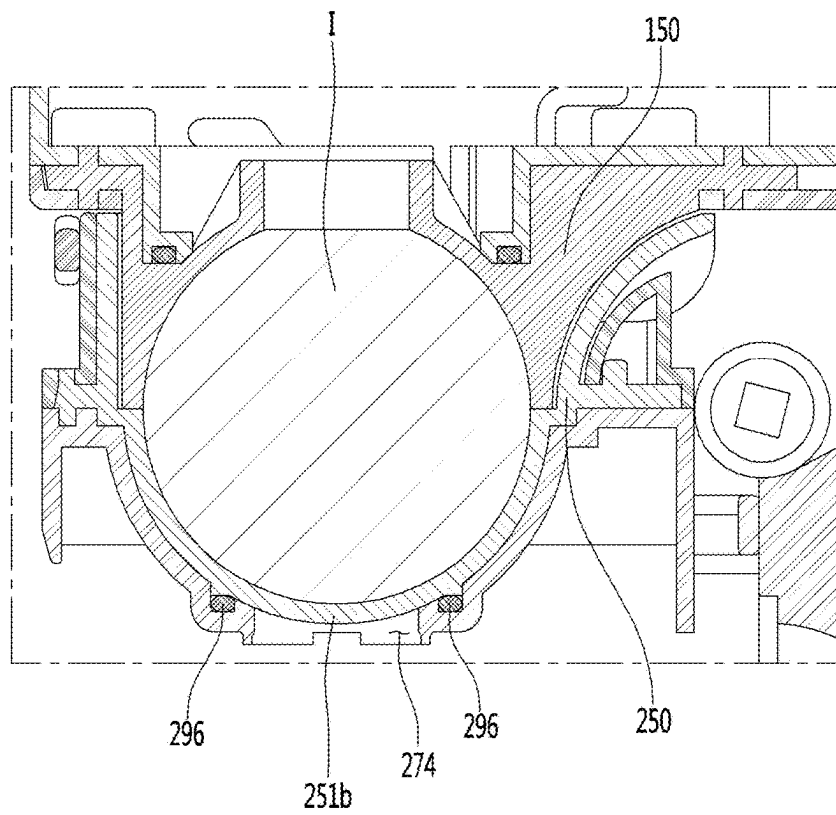


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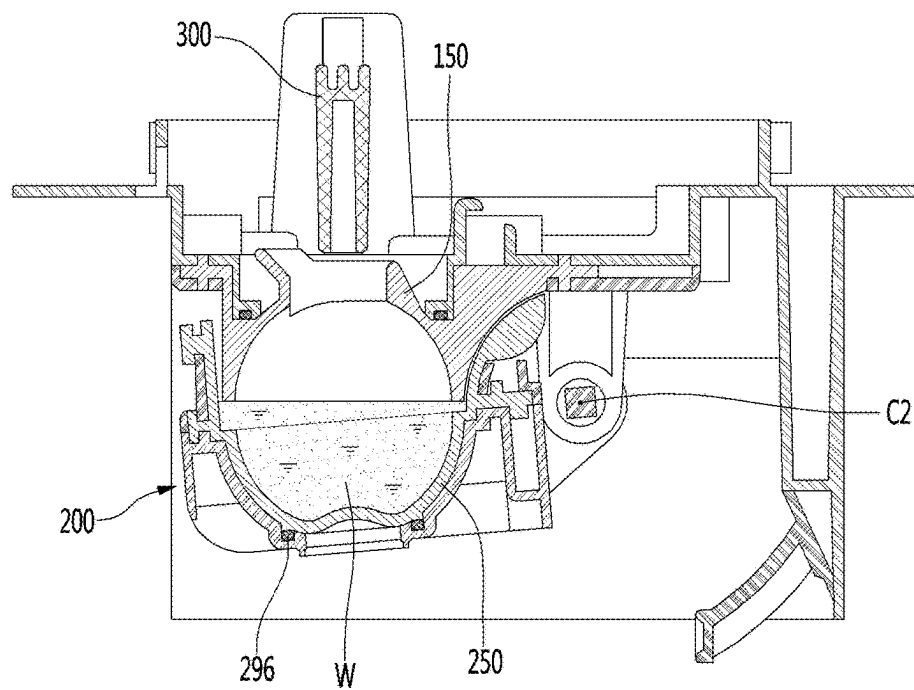


FIG. 34

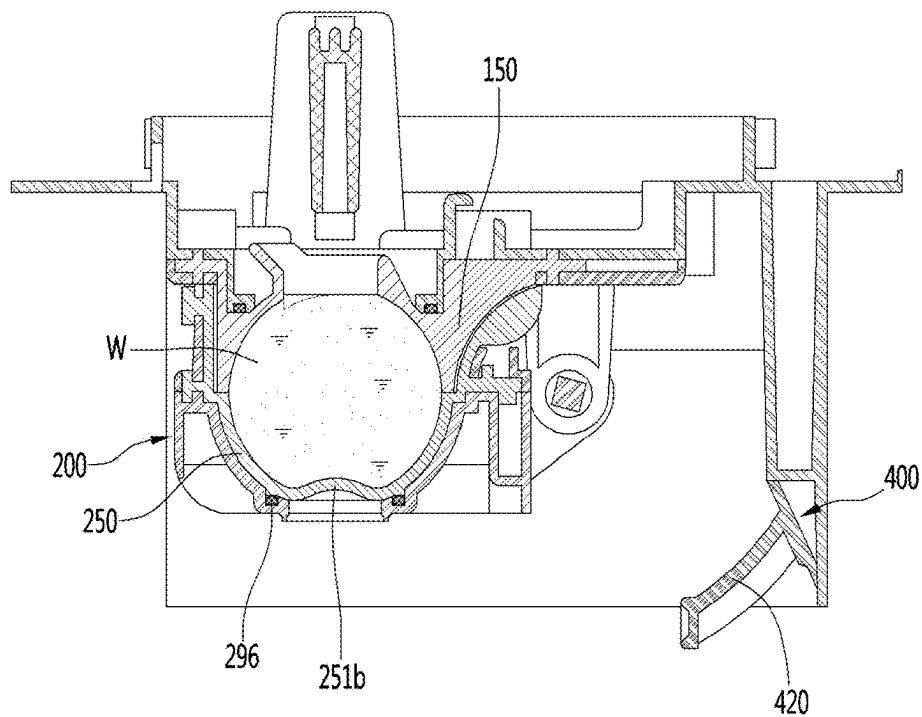


FIG. 35

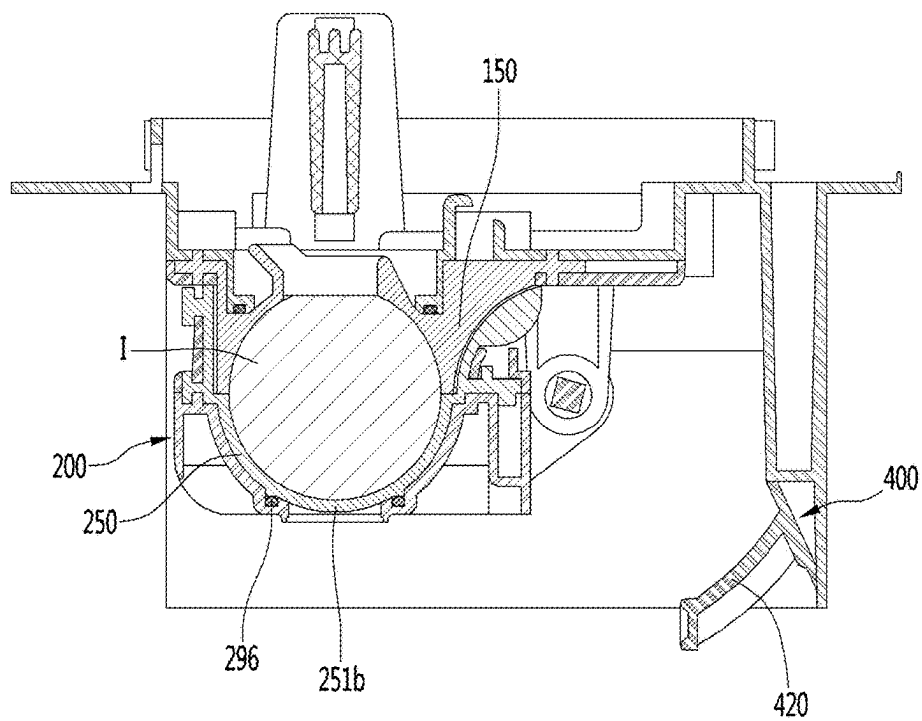


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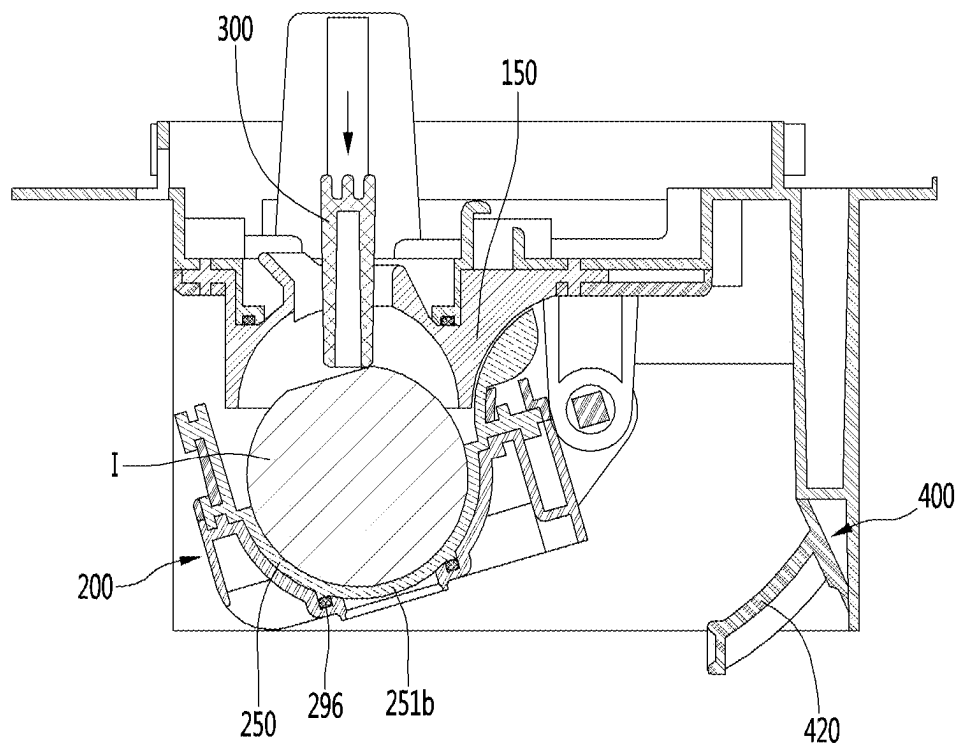


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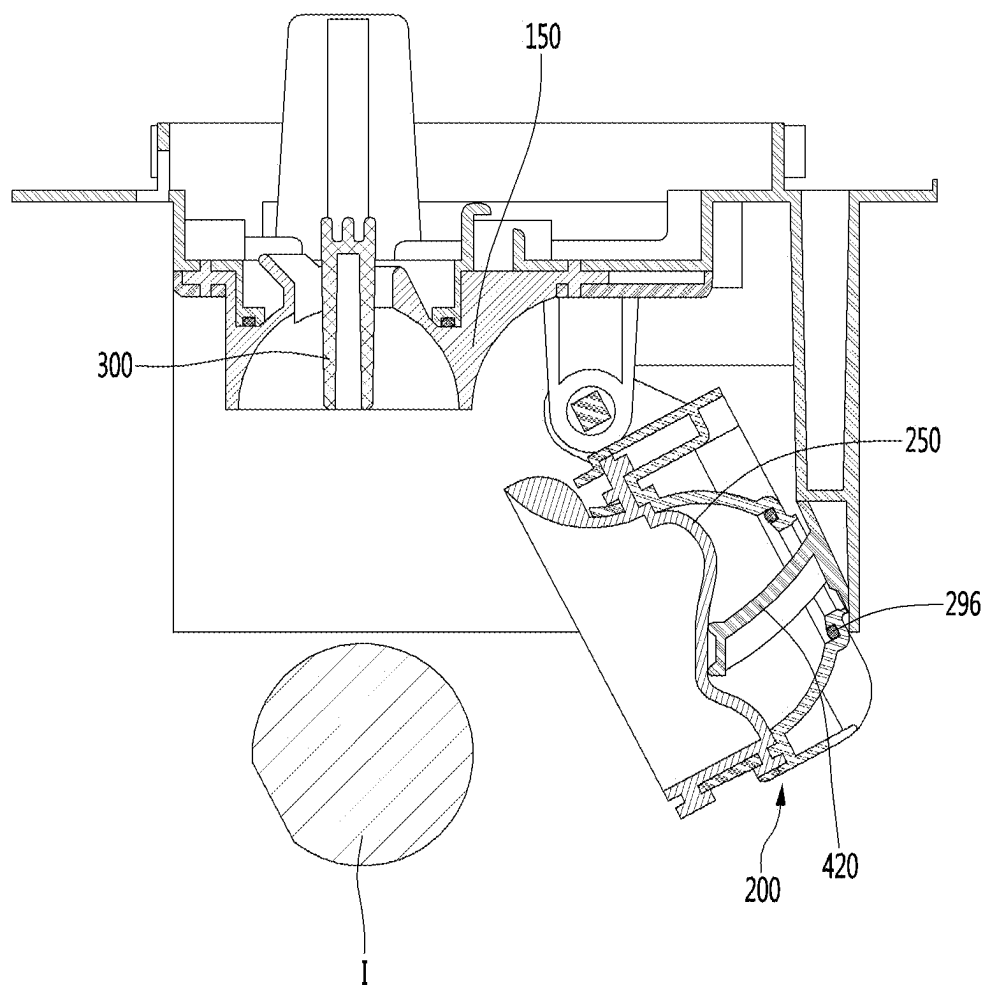


FIG. 38

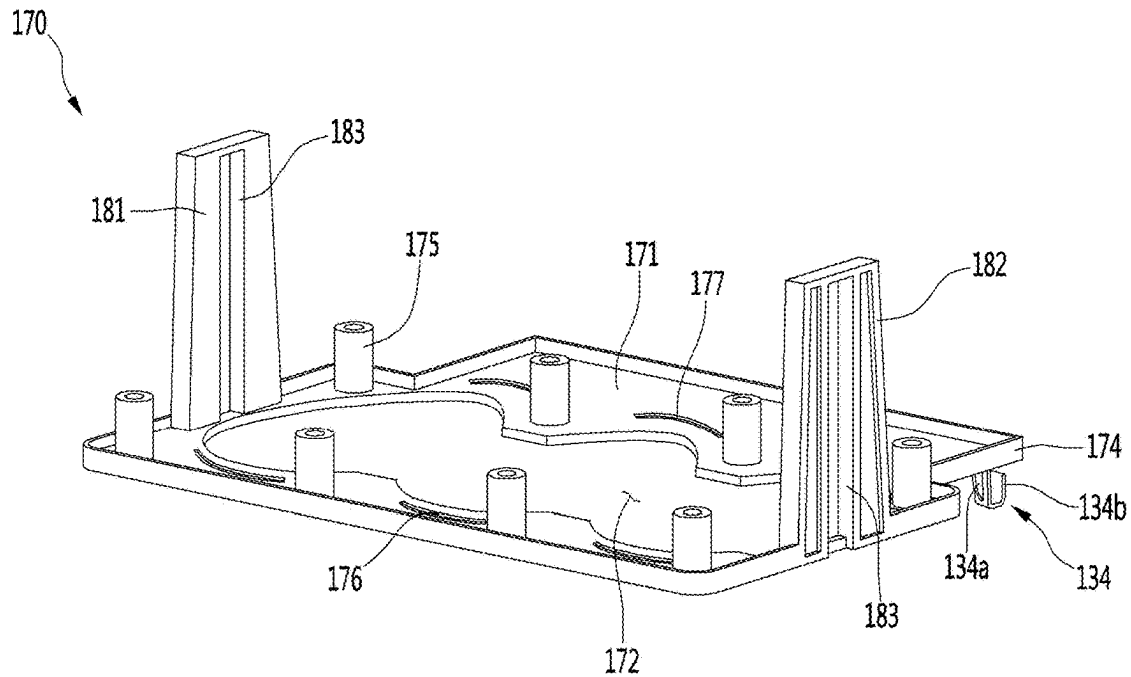


FIG. 39

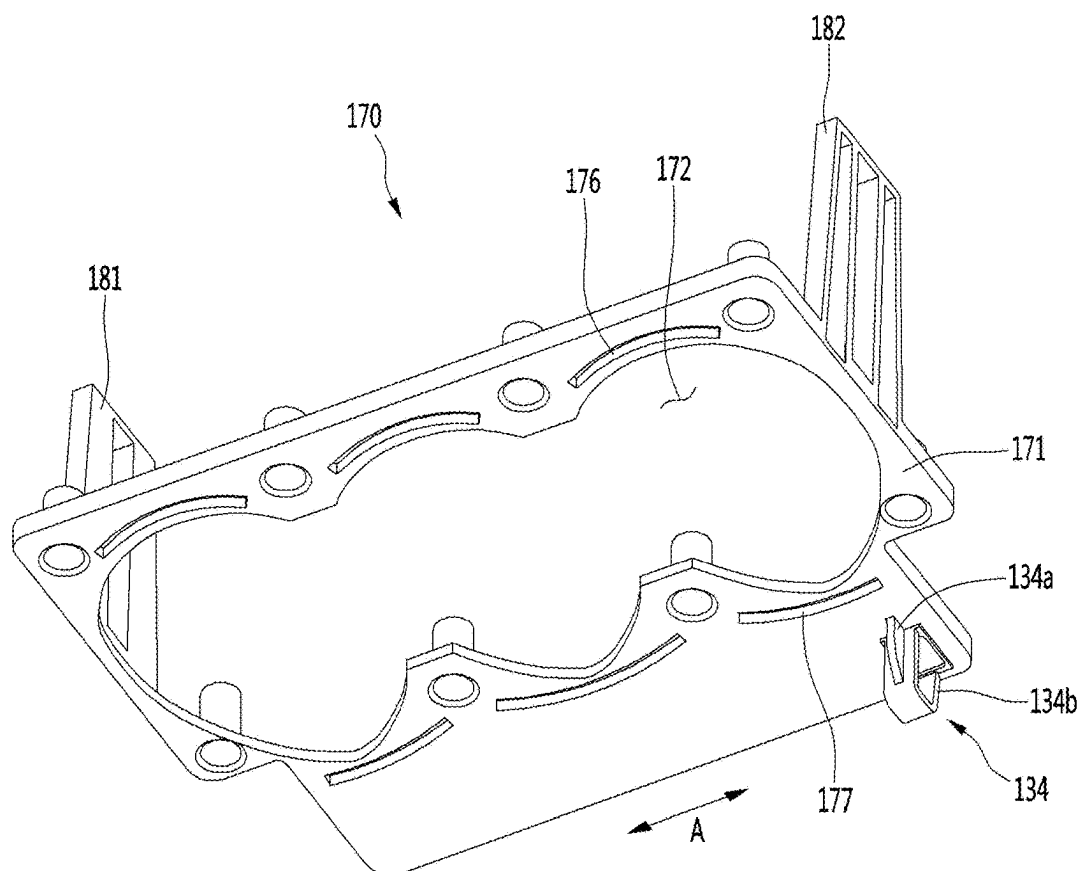


FIG. 40

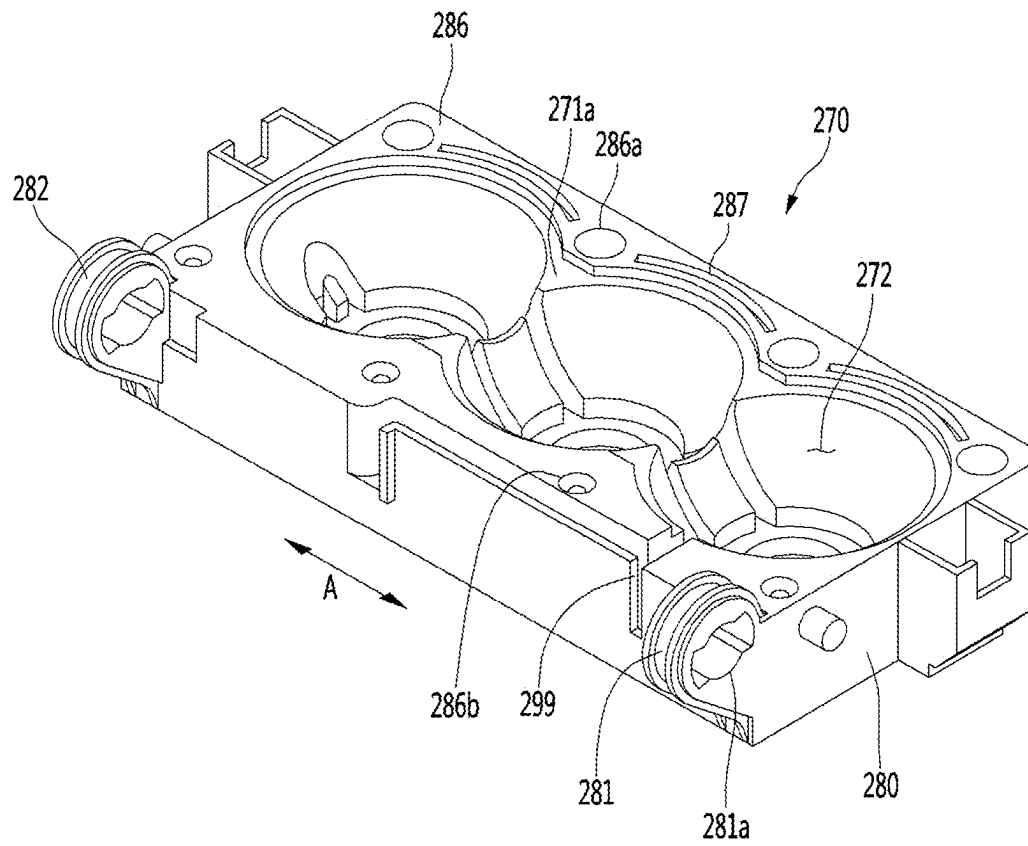


FIG. 41

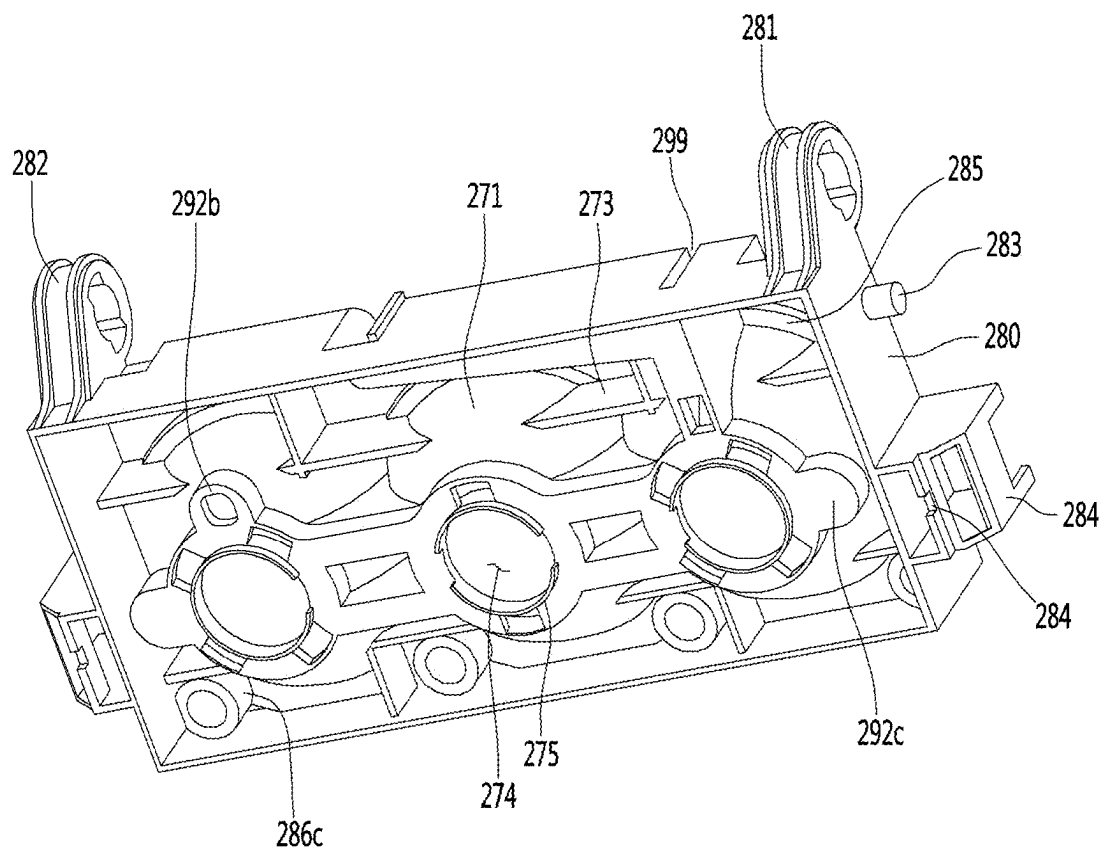


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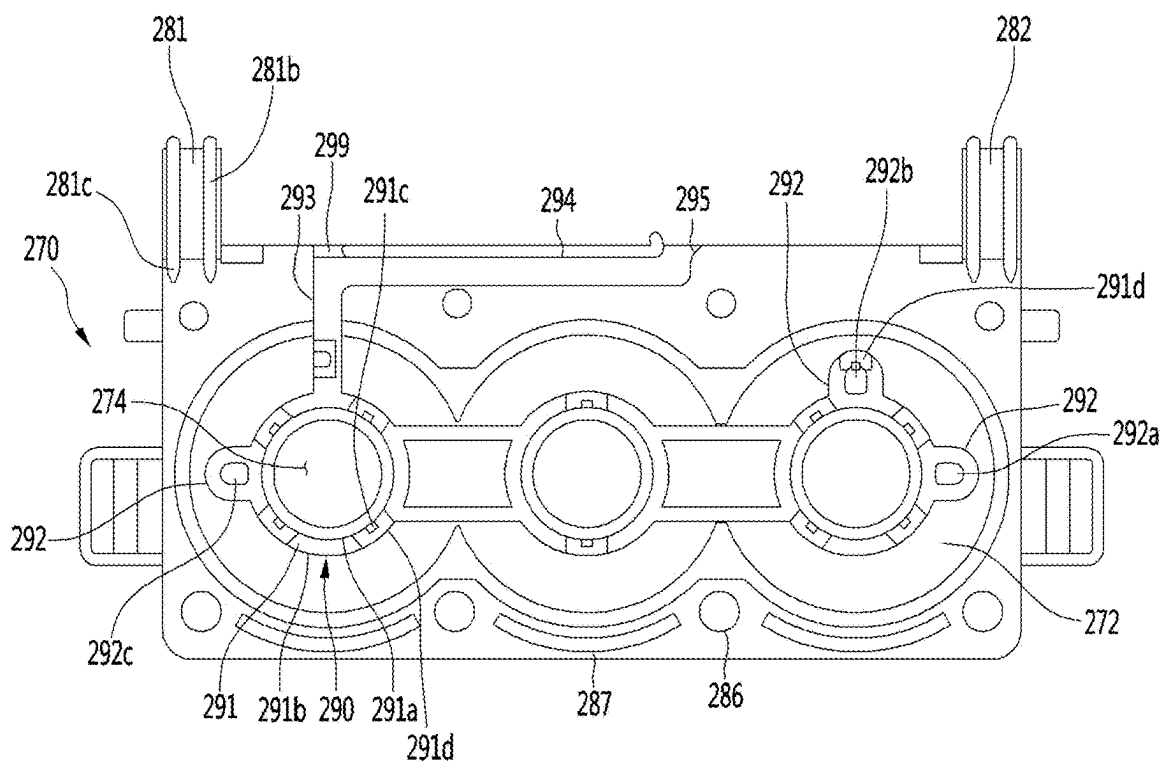


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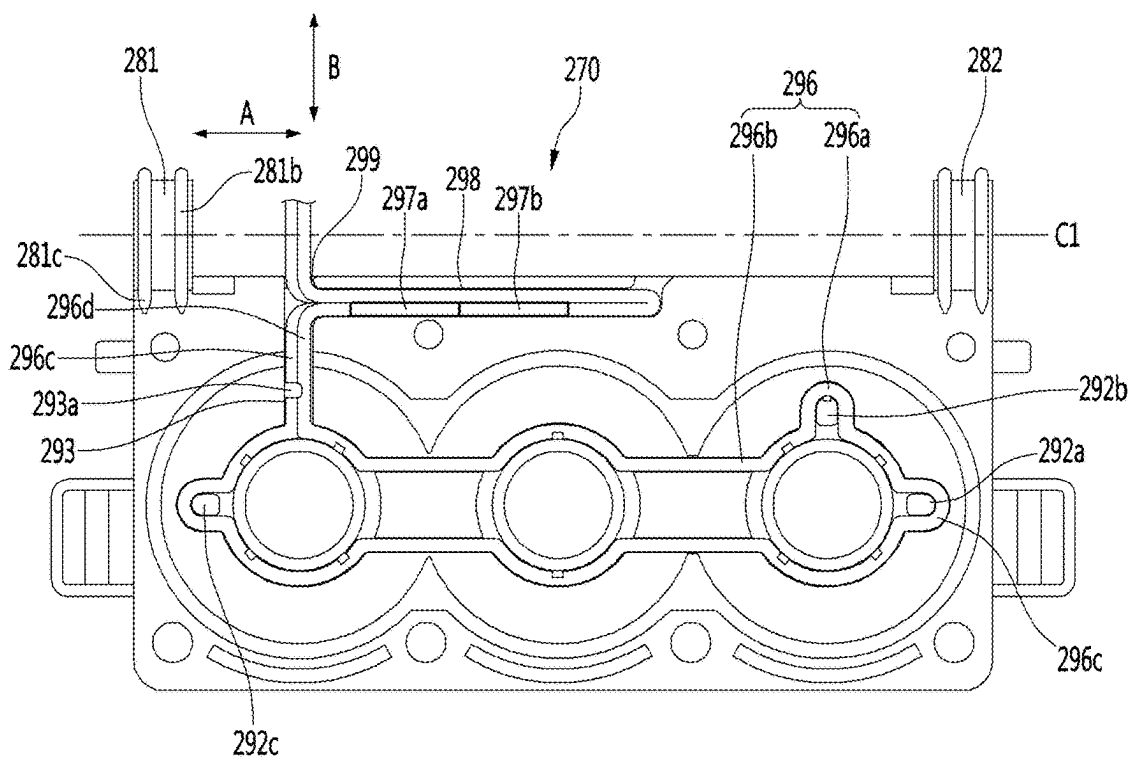


FIG. 44

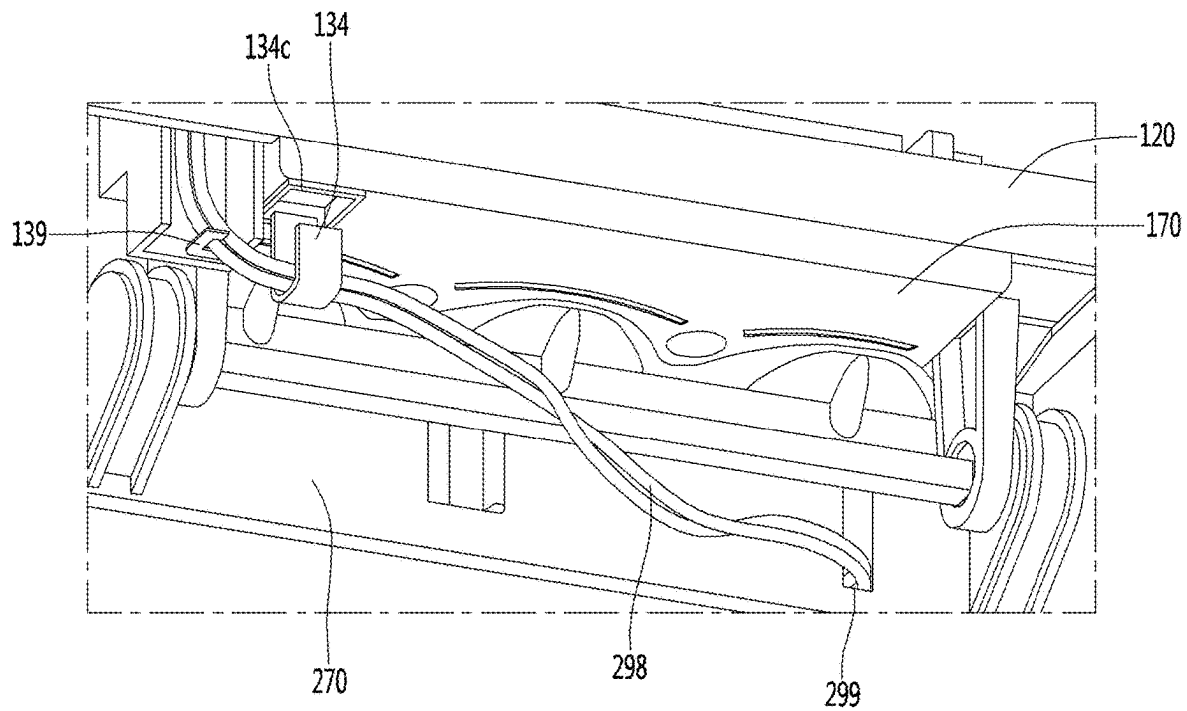
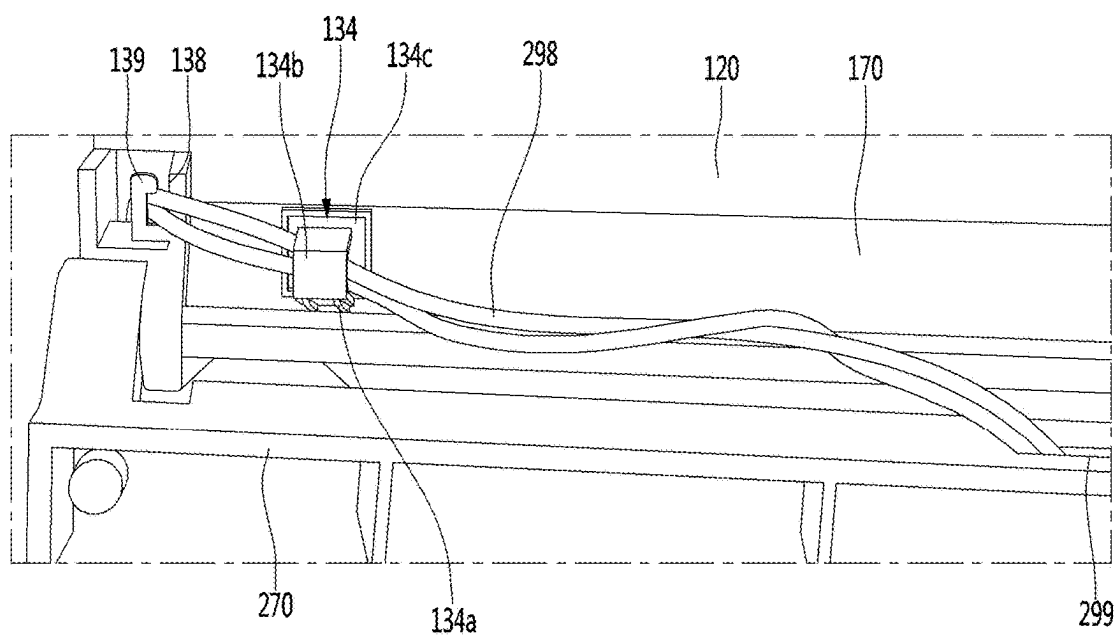


FIG. 45



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ICE MAKER AND REFRIGERATOR**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 16/685,410, filed on Nov. 15, 2019, which claims the benefit of priority to Korean Application No. 10-2018-0142116, filed on Nov. 16, 2018, Korean Application No. 10-2019-0033198, filed on Mar. 22, 2019, and Korean Application No. 10-2019-0088299, filed on Jul. 22, 2019. The disclosures of the prior applications are incorporated by reference in their entirety.

BACKGROUND

The present disclosure relates to an ice maker and a refrigerator.

In general, refrigerators are home appliances for storing foods at a low temperature in a storage space that is covered by a door.

The refrigerator may cool the inside of the storage space by using cold air to store the stored food in a refrigerated or frozen state.

Generally, an ice maker for making ice is provided in the refrigerator.

The ice maker is constructed so that water supplied from a water supply source or a water tank is accommodated in a tray to make ice.

Also, the ice maker is constructed to transfer the made ice from the ice tray in a heating manner or twisting manner.

As described above, the ice maker through which water is automatically supplied, and the ice automatically transferred may be opened upward so that the made ice is pumped up.

As described above, the ice made in the ice maker may have at least one flat surface such as crescent or cubic shape.

When the ice has a spherical shape, it is more convenient to ice the ice, and also, it is possible to provide different feeling of use to a user. Also, even when the made ice is stored, a contact area between the ice cubes may be minimized to minimize a mat of the ice cubes.

Prior art document 1, Korean Patent No. 10-1850918 provides an ice maker.

The ice maker of the prior art document 1 comprises an upper tray having a plurality of hemispherical upper cells arranged thereon and including a pair of link guide portions extending upward from both sides, a lower tray having a plurality of hemispherical lower cells arranged thereon and rotatably connected to the upper tray, and an ice separation heater for heating the upper tray.

In the prior document 1, since the ice separation heater is formed in a U-type shape and is placed on a top surface of the upper tray, heat may not be uniformly provided to the upper cells which the upper tray forms.

Since the ice separation heater contacts the upper tray at a higher position than the upper cells, the time required to transfer the heat of the ice separation heater to a surface of the upper cell may be increased. Therefore, there are disadvantage that it takes a long time to operate the ice separation heater, and accordingly, power consumption is increased.

In addition, since the ice separation heater contacts the upper tray at a higher position than the upper cell, there is a small amount of heat transmitted to a boundary between the upper tray and the lower tray. Therefore, since the

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boundary between the upper tray and the lower tray is not separated well, the lower tray may not be smoothly rotated for ice separation.

In addition, since the ice separation heater exists to be exposed outwards in a state that the ice separation heater is placed on the top surface of the upper tray, the heat of the ice separation heater is not concentrated on the upper tray and is emitted outside of the upper tray, thereby lowering heating efficiency.

In addition, in the prior art document 1, since the ice is frozen from each of the upper cell and the lower cell, bubbles are present in the completed ice, thereby creating opaque ice.

Prior art document 2, Japanese Laid-open Patent publication No. Hei 9-269172 discloses an ice making device.

The ice making device of the prior art document 2 includes an ice making dish, and a heater for heating a bottom of water supplied to the ice making dish.

The ice making dish includes a plurality of ice blocks, and the heater contacts one side and a bottom surface of the ice making block.

In the ice making process, the heat of the heater is transferred to one aspect and the bottom surface of the ice making block. A surface of water proceeds to coagulate, and convection in the water occurs, thereby creating transparent ice.

However, in the prior art document 2, since the ice making dish has a structure of being surrounded by an insulation component in a state that the ice making dish contacts the heater, it is difficult to apply a scheme using the heater of the prior art document 2 to the prior art document 1 having a type of rotating the upper tray.

Even if the heater of the prior art document 2 contacts the upper tray of the prior art document 1, the upper heater may be exposed outwards, and also, the heater is apprehended to interfere with a lower ejecting pin in a rotation process of the lower tray.

SUMMARY

The present embodiment provides an ice maker capable of rapidly providing heat of an upper heater to an upper chamber, and also transferring the heat to a boundary of an upper tray and a lower tray.

The present embodiment provides an ice maker capable of uniformly providing the heat of the upper heater for ice separation to upper chambers.

The present embodiment provides an ice maker of allowing bubbles caused by freezing ice from an upper side to be locally gathered at a lowermost side, thereby making ice transparent as a whole.

The present embodiment provides an ice maker of preventing lower heater for making transparent ice from interfering with a lower ejector in an ice separation process.

The present embodiment provides an ice maker capable of uniformly providing the heat of the upper heater to the lower chambers.

The present embodiment provides an ice maker of enabling the upper heater to be stably maintained in a fixed state.

The present embodiment provides an ice maker capable of reducing a tension of a wire by extending a length of the wire connected to the heater and preventing a disconnection.

The present embodiment provides an ice maker capable of preventing the disconnection of the wire by rotation of a lower assembly although the length of the wire is extended by adding a hook for guiding the wire.

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The present embodiment provides a refrigerator comprising the above-described ice maker.

An ice maker according to one aspect may comprise: an upper tray and a lower tray defining an upper chamber; and an upper heater for providing heat to the upper tray, and a lower heater for providing heat to the lower tray.

In the present invention, on the basis of a height of the ice chamber, a distance from a horizontal central line of the ice chamber to the upper heater may be shorter than a distance from the horizontal central line of the ice chamber to the lower heater.

The horizontal central line means a line passing through a contact surface of the upper tray and the lower tray.

The upper tray may define an upper chamber that is a portion of the ice chamber, and the lower tray may define a lower chamber that is another portion of the ice chamber. The lower chamber may be disposed under the upper chamber.

The upper tray may further comprise an upper opening communicating with the upper chamber and disposed in an upper side of the upper chamber.

The upper heater is disposed closer to the horizontal central line of the ice chamber than the upper opening so that the heat of the upper heater is smoothly transferred to a boundary of the upper tray and the lower tray. As an example, the horizontal central line is a line passing through a contact surface of the upper tray and the lower tray.

The upper heater is disposed in the same height as the height of a bisector of bisecting a distance between the upper opening and the horizontal central line or is higher than the bisector.

The upper heater may comprise an upper round portion surrounding the upper chamber, and a lower round portion surrounding the lower chamber.

A radius of curvature of the upper round portion of the upper heater may be greater than a radius of curvature of the lower round portion of the lower heater.

In this embodiment, the upper tray may include a plurality of upper chambers disposed in a line, and the lower tray may include a plurality of lower chambers disposed in a line.

For uniformly heating the plurality of upper chambers, the upper heater may be disposed to surround each of the plurality of upper chambers.

The lower round portion of the lower heater may be disposed to surround a vertical central line at a position of being spaced apart from the vertical central line of the ice chamber to prevent an interference of the lower ejector and the lower heater.

The vertical central line means a line vertical to the horizontal central line.

For uniformly heating the plurality of lower chambers, the lower heater may be disposed to surround each of the plurality of lower chambers.

The upper heater and the lower heater may be wire-type heaters.

The ice maker may further comprise an upper case contacted by the upper tray, and including a heater coupling part coupled to the upper heater.

The heater coupling part may be accommodated in an accommodation part in a state that the upper heater is coupled to the heater coupling part.

The heater coupling part may comprise a heater accommodation groove into which the upper heater is accommodated.

A diameter of the upper heater may be greater than a recessed depth of the heater accommodation groove to protrude outside of the heater accommodation groove.

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When the heater coupling part coupled to the upper heater is accommodated in the accommodation part, the heater may contact a bottom of the accommodation part.

The heater accommodation groove may include a plurality of rounded portions arranged to surround each of the upper chambers. Two rounded portions that are adjacent to each other may be connected by a linear portion.

The heater coupling part may include an inner wall and an outer wall for forming the heater accommodation groove, and the upper heater may be disposed between the inner wall and the outer wall.

At least one of the inner wall and the outer wall may include a separation prevention protrusion for preventing the upper heater from being separated.

The separation prevention protrusion may protrude from one to the other among the inner wall and the outer wall. A protrusion length of the separation prevention protrusion may be less than half an interval between the inner wall and the outer wall so that the upper heater can be smoothly accommodated in the heater accommodation groove.

The separation prevention protrusions may be disposed in each of the rounded portions of the heater accommodation grooves in order to efficiently prevent a separation of the upper heater curved in a horizontal direction.

A line of straightly connecting two separated points of the upper round portion or both ends of the upper round portion may pass through the upper chamber.

An opening for disposing a portion of the accommodated upper heater may be provided in the heater accommodation groove.

The ice maker according to another aspect may comprise: an upper tray defining an upper chamber that is a portion of an ice chamber; a lower tray rotatable relative to the upper tray and defining a lower chamber that is another portion of the ice chamber, wherein the lower chamber is disposed under the upper chamber; an upper heater disposed around the upper tray, for providing heat to the upper chamber; and a lower heater disposed around the lower tray, for providing heat to the lower chamber.

Based on the height of the ice chamber, a distance from a horizontal central line of the ice chamber to the upper heater may be shorter than a distance from the horizontal central line of the ice chamber to the lower heater.

The upper heater may comprise an upper round portion surrounding the upper chamber, and the lower heater may comprise a lower round portion surrounding the lower chamber.

Each of the upper round portion and the lower round portion of the upper heater may be disposed to vertically overlap the ice chamber.

A radius of curvature of the upper round portion of the upper heater may be greater than a radius of curvature of the lower round portion.

The lower round portion of the lower heater may be spaced apart from a vertical central line of the ice chamber and may be disposed to surround the vertical central line.

A distance between two points disposed in opposite sides based on the vertical central line of the ice chamber in the lower round portion of the lower heater may be smaller than a diameter of the ice chamber.

A distance between two points disposed in the opposite sides based on the vertical central line in the upper round portion of the upper heater may be greater than a distance between two points disposed in the opposite sides based on the vertical central line in the lower round portion of the lower heater.

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The upper tray may further comprise an upper opening communicating with the upper chamber and disposed in an upper side of the upper chamber. The upper heater is disposed closer to the horizontal central line of the ice chamber than the upper opening. The horizontal central line is a line passing through a contact surface of the upper tray and the lower tray.

The upper heater may be disposed in the same height as the height of a bisector of bisecting a distance between the upper opening and the horizontal central line or may be higher than the bisector.

The upper tray may include a plurality of upper chambers disposed in a line, and the lower tray may include a plurality of lower chambers disposed in a line. The upper heater may be disposed to surround each of the plurality of upper chambers, and the lower heater may be disposed to surround each of the plurality of lower chambers.

The upper heater may comprise upper round portions surrounding each of the plurality of upper chambers, and a linear portion connecting two upper round portions adjacent to each other.

The upper round portions may include a first upper round portion surrounding an upper chamber disposed in an outermost portion in the plurality of the upper chambers.

Both sides of the first upper round portion may be connected by a pair of upper linear portions, and a distance between the pair of upper linear portions may less than double in a radius of curvature of the first upper round portion.

A distance between the pair of upper linear portions may be equal to or greater than the radius of curvature of the first upper round portion.

The lower heater may comprise lower round portions surrounding each of the plurality of lower chambers, and a linear portion connecting two lower round portions adjacent to each other.

The lower round portions may comprise a first rounded portion surrounding a lower chamber arranged in an outermost portion in the plurality of lower chambers.

Both sides of the first upper round portion may be connected by a pair of linear portions, and a distance between the pair of linear portions may less than double in a radius of curvature of the first upper round portion.

A distance between the pair of linear portions may be equal to or greater than the radius of curvature of the first upper round portion.

A refrigerator according to another aspect may comprise: a cabinet provided with a storage space; a door opening or closing the storage space; and an ice maker for making ice by cold air of the storage space.

The ice maker comprises: an upper tray defining a portion of an ice chamber having a spherical shape; a lower tray defining another portion of the ice chamber; an upper heater for providing heat to the upper tray; and a lower heater for providing heat to the lower tray.

At least a portion of each of the upper heater and the lower heater may be disposed to vertically overlap the ice chamber. The upper tray may comprise an upper opening. A line of bisecting a distance between the upper opening and the horizontal central line of the ice chamber may be named a bisector.

The upper heater may be disposed between the bisector and the upper opening.

The ice maker according to another aspect may comprise an upper assembly including an upper tray with an upper chamber recessed upwards so as to define an upper side of an ice chamber in which water is fully entered to make ice,

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and a lower assembly including a lower tray with a lower chamber recessed downwards so as to define a lower side of an ice chamber.

The lower assembly may include may comprise a lower support supporting a lower side of the lower tray and including a heater coupling part.

The ice maker may include a heater coupled to the heater coupling part of the lower support, and capable of providing heat to a plurality of lower chambers to make the made ice transparent.

The heater may be operated in an ice making process. When the heater is operated, the ice may be gradually made in the upper chamber.

The lower assembly may comprise an upper support contacting one side of the upper tray. The upper support may comprise a wire guiding hook extending downwards, for guiding a wire connected to the heater.

The upper support may comprise a plurality of chamber accommodation parts for accommodating the plurality of lower chambers. The heater coupling part may comprise heater accommodation grooves recessed in the plurality of chamber accommodation parts.

A diameter of the heater may be greater than a recessed depth of the heater accommodation groove. Therefore, the heater may contact the lower tray.

The heater accommodation groove may comprise a plurality of rounded portions disposed to surround each of the lower chambers, and a linear portion connected to the plurality of rounded portions.

The heater is a wire-type heater, and when the heater is accommodated in a plurality of rounded portions of the heater accommodation groove, the heater may be curved in a shape corresponding to the plurality of rounded portions.

The heater coupling part may comprise an inner wall and an outer wall for forming the heater accommodation groove. The heater may be accommodated between the inner wall and the outer wall, and at least one of the inner wall and the outer wall may include a separation prevention protrusion for preventing the upper heater from being separated.

The separation prevention protrusion may protrude from one to the other among the inner wall and the outer wall. A protrusion length of the separation prevention protrusion may be less than half an interval between the inner wall and the outer wall.

A penetration opening may be provided in the heater accommodation groove so that a portion of the accommodated heater is positioned therein.

In this embodiment, the lower tray body may include a heater contact part protruding such that the heater is contacted. A bottom surface of the heater contact part is a plane, and the bottom surface may be contacted by the heater.

The heater may be lower than an intermediate point of a height of the lower chamber in a state that the heater contacts the lower tray.

The lower support may comprise a first guide groove extending from one lower chamber among the plurality of lower chambers and accommodating the heater, and a second guide groove extending in a direction of intersecting at the first guide home and guiding a wire connected to the heater.

The lower support may be rotatable on the basis of a rotational central axis, and the second guide groove may extend in a direction of aligning the rotational central axis.

A power input terminal and a power output terminal of the heater may be disposed in the first guide groove. The power input terminal and the power output terminal may be con-

nected to a first connector. The first connector may be connected to a second connector connected to the wire.

The first connector and the second connector may be disposed in the second guide groove.

The plurality of lower chambers may be disposed in a line, and the one lower chamber and the other lower chamber disposed farthest among the plurality of lower chambers may further comprise a detour accommodation groove extending from the heater accommodation groove.

The wire guiding hook may comprise a curving part formed to be curved one or more times and a support part extending to a bottom surface of the upper support for supporting the curving part.

The wire may be withdrawn outside of the lower support through a withdrawing slot included in an end of the second guide groove by reciprocating the second guide groove.

A refrigerator according to another aspect may comprise a cabinet provided in a freezer; a housing provided in the freezer; and an ice maker installed in the housing.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a refrigerator according to one embodiment of the present disclosure.

FIG. 2 is a view showing a state in which a door of the refrigerator of FIG. 1 is opened.

FIG. 3 and FIG. 4 is a perspective view of an ice maker according to one embodiment of the present disclosure.

FIG. 5 is an exploded perspective view of an ice maker according to one embodiment of the present disclosure.

FIG. 6 is a top perspective view of an upper case according to one embodiment of the present disclosure.

FIG. 7 is a bottom perspective view of an upper case according to one embodiment of the present disclosure.

FIG. 8 is a top perspective view of an upper tray according to one embodiment of the present disclosure.

FIG. 9 is a bottom perspective view of an upper tray according to one embodiment of the present disclosure.

FIG. 10 is a side elevation view of an upper tray according to one embodiment of the present disclosure.

FIG. 11 is a top perspective view of an upper support according to one embodiment of the present disclosure.

FIG. 12 is a bottom perspective view of an upper support according to one embodiment of the present disclosure.

FIG. 13 is an enlarged view showing a heater coupling portion in the upper case of FIG. 6.

FIG. 14 is a view showing a state in which a heater is coupled to the upper case of FIG. 6.

FIG. 15 is a view showing a layout of a wire connected to the heater in the upper case.

FIG. 16 is a sectional view showing a state in which the upper assembly has been assembled.

FIG. 17 is a perspective view of a lower assembly according to one embodiment of the present disclosure.

FIG. 18 is a top perspective view of a lower case according to one embodiment of the present disclosure.

FIG. 19 is a bottom perspective view of a lower case according to one embodiment of the present disclosure.

FIG. 20 is a top perspective view of a lower tray according to one embodiment of the present disclosure.

FIG. 21 and FIG. 22 are bottom perspective views of a lower tray according to one embodiment of the present disclosure.

FIG. 23 is a side elevation view of a lower tray according to one embodiment of the present disclosure.

FIG. 24 is a top perspective view of a lower support according to one embodiment of the present disclosure.

FIG. 25 is a bottom perspective view of a lower support according to one embodiment of the present disclosure.

FIG. 26 is a cross-sectional view taken along line D-D of FIG. 17 for showing a state that a lower assembly is assembled.

FIG. 27 is a plan view of a lower support according to one embodiment of the present disclosure.

FIG. 28 is a perspective view showing a state in which a lower heater is coupled to a lower support of FIG. 27.

FIG. 29 is a view showing a state in which a lower assembly is coupled to an upper assembly and, at the same time, a wire connected to a lower heater penetrates an upper case.

FIG. 30 is a sectional view showing a state in which a lower heater is installed on a lower support.

FIG. 31 is a cross-sectional view taken along line A-A of FIG. 3.

FIG. 32 is a view showing a state in which ice generation is completed in FIG. 31.

FIG. 33 is a cross-sectional view taken along line B-B of FIG. 3 in a water supplied state.

FIG. 34 is a cross-sectional view taken along line B-B of FIG. 3 in the ice-making state.

FIG. 35 is a cross-sectional view taken along line B-B of FIG. 3 in the ice-making completed state.

FIG. 36 is a cross-sectional view taken along line B-B of FIG. 3 in an initial state of ice separation.

FIG. 37 is a cross-sectional view taken along line B-B of FIG. 3 in an ice-separation completed state.

FIG. 38 is an upper perspective view of an upper support according to another embodiment of the present invention.

FIG. 39 is a lower perspective view of the upper support according to another embodiment of the present invention.

FIG. 40 is an upper perspective view of a lower support according to another embodiment of the present invention.

FIG. 41 is a lower perspective view of the lower support according to another embodiment of the present invention.

FIG. 42 is a top plan view of the lower support according to another embodiment of the present invention.

FIG. 43 is a perspective view that the lower heater is coupled to the lower support of FIG. 42.

FIG. 44 is a view showing a state in which a wire connected to the lower heater penetrates an upper case in a state that a lower assembly is coupled to an upper assembly.

FIG. 45 is a bottom view showing a state in which a wire connected to the lower heater penetrates an upper case in a state that a lower assembly is coupled to an upper assembly.

DETAILED DESCRIPTION

FIG. 1 is a perspective view of a refrigerator according to an embodiment, and FIG. 2 is a view illustrating a state in which a door of the refrigerator of FIG. 1 is opened.

Referring to FIGS. 1 and 2, a refrigerator 1 according to an embodiment may include a cabinet 2 defining a storage space and a door that opens and closes the storage space.

In detail, the cabinet 2 may define the storage space that is vertically divided by a barrier. Here, a refrigerating compartment 3 may be defined at an upper side, and a freezing compartment 4 may be defined at a lower side.

Accommodation members such as a drawer, a shelf, a basket, and the like may be provided in the refrigerating compartment 3 and the freezing compartment 4.

The door may include a refrigerating compartment door 5 opening/closing the refrigerating compartment 3 and a freezing compartment door 6 opening/closing the freezing compartment 4.

The refrigerating compartment door **5** may be constituted by a pair of left and right doors and be opened and closed through rotation thereof. Also, the freezing compartment door **6** may be inserted and withdrawn in a drawer manner.

Alternatively, the arrangement of the refrigerating compartment **3** and the freezing compartment **4** and the shape of the door may be changed according to kinds of refrigerators, but are not limited thereto. For example, the embodiments may be applied to various kinds of refrigerators. For example, the freezing compartment **4** and the refrigerating compartment **3** may be disposed at left and right sides, or the freezing compartment **4** may be disposed above the refrigerating compartment **3**.

An ice maker **100** may be provided in the freezing compartment **4**. The ice maker **100** is constructed to make ice by using supplied water. Here, the ice may have a spherical shape.

Also, an ice bin **102** in which the ice is stored after being transferred from the ice maker **100** may be further provided below the ice maker **100**.

The ice maker **100** and the ice bin **102** may be mounted in the freezing compartment **4** in a state of being respectively mounted in separate housings **101**.

A user may open the refrigerating compartment door **6** to approach the ice bin **102**, thereby obtaining the ice.

In another example, a dispenser for dispensing purified water or the made ice to the outside may be provided in the refrigerating compartment door **5**.

Also, the ice made in the ice maker **100** or the ice stored in the ice bin **102** after being made in the ice maker **100** may be transferred to the dispenser by a transfer unit. Thus, the user may obtain the ice from the dispenser.

Hereinafter, the ice maker will be described in detail with reference to the accompanying drawings.

FIGS. **3** and **4** are perspective views of the ice maker according to an embodiment, and FIG. **5** is an exploded perspective view of the ice maker according to an embodiment.

Referring to FIGS. **3** to **5**, the ice maker **100** may include an upper assembly **110** and a lower assembly **200**.

The lower assembly **200** may rotate with respect to the upper assembly **110**. For example, the lower assembly **200** may be connected to be rotatable with respect to the upper assembly **110**.

In a state in which the lower assembly **200** contacts the upper assembly **110**, the lower assembly **200** together with the upper assembly **110** may make spherical ice.

That is, the upper assembly **110** and the lower assembly **200** may define an ice chamber **111** for making the spherical ice. The ice chamber **111** may have a chamber having a substantially spherical shape.

As used herein, a term "spherical or hemisphere form" not only includes a geometrically complete sphere or hemisphere form but also a geometrically complete sphere-like or geometrically complete hemisphere-like form.

The upper assembly **110** and the lower assembly **200** may define a plurality of ice chambers **111**.

Hereinafter, a structure in which three ice chambers are defined by the upper assembly **110** and the lower assembly **200** will be described as an example, and also, the embodiments are not limited to the number of ice chambers **111**.

In the state in which the ice chamber **111** is defined by the upper assembly **110** and the lower assembly **200**, water is supplied to the ice chamber **111** through a water supply part **190**.

The water supply part **190** is coupled to the upper assembly **110** to guide water supplied from the outside to the ice chamber **111**.

After the ice is made, the lower assembly **200** may rotate in a forward direction. Thus, the spherical ice made between the upper assembly **110** and the lower assembly **200** may be separated from the upper assembly **110** and the lower assembly **200**.

The ice maker **100** may further include a driving unit **180** so that the lower assembly **200** is rotatable with respect to the upper assembly **110**.

The driving unit **180** may include a driving motor and a power transmission part for transmitting power of the driving motor to the lower assembly **200**. The power transmission part may include one or more gears.

The driving motor may be a bi-directional rotatable motor. Thus, the lower assembly **200** may rotate in both directions.

The ice maker **100** may further include an upper ejector **300** so that the ice is capable of being separated from the upper assembly **110**.

The upper ejector **300** may be constructed so that the ice closely attached to the upper assembly **110** is separated from the upper assembly **110**.

The upper ejector **300** may include an ejector body **310** and a plurality of upper ejecting pins **320** extending in a direction crossing the ejector body **310**.

The upper ejecting pins **320** may be provided in the same number of ice chambers **111**.

A separation prevention protrusion **312** for preventing a connection unit **350** from being separated in the state of being coupled to the connection unit **350** that will be described later may be provided on each of both ends of the ejector body **310**.

For example, the pair of separation prevention protrusions **312** may protrude in opposite directions from the ejector body **310**.

While the upper ejecting pin **320** passing through the upper assembly **110** and inserted into the ice chamber **111**, the ice within the ice chamber **111** may be pressed.

The ice pressed by the upper ejecting pin **320** may be separated from the upper assembly **110**.

Also, the ice maker **100** may further include a lower ejector **400** so that the ice closely attached to the lower assembly **200** is capable of being separated.

The lower ejector **400** may press the lower assembly **200** to separate the ice closely attached to the lower assembly **200** from the lower assembly **200**. For example, the lower ejector **400** may be fixed to the upper assembly **110**.

The lower ejector **400** may include an ejector body **410** and a plurality of lower ejecting pins **420** protruding from the ejector body **410**. The lower ejecting pins **420** may be provided in the same number of ice chambers **111**.

While the lower assembly **200** rotates to transfer the ice, rotation force of the lower assembly **200** may be transmitted to the upper ejector **300**.

For this, the ice maker **100** may further include the connection unit **350** connecting the lower assembly **200** to the upper ejector **300**. The connection unit **350** may include one or more links.

For example, when the lower assembly **200** rotates in one direction, the upper ejector **300** may descend by the connection unit **350** to allow the upper ejector pin **320** to press the ice.

On the other hand, when the lower assembly **200** rotates in the other direction, the upper ejector **300** may ascend by the connection unit **350** to return to its original position.

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Hereinafter, the upper assembly **110** and the lower assembly **200** will be described in more detail.

The upper assembly **110** may include an upper tray **150** defining a portion of the ice chamber **111** making the ice. For example, the upper tray **150** may define an upper portion of the ice chamber **111**.

The upper assembly **110** may further include an upper support **170** fixing a position of the upper tray **150**.

The upper support **170** may restrict downward movement of the upper tray **150**.

The upper assembly **110** may further include an upper case **120** fixing a position of the upper tray **150**.

The upper tray **150** may be disposed below the upper case **120**.

As described above, the upper case **120**, the upper tray **150**, and the upper support **170**, which are vertically aligned, may be coupled to each other through a coupling member.

That is, the upper tray **150** may be fixed to the upper case **120** through coupling of the coupling member.

For example, the water supply part **190** may be fixed to the upper case **120**.

The ice maker **100** may further include a temperature sensor **500** detecting a temperature of the ice chamber **111**.

In one example, the temperature sensor **500** detects the temperature of the upper tray **150** thus to indirectly detect the temperature of the water or the temperature of the ice in the ice chamber **111**.

For example, the temperature sensor **500** may be mounted on the upper case **120**. Also, when the upper tray **150** is fixed to the upper case **120**, the temperature sensor **500** may contact the upper tray **150**.

The lower assembly **200** may include a lower tray **250** defining the other portion of the ice chamber **111** making the ice. For example, the lower tray **250** may define a lower portion of the ice chamber **111**.

The lower assembly **200** may further include a lower support **270** supporting a lower portion of the lower tray **250**.

The lower assembly **200** may further include a lower case **210** of which at least a portion covers an upper side of the lower tray **250**.

The lower case **210**, the lower tray **250**, and the lower support **270** may be coupled to each other through a coupling member.

The ice maker **100** may further include a switch for turning on/off the ice maker **100**. When the user turns on the switch **600**, the ice maker **100** may make ice.

That is, when the switch **600** is turned on, water may be supplied to the ice maker **100**. Then, an ice making process of making ice by using cold air and an ice separating process of transferring the ice through the rotation of the lower assembly **200**.

On the other hand, when the switch **600** is manipulated to be turned off, the making of the ice through the ice maker **100** may be impossible. For example, the switch **600** may be provided in the upper case **120**.

<Upper Case>

FIG. **6** is a top perspective view of the upper case according to an embodiment, and FIG. **7** is a bottom perspective view of the upper case according to an embodiment.

Referring to FIGS. **6** and **7**, the upper case **120** may be fixed to a housing **101** within the freezing compartment **4** in a state in which the upper tray **150** is fixed.

The upper case **120** may include an upper plate for fixing the upper tray **150**.

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The upper tray **150** may be fixed to the upper plate **121** in a state in which a portion of the upper tray **150** contacts a bottom surface of the upper plate **121**.

An opening **123** through which a portion of the upper tray **150** passes may be defined in the upper plate **121**.

For example, when the upper tray **150** is fixed to the upper plate **121** in a state in which the upper tray **150** is disposed below the upper plate **121**, a portion of the upper tray **150** may protrude upward from the upper plate **121** through the opening **123**.

Alternatively, the upper tray **150** may not protrude upward from the upper plate **121** through opening **123** but protrude downward from the upper plate **121** through the opening **123**.

The upper plate **121** may include a recess **122** that is recessed downward. The opening **123** may be defined in a bottom surface **122a** of the recess **122**.

Thus, the upper tray **150** passing through the opening **123** may be disposed in a space defined by the recess **122**.

A heater coupling part **124** for coupling an upper heater (see reference numeral **148** of FIG. **13**) that heats the upper tray **150** so as to transfer the ice may be provided in the upper case **120**.

For example, the heater coupling part **124** may be provided on the upper plate **121**. The heater coupling part **124** may be disposed below the recess **122**.

The upper case **120** may further include a plurality of installation ribs **128** and **129** for installing the temperature sensor **500**.

The pair of installation ribs **128** and **129** may be disposed to be spaced apart from each other in a direction of an arrow B of FIG. **7**. The pair of installation ribs **128** and **129** may be disposed to face each other, and the temperature sensor **500** may be disposed between the pair of installation ribs **128** and **129**.

The pair of installation ribs **128** and **129** may be provided on the upper plate **121**.

A plurality of slots **131** and **132** coupled to the upper tray **150** may be provided in the upper plate **121**.

A portion of the upper tray **150** may be inserted into the plurality of slots **131** and **132**.

The plurality of slots **131** and **132** may include a first upper slot **131** and a second upper slot **132** disposed at an opposite side of the first upper slot **131** with respect to the opening **123**.

The opening **123** may be defined between the first upper slot **131** and the second upper slot **132**.

The first upper slot **131** and the second upper slot **132** may be spaced apart from each other in a direction of an arrow B of FIG. **7**.

Although not limited, the plurality of first upper slots **131** may be arranged to be spaced apart from each other in a direction of an arrow A (hereinafter, referred to as a first direction) that a direction crossing a direction of an arrow B (hereinafter, referred to as a second direction).

Also, the plurality of second upper slots **132** may be arranged to be spaced apart from each other in the direction of the arrow A.

In this specification, the direction of the arrow A may be the same direction as the arranged direction of the plurality of ice chambers **111**.

For example, the first upper slot **131** may be defined in a curved shape. Thus, the first upper slot **131** may increase in length.

For example, the second upper slot **132** may be defined in a curved shape. Thus, the second upper slot **133** may increase in length.

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When each of the upper slots **131** and **132** increases in length, a protrusion (that is disposed on the upper tray) inserted into each of the upper slots **131** and **132** may increase in length to improve coupling force between the upper tray **150** and the upper case **120**.

A distance between the first upper slot **131** and the opening **123** may be different from that between the second upper slot **132** and the opening **123**. For example, the distance between the first upper slot **131** and the opening **123** may be greater than that between the second upper slot **132** and the opening **123**.

Also, when viewed from the opening **123** toward each of the upper slots **131**, a shape that is convexly rounded from each of the slots **131** toward the outside of the opening **123** may be provided.

The upper plate **121** may further include a sleeve **133** into which a coupling boss of the upper support, which will be described later, is inserted.

The sleeve **133** may have a cylindrical shape and extend upward from the upper plate **121**.

For example, a plurality of sleeves **133** may be provided on the upper plate **121**. The plurality of sleeves **133** may be arranged to be spaced apart from each other in the direction of the arrow A. Also, the plurality of sleeves **133** may be arranged in a plurality of rows in the direction of the arrow B.

A portion of the plurality of sleeves may be disposed between the two first upper slots **131** adjacent to each other.

The other portion of the plurality of sleeves may be disposed between the two second upper slots **132** adjacent to each other or be disposed to face a region between the two second upper slots **132**.

The upper case **120** may further include a plurality of hinge supports **135** and **136** allowing the lower assembly **200** to rotate.

The plurality of hinge supports **135** and **136** may be disposed to be spaced apart from each other in the direction of the arrow A with respect to FIG. 7. Also, a first hinge hole **137** may be defined in each of the hinge supports **135** and **136**.

For example, the plurality of hinge supports **135** and **136** may extend downward from the upper plate **121**.

The upper case **120** may further include a vertical extension part **140** vertically extending along a circumference of the upper plate **121**. The vertical extension part **140** may extend upward from the upper plate **121**.

The vertical extension part **140** may include one or more coupling hooks **140a**. The upper case **120** may be hook-coupled to the housing **101** by the coupling hooks **140a**.

The water supply part **190** may be coupled to the vertical extension part **140**.

The upper case **120** may further include a horizontal extension part **142** horizontally extending to the outside of the vertical extension part **140**.

A screw coupling part **142a** protruding outward to screw-couple the upper case **120** to the housing **101** may be provided on the horizontal extension part **142**.

The upper case **120** may further include a side circumferential part **143**. The side circumferential part **143** may extend downward from the horizontal extension part **142**.

The side circumferential part **143** may be disposed to surround a circumference of the lower assembly **200**. That is, the side circumferential part **143** may prevent the lower assembly **200** from being exposed to the outside.

Although the upper case is coupled to the separate housing **101** within the freezing compartment **4** as described above, the embodiment is not limited thereto. For example,

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the upper case **120** may be directly coupled to a wall defining the freezing compartment **4**.

<Upper Tray>

FIG. 8 is a top perspective view of the upper tray according to an embodiment, FIG. 9 is a bottom perspective view of the upper tray according to an embodiment, and FIG. 10 is a side view of the upper tray according to an embodiment.

Referring to FIGS. 8 to 10, the upper tray **150** may be made of a non-metal material and a flexible material that is capable of being restored to its original shape after being deformed by an external force.

For example, the upper tray **150** may be made of a silicone material. Like this embodiment, when the upper tray **150** is made of the silicone material, even though external force is applied to deform the upper tray **150** during the ice separating process, the upper tray **150** may be restored to its original shape. Thus, in spite of repetitive ice making, spherical ice may be made.

If the upper tray **150** is made of a metal material, when the external force is applied to the upper tray **150** to deform the upper tray **150** itself, the upper tray **150** may not be restored to its original shape any more.

In this case, after the upper tray **150** is deformed in shape, the spherical ice may not be made. That is, it is impossible to repeatedly make the spherical ice.

On the other hand, like this embodiment, when the upper tray **150** is made of the flexible material that is capable of being restored to its original shape, this limitation may be solved.

Also, when the upper tray **150** is made of the silicone material, the upper tray **150** may be prevented from being melted or thermally deformed by heat provided from an upper heater that will be described later.

The upper tray **150** may include an upper tray body **151** defining an upper chamber **152** that is a portion of the ice chamber **111**.

The upper tray body **151** may define a plurality of upper chambers **152**.

For example, the plurality of upper chambers **152** may define a first upper chamber **152a**, a second upper chamber **152b**, and a third upper chamber **152c**.

The upper tray body **151** may include three chamber walls **153** defining three independent upper chambers **152a**, **152b**, and **152c**. The three chamber walls **153** may be connected to each other to form one body.

The first upper chamber **152a**, the second upper chamber **152b**, and the third upper chamber **152c** may be arranged in a line. For example, the first upper chamber **152a**, the second upper chamber **152b**, and the third upper chamber **152c** may be arranged in a direction of an arrow A with respect to FIG. 9. The direction of the arrow A of FIG. 9 may be the same direction as the direction of the arrow A of FIG. 7.

The upper chamber **152** may have a hemispherical shape. That is, an upper portion of the spherical ice may be made by the upper chamber **152**.

An upper opening **154** may be defined in an upper side of the upper tray body **151**. The upper opening **154** may be communicated with the upper chamber **152**.

For example, three upper openings **154** may be defined in the upper tray body **151**.

Cold air may be guided into the ice chamber **111** through the upper opening **154**. Further, water may be supplied into the ice chamber **111** through the upper opening **154**.

In the ice separating process, the upper ejector **300** may be inserted into the upper chamber **152** through the upper opening **154**.

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While the upper ejector **300** is inserted through the upper opening **154**, an inlet wall **155** may be provided on the upper tray **150** to minimize deformation of the upper opening **154** in the upper tray **150**.

The inlet wall **155** may be disposed along a circumference of the upper opening **154** and extend upward from the upper tray body **151**.

The inlet wall **155** may have a cylindrical shape. Thus, the upper ejector **30** may pass through the upper opening **154** via an inner space of the inlet wall **155**.

One or more first connection ribs **155a** may be provided along a circumference of the inlet wall **155** to prevent the inlet wall **155** from being deformed while the upper ejector **300** is inserted into the upper opening **154**.

The first connection rib **155a** may connect the inlet wall **155** to the upper tray body **151**. For example, the first connection rib **155a** may be integrated with the circumference of the inlet wall **155** and an outer face of the upper tray body **151**.

Although not limited, the plurality of connection ribs **155a** may be disposed along the circumference of the inlet wall **155**.

The two inlet walls **155** corresponding to the second upper chamber **152b** and the third upper chamber **152c** may be connected to each other through the second connection rib **162**. The second connection rib **162** may also prevent the inlet wall **155** from being deformed.

A water supply guide **156** may be provided in the inlet wall **155** corresponding to one of the three upper chambers **152a**, **152b**, and **152c**.

Although not limited, the water supply guide **156** may be provided in the inlet wall corresponding to the second upper chamber **152b**.

The water supply guide **156** may be inclined upward from the inlet wall **155** in a direction which is away from the second upper chamber **152b**.

The upper tray **150** may further include a first accommodation part **160**. The heater coupling part **124** of the upper case **120** may be accommodated in the first accommodation part **160**.

An upper heater (see reference numeral **148** of FIG. **14**) may be provided in the heater coupling part **124**. Thus, it may be understood that the upper heater (see reference numeral **148** of FIG. **14**) is accommodated in the first accommodation part **160**.

The first accommodation part **160** may be disposed in a shape that surrounds the upper chambers **152a**, **152b**, and **152c**. The first accommodation part **160** may be provided by recessing a top surface of the upper tray body **151** downward.

The first accommodation part **160** may be lower than the upper opening **154**.

The upper tray **150** may further include a second accommodation part **161** (or referred to as a sensor accommodation part) in which the temperature sensor **500** is accommodated.

For example, the second accommodation part **161** may be provided in the upper tray body **151**. Although not limited, the second accommodation part **161** may be provided by recessing a bottom surface of the first accommodation part **160** downward.

Also, the second accommodation part **161** may be disposed between the two upper chambers adjacent to each other. For example, the second accommodation part **161** may be disposed between the first upper chamber **152a** and the second upper chamber **152b**.

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Thus, an interference between the upper heater (see reference numeral **148** of FIG. **14**) accommodated in the first accommodation part **160** and the temperature sensor **500** may be prevented.

In the state in which the temperature sensor **500** is accommodated in the second accommodation part **161**, the temperature sensor **500** may contact an outer face of the upper tray body **151**.

The chamber wall **153** of the upper tray body **151** may include a vertical wall **153a** and a curved wall **153b**.

The curved wall **153b** may be rounded upward in a direction that is away from the upper chamber **152**.

The upper tray **150** may further include a horizontal extension part **164** horizontally extending from the circumference of the upper tray body **151**. For example, the horizontal extension part **164** may extend along a circumference of an upper edge of the upper tray body **151**.

The horizontal extension part **164** may contact the upper case **120** and the upper support **170**.

For example, a bottom surface **164b** (or referred to as a "first surface") of the horizontal extension part **164** may contact the upper support **170**, and a top surface **164a** (or referred to as a "second surface") of the horizontal extension part **164** may contact the upper case **120**.

At least a portion of the horizontal extension part **164** may be disposed between the upper case **120** and the upper support **170**.

The horizontal extension part **164** may include a plurality of upper protrusions **165** and **166** respectively inserted into the plurality of upper slots **131** and **132**.

The plurality of upper protrusions **165** and **166** may include a first upper protrusion **165** and a second upper protrusion **166** disposed at an opposite side of the first upper protrusion **165** with respect to the upper opening **154**.

The first upper protrusion **165** may be inserted into the first upper slot **131**, and the second upper protrusion **166** may be inserted into the second upper slot **132**.

The first upper protrusion **165** and the second upper protrusion **166** may protrude upward from the top surface **164a** of the horizontal extension part **164**.

The first upper protrusion **165** and the second upper protrusion **166** may be spaced apart from each other in the direction of the arrow B of FIG. **9**. The direction of the arrow B of FIG. **9** may be the same direction as the direction of the arrow B of FIG. **7**.

Although not limited, the plurality of first upper protrusions **165** may be arranged to be spaced apart from each other in the direction of the arrow A.

The plurality of second upper protrusions **166** may be arranged to be spaced apart from each other in the direction of the arrow A.

For example, the first upper protrusion **165** may be provided in a curved shape. Also, for example, the second upper protrusion **166** may be provided in a curved shape.

In this embodiment, each of the upper protrusions **165** and **166** may be constructed so that the upper tray **150** and the upper case **120** are coupled to each other, and also, the horizontal extension part is prevented from being deformed during the ice making process or the ice separating process.

Here, when each of the upper protrusions **165** and **166** is provided in the curved shape, distances between the upper protrusions **165** and **166** and the upper chamber **152** in a longitudinal direction of the upper protrusions **165** and **166** may be equal or similar to each other to effectively prevent the horizontal extension parts **264** from being deformed.

For example, the deformation in the horizontal direction of the horizontal extension part **264** may be minimized to

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prevent the horizontal extension part **264** from being plastic-deformed. If when the horizontal extension part **264** is plastic-deformed, since the upper tray body is not positioned at the correct position during the ice making, the shape of the ice may not close to the spherical shape.

The horizontal extension part **164** may further include a plurality of lower protrusions **167** and **168**. The plurality of lower protrusions **167** and **168** may be inserted into a lower slot of the upper support **170**, which will be described below.

The plurality of lower protrusions **167** and **168** may include a first lower protrusion **167** and a second lower protrusion **168** disposed at an opposite side of the first lower protrusion **167** with respect to the upper chamber **152**.

The first lower protrusion **167** and the second lower protrusion **168** may protrude downward from the bottom surface **164b** of the horizontal extension part **164**.

The first lower protrusion **167** may be disposed at an opposite to the first upper protrusion **165** with respect to the horizontal extension part **164**. The second lower protrusion **168** may be disposed at an opposite side of the second upper protrusion **166** with respect to the horizontal extension part **164**.

The first lower protrusion **167** may be spaced apart from the vertical wall **153a** of the upper tray body **151**. The second lower protrusion **168** may be spaced apart from the curved wall **153b** of the upper tray body **151**.

Each of the plurality of lower protrusions **167** and **168** may also be provided in a curved shape. Since the protrusions **165**, **166**, **167**, and **168** are disposed on each of the top and bottom surfaces **164a** and **164b** of the horizontal extension part **164**, the deformation in the horizontal direction of the horizontal extension part **164** may be effectively prevented.

A through-hole **169** through which the coupling boss of the upper support **170**, which will be described later, may be provided in the horizontal extension part **164**.

For example, a plurality of through-holes **169** may be provided in the horizontal extension part **164**.

A portion of the plurality of through-holes **169** may be disposed between the two first upper protrusions **165** adjacent to each other or the two first lower protrusions **167** adjacent to each other.

The other portion of the plurality of through-holes **169** may be disposed between the two second lower protrusions **168** adjacent to each other or be disposed to face a region between the two second lower protrusions **168**.

<Upper Support>

FIG. **11** is a top perspective view of the upper support according to an embodiment, and FIG. **12** is a bottom perspective view of the upper support according to an embodiment.

Referring to FIGS. **11** and **12**, the upper support **170** may include a support plate **171** contacting the upper tray **150**.

For example, a top surface of the support plate **171** may contact the bottom surface **164b** of the horizontal extension part **164** of the upper tray **150**.

A plate opening **172** through which the upper tray body **151** passes may be defined in the support plate **171**.

A circumferential wall **174** that is bent upward may be provided on an edge of the support plate **171**. For example, the circumferential wall **174** may contact at least a portion of a circumference of a side surface of the horizontal extension part **164**.

Also, a top surface of the circumferential wall **174** may contact a bottom surface of the upper plate **121**.

The support plate **171** may include a plurality of lower slots **176** and **177**.

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The plurality of lower slots **176** and **177** may include a first lower slot **176** into which the first lower protrusion **167** is inserted and a second lower slot **177** into which the second lower protrusion **168** is inserted.

The plurality of first lower slots **176** may be disposed to be spaced apart from each other in the direction of the arrow A on the support plate **171**. Also, the plurality of second lower slots **177** may be disposed to be spaced apart from each other in the direction of the arrow A on the support plate **171**.

The support plate **171** may further include a plurality of coupling bosses **175**. The plurality of coupling bosses **175** may protrude upward from the top surface of the support plate **171**.

Each of the coupling bosses **175** may pass through the through-hole **169** of the horizontal extension part **164** and be inserted into the sleeve **133** of the upper case **120**.

In the state in which the coupling boss **175** is inserted into the sleeve **133**, a top surface of the coupling boss **175** may be disposed at the same height as a top surface of the sleeve **133** or disposed at a height lower than that of the top surface of the sleeve **133**.

A coupling member coupled to the coupling boss **175** may be, for example, a bolt (see reference symbol B1 of FIG. **3**). The bolt B1 may include a body part and a head part having a diameter greater than that of the body part. The bolt B1 may be coupled to the coupling boss **175** from an upper side of the coupling boss **175**.

While the body part of the bolt B1 is coupled to the coupling boss **175**, when the head part contacts the top surface of the sleeve **133**, and the head part contacts the top surface of the sleeve **133** and the top surface of the coupling boss **175**, assembling of the upper assembly **110** may be completed.

The upper support **170** may further include a plurality of unit guides **181** and **182** for guiding the connection unit **350** connected to the upper ejector **300**.

The plurality of unit guides **181** and **182** may be, for example, disposed to be spaced apart from each other in the direction of the arrow A with respect to FIG. **12**.

The unit guides **181** and **182** may extend upward from the top surface of the support plate **171**. Each of the unit guides **181** and **182** may be connected to the circumferential wall **174**.

Each of the unit guides **181** and **182** may include a guide slot **183** vertically extends.

In a state in which both ends of the ejector body **310** of the upper ejector **300** pass through the guide slot **183**, the connection unit **350** is connected to the ejector body **310**.

Thus, when the rotation force is transmitted to the ejector body **310** by the connection unit **350** while the lower assembly **200** rotates, the ejector body **310** may vertically move along the guide slot **183**.

<Upper heater Coupling Structure>

FIG. **13** is an enlarged view of the heater coupling part in the upper case of FIG. **6**, FIG. **14** is a view illustrating a state in which a heater is coupled to the upper case of FIG. **6**, and FIG. **15** is a view illustrating an arrangement of a wire connected to the heater in the upper case.

Referring to FIGS. **9**, **13** to **15**, the heater coupling part **124** may include a heater accommodation groove **124a** accommodating the upper heater **148**.

For example, the heater accommodation groove **124a** may be defined by recessing a portion of a bottom surface of the recess **122** of the upper case **120** upward.

The heater accommodation groove **124a** may extend along a circumference of the opening **123** of the upper case **120**.

For example, the upper heater **148** may be a wire-type heater. Thus, the upper heater **148** may be bendable. The upper heater **148** may be bent to correspond to a shape of the heater accommodation groove **124a** so as to accommodate the upper heater **148** in the heater accommodation groove **124a**.

The upper heater **148** may be a DC heater receiving DC power. The upper heater **148** may be turned on to transfer ice.

When heat of the upper heater **148** is transferred to the upper tray **150**, ice may be separated from a surface (inner face) of the upper tray **150**.

If the upper tray **150** is made of a metal material, and the heat of the upper heater **148** has a high temperature, a portion of the ice, which is heated by the upper heater **148**, may be adhered again to the surface of the upper tray after the upper heater **148** is turned off. As a result, the ice may be opaque.

That is, an opaque band having a shape corresponding to the upper heater may be formed around the ice.

However, in this embodiment, since the DC heater having low output is used, and the upper tray **150** is made of the silicone material, an amount of heat transferred to the upper tray **150** may be reduced, and thus, the upper tray itself may have low thermal conductivity.

Thus, the heat may not be concentrated into the local portion of the ice, and a small amount of heat may be slowly applied to prevent the opaque band from being formed around the ice because the ice is effectively separated from the upper tray.

The upper heater **148** may be disposed to surround the circumference of each of the plurality of upper chambers **152** so that the heat of the upper heater **148** is uniformly transferred to the plurality of upper chambers **152** of the upper tray **150**.

Also, the upper heater **148** may contact the circumference of each of the chamber walls **153** respectively defining the plurality of upper chambers **152**. Here, the upper heater **148** may be disposed at a position that is lower than that of the upper opening **154**.

Since the heater accommodation groove **124a** is recessed from the recess **122**, the heater accommodation groove **124a** may be defined by an outer wall **124b** and an inner wall **124c**.

The upper heater **148** may have a diameter greater than that of the heater accommodation groove **124a** so that the upper heater **148** protrudes to the outside of the heater coupling part **124** in the state in which the upper heater **148** is accommodated in the heater accommodation groove **124a**.

Since a portion of the upper heater **148** protrudes to the outside of the heater accommodation groove **124a** in the state in which the upper heater **148** is accommodated in the heater accommodation groove **124a**, the upper heater **148** may contact the upper tray **150**.

A separation prevention protrusion **124d** may be provided on at least one of the outer wall **124b** and the inner wall **124c** to prevent the upper heater **148** accommodated in the heater accommodation groove **124a** from being separated from the heater accommodation groove **124a**.

In FIG. **13**, for example, a plurality of separation prevention protrusions **124d** are provided on the inner wall **124c**.

The separation prevention protrusion **124d** may protrude from an end of the inner wall **124c** toward the outer wall **124b**.

Here, a protruding length of the separation prevention protrusion **124d** may be less than about $\frac{1}{2}$ of a distance between the outer wall **124b** and the inner wall **124c** to prevent the upper heater **148** from being easily separated from the heater accommodation groove **124a** without interfering with the insertion of the upper heater **148** by the separation prevention protrusion **124d**.

As illustrated in FIG. **14**, in the state in which the upper heater **148** is accommodated in the heater accommodation groove **124a**, the upper heater **148** may be divided into an upper round portion **148c** and a linear portion **148d**.

That is, the heater accommodation groove **124a** may include an upper round portion and an upper linear portion. Thus, the upper heater **148** may be divided into the upper round portion **148c** and the upper linear portion **148d** to correspond to the upper round portion and the linear portion of the heater accommodation groove **124a**.

The upper round portion **148c** may be a portion disposed along the circumference of the upper chamber **152** and also a portion that is bent to be rounded in a horizontal direction.

The upper round portion **148c** may comprise a first upper round portion **148e** corresponding to first and third **152a**, **152c** of both sides of an outermost section among a plurality of upper chambers **152**.

The first upper round portion **148e** may be connected by a pair of upper linear portions **148d**. That is, the pair of upper linear portions **148d** each may be connected to both ends of one first upper round portion **148e**.

A length of the first rounded portion **148e** is longer than lengths of each of the pair of upper linear portions **148d**. The pair of upper linear portions **148d** connected to both ends of the first upper round portion **148e** may be disposed substantially in parallel.

A distance (R2) between the pair of upper linear portions **148d** is smaller than double ($2 \times R1$) in a radius of curvature of the first upper round portion (**148e**).

As the distance (R2) between the pair of upper linear portions **148d** gets longer, the pair of upper linear portions **148d** moves away from the upper chamber **152**, and accordingly, it takes a long time to transfer the heat of the pair of upper linear portions **148d** to the upper chamber **152**.

However, according to this embodiment, since the distance (R2) between the pair of upper linear portions **148d** is smaller than double in a radius of curvature of the first upper round portion **148e**, an interval between the pair of upper linear portions **148d** and the upper chamber **152** may be reduced to rapidly transfer the heat of the upper linear portion **148d** to the upper chamber **152**.

The distance (R2) between the pair of upper linear portions **148d** may be equal to or larger than a radius of curvature (R1) of the first upper round portion **148e**.

As the distance (R2) between the pair of upper linear portions **148d** is reduced, there is a large degree of bending in a boundary between the pair of upper linear portions **148d** and the first upper round portion **148e**, thereby providing a lot of concern for a disconnection, and also, heat between two upper chambers that are adjacent to each other may be unnecessarily concentrated.

However, according to this embodiment, if the distance (R2) between the pair of upper linear portions **148d** is equal to or larger than the radius of curvature (R1) of the first upper round portion **148e**, the above-described problem can be prevented.

The upper round portion **148c** may further comprise a second upper round portion **148f** corresponding to the second upper chamber **152b** disposed between first and third

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upper chambers **152a**, **152c** at both sides of an outermost section among the plurality of upper chambers **152**.

As an example, a pair of second upper round portions **148f** may be spaced apart from each other. This is because each of the pair of second upper round portions **148f** has to be connected to the first upper round portion **148e** by the upper linear part **148d** of both sides.

A length of the second upper round portion **148f** may be shorter than a length of the first upper round portion **148e**. The upper linear portions **148d** at both sides of the second upper round portion **148f** may be connected.

The upper liner portion **148d** may be a portion connecting the upper round portions **148c** corresponding to the upper chambers **152** to each other.

Since the upper heater **148** is disposed at a position lower than that of the upper opening **154**, a line connecting two points of the upper round portions, which are spaced apart from each other, to each other may pass through upper chamber **152**.

Since the upper round portion **148c** of the upper heater **148** may be separated from the heater accommodation groove **124a**, the separation prevention protrusion **124d** may be disposed to contact the upper round portion **148c**.

A through-opening **124e** may be defined in a bottom surface of the heater accommodation groove **124a**. When the upper heater **148** is accommodated in the heater accommodation groove **124a**, a portion of the upper heater **148** may be disposed in the through-opening **124e**. For example, the through-opening **124e** may be defined in a portion of the upper heater **148** facing the separation prevention protrusion **124d**.

When the upper heater **148** is bent to be horizontally rounded, tension of the upper heater **148** may increase to cause disconnection, and also, the upper heater **148** may be separated from the heater accommodation groove **124a**.

However, when the through-opening **124e** is defined in the heater accommodation groove **124a** like this embodiment, a portion of the upper heater **148** may be disposed in the through-opening **124e** to reduce the tension of the upper heater **148**, thereby preventing the heater accommodation groove **124a** from being separated from the upper heater **148**.

As illustrated in FIG. 15, in a state in which a power input terminal **148a** and a power output terminal **148b** of the upper heater **148** are disposed in parallel to each other, the upper heater **148** may pass through a heater through-hole **125** defined in the upper case **120**.

Since the upper heater **148** is accommodated from a lower side of the upper case **120**, the power input terminal **148a** and the power output terminal **148b** of the upper heater **148** may extend upward to pass through the heater through-hole **125**.

The power input terminal **148a** and the power output terminal **148b** passing through the heater through-hole **125** may be connected to one first connector **129a**.

Also, a second connector **129c** to which two wires **129d** connected to correspond to the power input terminal **148a** and the power output terminal **148b** are connected may be connected to the first connector **129a**.

A first guide part **126** guiding the upper heater **148**, the first connector **129a**, the second connector **129c**, and the wire **129d** may be provided on the upper plate **121** of the upper case **120**.

In FIG. 15, for example, a structure in which the first guide part **126** guides the first connector **129a** is illustrated.

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The first guide part **126** may extend upward from the top surface of the upper plate **121** and have an upper end that is bent in the horizontal direction.

Thus, the upper bent portion of the first guide part **126** may limit upward movement of the first connector **126**.

The wire **129d** may be led out to the outside of the upper case **120** after being bent in an approximately "U" shape to prevent interference with the surrounding structure.

Since the wire **129d** is bent at least once, the upper case **120** may further include wire guides **127** and **128** for fixing a position of the wire **129d**.

The wire guides **127** and **128** may include a first guide **127** and a second guide **128**, which are disposed to be spaced apart from each other in the horizontal direction. The first guide **127** and the second guide **128** may be bent in a direction corresponding to the bending direction of the wire **129d** to minimize damage of the wire **129d** to be bent.

That is, each of the first guide **127** and the second guide **128** may include a curved portion.

To limit upward movement of the wire **129d** disposed between the first guide **127** and the second guide **128**, at least one of the first guide **127** and the second guide **128** may include an upper guide **127a** extending toward the other guide.

FIG. 16 is a cross-sectional view illustrating a state in which an upper assembly is assembled.

Referring to FIGS. 14 and 16, in the state in which the upper heater **148** is coupled to the heater coupling part **124** of the upper case **120**, the upper case **120**, the upper tray **150**, and the upper support **170** may be coupled to each other.

The first upper protrusion **165** of the upper tray **150** may be inserted into the first upper slot **131** of the upper case **120**. Also, the second upper protrusion **166** of the upper tray **150** may be inserted into the second upper slot **132** of the upper case **120**.

Then, the first lower protrusion **167** of the upper tray **150** may be inserted into the first lower slot **176** of the upper support **170**, and the second lower protrusion **168** of the upper tray **150** may be inserted into the second lower slot **177** of the upper support **170**.

Thus, the coupling boss **175** of the upper support **170** may pass through the through-hole of the upper tray **150** and then be accommodated in the sleeve **133** of the upper case **120**. In this state, the bolt **B1** may be coupled to the coupling boss **175** from an upper side of the coupling boss **175**.

In the state in which the bolt **B1** is coupled to the coupling boss **175**, the head part of the bolt **B1** may be disposed at a position higher than that of the upper plate **121**.

On the other hand, since the hinge supports **135** and **136** are disposed lower than the upper plate **121**, while the lower assembly **200** rotates, the upper assembly **110** or the connection unit **350** may be prevented from interfering with the head part of the bolt **B1**.

While the upper assembly **110** is assembled, a plurality of unit guides **181** and **182** of the upper support **170** may protrude upward from the upper plate **121** through the through-opening (see reference numerals **139a** and **139b** of FIG. 6) defined in both sides of the upper plate **121**.

As described above, the upper ejector **300** passes through the guide slots **183** of the unit guides **181** and **182** protruding upward from the upper plate **121**.

Thus, the upper ejector **300** may descend in the state of being disposed above the upper plate **121** and be inserted into the upper chamber **152** to separate ice of the upper chamber **152** from the upper tray **150**.

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When the upper assembly **110** is assembled, the heater coupling part **124** to which the upper heater **148** is coupled may be accommodated in the first accommodation part **160** of the upper tray **150**.

In the state in which the heater coupling part **124** is accommodated in the first accommodation part **160**, the upper heater **148** may contact the bottom surface **160a** of the first accommodation part **160**.

Like this embodiment, when the upper heater **148** is accommodated in the heater coupling part **124** having the recessed shape to contact the upper tray body **151**, heat of the upper heater **148** may be minimally transferred to other portion except for the upper tray body **151**. At least a portion of the upper heater **148** may be disposed to vertically overlap the upper chamber **152** so that the heat of the upper heater **148** is smoothly transferred to the upper chamber **152**.

In this embodiment, the upper round portion **148c** of the upper heater **148** may vertically overlap the upper chamber **152**.

As an example, the radius of curvature (R1) of the upper round portion **148c** is smaller than a radius of the upper chamber **152**.

<Lower Case>

FIG. **17** is a perspective view of a lower assembly according to an embodiment, FIG. **18** is a top perspective view of a lower case according to an embodiment, and FIG. **19** is a bottom perspective view of the lower case according to an embodiment.

Referring to FIGS. **17** to **19**, the lower assembly **200** may include a lower tray **250**. The lower tray **250** defines the ice chamber **121** together with the upper tray **150**.

The lower assembly **200** may further include a lower support **270** that supports the lower tray **250**. The lower support **270** and the lower tray **250** may rotate together while the lower tray **250** is seated on the lower support **270**.

The lower assembly **200** may further include a lower case **210** for fixing a position of the lower tray **250**.

The lower case **210** may surround the circumference of the lower tray **250**, and the lower support **270** may support the lower tray **250**.

The connection unit **350** may be coupled to the lower support **270**.

The connection unit **350** may include a first link **352** that receives power of the driving unit **180** to allow the lower support **270** to rotate and a second link **356** connected to the lower support **270** to transmit rotation force of the lower support **270** to the upper ejector **300** when the lower support **270** rotates.

The first link **352** and the lower support **270** may be connected to each other by an elastic member **360**. For example, the elastic member **360** may be a coil spring.

The elastic member **360** may have one end connected to the first link **362** and the other end connected to the lower support **270**.

The elastic member **360** provide elastic force to the lower support **270** so that contact between the upper tray **150** and the lower tray **250** is maintained.

In this embodiment, the first link **352** and the second link **356** may be disposed on both sides of the lower support **270**, respectively.

One of the two first links may be connected to the driving unit **180** to receive the rotation force from the driving unit **180**.

The two first links **352** may be connected to each other by a connection shaft (see reference numeral **370** of FIG. **5**).

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A hole **358** through which the ejector body **310** of the upper ejector **300** passes may be defined in an upper end of the second link **356**.

The lower case **210** may include a lower plate **211** for fixing the lower tray **250**.

A portion of the lower tray **250** may be fixed to contact a bottom surface of the lower plate **211**.

An opening **212** through which a portion of the lower tray **250** passes may be defined in the lower plate **211**.

For example, when the lower tray **250** is fixed to the lower plate **211** in a state in which the lower tray **250** is disposed below the lower plate **211**, a portion of the lower tray **250** may protrude upward from the lower plate **211** through the opening **212**.

The lower case **210** may further include a circumferential wall **214** surrounding the lower tray **250** passing through the lower plate **211**.

The circumferential wall **214** may include a vertical wall **214a** and a curved wall **215**.

The vertical wall **214a** is a wall vertically extending upward from the lower plate **211**. The curved wall **215** is a wall that is rounded in a direction that is away from the opening **212** upward from the lower plate **211**.

The vertical wall **214a** may include a first coupling slit **214b** coupled to the lower tray **250**. The first coupling slit **214b** may be defined by recessing an upper end of the vertical wall downward.

The curved wall **215** may include a second coupling slit **215a** to the lower tray **250**.

The second coupling slit **215a** may be defined by recessing an upper end of the curved wall **215** downward.

The lower case **210** may further include a first coupling boss **216** and a second coupling boss **217**.

The first coupling boss **216** may protrude downward from the bottom surface of the lower plate **211**. For example, the plurality of first coupling bosses **216** may protrude downward from the lower plate **211**.

The plurality of first coupling bosses **216** may be arranged to be spaced apart from each other in the direction of the arrow A with respect to FIG. **17**.

The second coupling boss **217** may protrude downward from the bottom surface of the lower plate **211**. For example, the plurality of second coupling bosses **217** may protrude from the lower plate **211**. The plurality of first coupling bosses **217** may be arranged to be spaced apart from each other in the direction of the arrow A with respect to FIG. **17**.

The first coupling boss **216** and the second coupling boss **217** may be disposed to be spaced apart from each other in the direction of the arrow B.

In this embodiment, a length of the first coupling boss **216** and a length of the second coupling boss **217** may be different from each other. For example, the first coupling boss **216** may have a length less than that of the second coupling boss **217**.

The first coupling member may be coupled to the first coupling boss **216** at an upper portion of the first coupling boss **216**. On the other hand, the second coupling member may be coupled to the second coupling boss **217** at a lower portion of the second coupling boss **217**.

A groove **215b** for movement of the coupling member may be defined in the curved wall **215** to prevent the first coupling member from interfering with the curved wall **215** while the first coupling member is coupled to the first coupling boss **216**.

The lower case **210** may further include a slot **218** coupled to the lower tray **250**.

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A portion of the lower tray **250** may be inserted into the slot **218**. The slot **218** may be disposed adjacent to the vertical wall **214a**.

For example, a plurality of slots **218** may be defined to be spaced apart from each other in the direction of the arrow A of FIG. **18**. Each of the slots **218** may have a curved shape.

The lower case **210** may further include an accommodation groove **218a** into which a portion of the lower tray **250** is inserted.

The accommodation groove **218a** may be defined by recessing a portion of the lower tray **211** toward the curved wall **215**.

The lower case **210** may further include an extension wall **219** contacting a portion of the circumference of the side surface of the lower plate **212** in the state of being coupled to the lower tray **250**. The extension wall **219** may linearly extend in the direction of the arrow A.

<Lower Tray>

FIG. **20** is a top perspective view of the lower tray according to an embodiment, FIGS. **21** and **22** are bottom perspective views of the lower tray according to an embodiment, and FIG. **23** is a side view of the lower tray according to an embodiment.

Referring to FIGS. **20** to **23**, the lower tray **250** may be made of a flexible material that is capable of being restored to its original shape after being deformed by an external force.

For example, the lower tray **250** may be made of a silicone material. Like this embodiment, when the lower tray **250** is made of a silicone material, the lower tray **250** may be restored to its original shape even through external force is applied to deform the lower tray **250** during the ice separating process. Thus, in spite of repetitive ice making, spherical ice may be made.

If the lower tray **250** is made of a metal material, when the external force is applied to the lower tray **250** to deform the lower tray **250** itself, the lower tray **250** may not be restored to its original shape any more.

In this case, after the lower tray **250** is deformed in shape, the spherical ice may not be made. That is, it is impossible to repeatedly make the spherical ice.

On the other hand, like this embodiment, when the lower tray **250** is made of the flexible material that is capable of being restored to its original shape, this limitation may be solved.

Also, when the lower tray **250** is made of the silicone material, the lower tray **250** may be prevented from being melted or thermally deformed by heat provided from an upper heater that will be described later.

The lower tray **250** may include a lower tray body **251** defining a lower chamber **252** that is a portion of the ice chamber **111**.

The lower tray body **251** may define a plurality of lower chambers **252**.

For example, the plurality of lower chambers **252** may include a first lower chamber **252a**, a second lower chamber **252b**, and a third lower chamber **252c**.

The lower tray body **251** may include three chamber walls **252d** defining three independent lower chambers **252a**, **252b**, and **252c**. The three chamber walls **252d** may be integrated in one body to form the lower tray body **251**.

The first lower chamber **252a**, the second lower chamber **252b**, and the third lower chamber **252c** may be arranged in a line. For example, the first lower chamber **252a**, the second lower chamber **252b**, and the third lower chamber **252c** may be arranged in a direction of an arrow A with respect to FIG. **20**.

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Accordingly, the lower chamber **252** may have a hemispherical shape or a shape similar to the hemispherical shape. That is, a lower portion of the spherical ice may be made by the lower chamber **252**.

The lower tray **250** may further include a first extension part **253** horizontally extending from an edge of an upper end of the lower tray body **251**. The first extension part **253** may be continuously formed along the circumference of the lower tray body **251**.

The lower tray **250** may further include a circumferential wall **260** extended upward from an upper surface of the first extension part **253**.

In this embodiment, since the first extension part **253** extends from the lower tray **250** and the circumferential wall **260** extends from the first extension part **253**, a bottom surface of the upper tray body **151** may contact a top surface **251e** of the lower tray body **251**.

The circumferential wall **260** may surround the upper tray body **251** seated on the top surface **251e** of the lower tray body **251**.

The circumferential wall **260** may include a first wall **260a** surrounding the vertical wall **153a** of the upper tray body **151** and a second wall **260b** surrounding the curved wall **153b** of the upper tray body **151**.

The first wall **260a** is a vertical wall vertically extending from the top surface of the first extension part **253**. The second wall **260b** is a curved wall having a shape corresponding to that of the upper tray body **151**. That is, the second wall **260b** may be rounded upward from the first extension part **253** in a direction that is away from the lower chamber **252**.

The lower tray **250** may further include a second extension part **254** horizontally extending from the circumferential wall **260**.

The second extension part **254** may be disposed higher than the first extension part **253**. Thus, the first extension part **253** and the second extension part **254** may be stepped with respect to each other.

The second extension part **254** may include a first upper protrusion **255** inserted into the slot **218** of the lower case **210**. The first upper protrusion **255** may be disposed to be horizontally spaced apart from the circumferential wall **260**.

For example, the first upper protrusion **255** may protrude upward from a top surface of the second extension part **254** at a position adjacent to the first wall **260a**.

Although not limited, a plurality of first upper protrusions **255** may be arranged to be spaced apart from each other in the direction of the arrow A with respect to FIG. **20**. The first upper protrusion **255** may extend, for example, in a curved shape. That is, the first upper protrusion **255** is curved in a horizontal direction.

The second extension part **254** may include a first lower protrusion **257** inserted into a protrusion groove of the lower case **270**, which will be described later. The first lower protrusion **257** may protrude downward from a bottom surface of the second extension part **254**.

Although not limited, the plurality of first lower protrusions **257** may be arranged to be spaced apart from each other in the direction of arrow A.

That is, the first lower protrusion **257** is curved in a horizontal direction.

The first upper protrusion **255** and the first lower protrusion **257** may be disposed at opposite sides with respect to a vertical direction of the second extension part **254**. At least a portion of the first upper protrusion **255** may vertically overlap the second lower protrusion **257**.

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A plurality of through-holes may be defined in the second extension part **254**.

The plurality of through-holes **256** may include a first through-hole **256a** through which the first coupling boss **216** of the lower case **210** passes and a second through-hole **256b** through which the second coupling boss **217** of the lower case **210** passes.

For example, the plurality of through-holes **256a** may be defined to be spaced apart from each other in the direction of the arrow A of FIG. 20.

Also, the plurality of second through-holes **256b** may be disposed to be spaced apart from each other in the direction of the arrow A of FIG. 20.

The plurality of first through-holes **256a** and the plurality of second through-holes **256b** may be disposed at opposite sides with respect to the lower chamber **252**.

A portion of the plurality of second through-holes **256b** may be defined between the two first upper protrusions **255**. Also, a portion of the plurality of second through-holes **256b** may be defined between the two first lower protrusions **257**.

The second extension part **254** may further a second upper protrusion **258**. The second upper protrusion **258** may be disposed at an opposite side of the first upper protrusion **255** with respect to the lower chamber **252**.

The second upper protrusion **258** may be disposed to be horizontally spaced apart from the circumferential wall **260**. For example, the second upper protrusion **258** may protrude upward from a top surface of the second extension part **254** at a position adjacent to the second wall **260b**.

Although not limited, the plurality of second upper protrusions **258** may be arranged to be spaced apart from each other in the direction of the arrow A of FIG. 20.

The second upper protrusion **258** may be accommodated in the accommodation groove **218a** of the lower case **210**. In the state in which the second upper protrusion **258** is accommodated in the accommodation groove **218a**, the second upper protrusion **258** may contact the curved wall **215** of the lower case **210**.

The circumferential wall **260** of the lower tray **250** may include a first coupling protrusion **262** coupled to the lower case **210**.

The first coupling protrusion **262** may horizontally protrude from the first wall **260a** of the circumferential wall **260**. The first coupling protrusion **262** may be disposed on an upper portion of a side surface of the first wall **260a**.

The first coupling protrusion **262** may include a neck part **262a** having a relatively less diameter when compared to those of other portions. The neck part **262a** may be inserted into a first coupling slit **214b** defined in the circumferential wall **214** of the lower case **210**.

The circumferential wall **260** of the lower tray **250** may further include a second coupling protrusion **262c** coupled to the lower case **210**.

The second coupling protrusion **262c** may horizontally protrude from the second wall **260a** of the circumferential wall **260**. The second coupling protrusion **262c** is positioned lower than a top end of the circumferential wall **260**.

The second coupling protrusion **260c** may be inserted into a second coupling slit **215a** defined in the circumferential wall **214** of the lower case **210**.

The second extension part **254** may include a second lower protrusion **266**. The second lower protrusion **266** may be disposed at an opposite side of the second lower protrusion **257** with respect to the lower chamber **252**.

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The second lower protrusion **266** may protrude downward from a bottom surface of the second extension part **254**. For example, the second lower protrusion **266** may linearly extend.

A portion of the plurality of first through-holes **256a** may be defined between the second lower protrusion **266** and the lower chamber **252**.

The second lower protrusion **266** may be accommodated in a guide groove defined in the lower support **270**, which will be described later.

The second extension part **254** may further a side restriction part **264**. The side restriction part **264** restricts horizontal movement of the lower tray **250** in the state in which the lower tray **250** is coupled to the lower case **210** and the lower support **270**.

The side restriction part **264** laterally protrudes from the second extension part **254** and has a vertical length greater than a thickness of the second extension part **254**. For example, one portion of the side restriction part **264** may be disposed higher than the top surface of the second extension part **254**, and the other portion of the side restriction part **264** may be disposed lower than the bottom surface of the second extension part **254**.

Thus, the one portion of the side restriction part **264** may contact a side surface of the lower case **210**, and the other portion may contact a side surface of the lower support **270**. In one example, the lower tray body **251** may have a heater contact portion **251a** which the lower heater **296** contacts. In one example, the heater contact portion **251a** may be formed on each of the chamber walls **252d**. The heater contact portion **251a** may protrude from the respective chamber wall **252d**. In one example, the heater contact portion **251a** may be formed in a circular ring shape.

<Lower Support>

FIG. 24 is a top perspective view of the lower support according to an embodiment, FIG. 25 is a bottom perspective view of the lower support according to an embodiment, and FIG. 26 is a cross-sectional view taken along line D-D of FIG. 17 for showing a state that a lower assembly is assembled.

Referring to FIGS. 24 to 26, the lower support **270** may include a support body **271** supporting the lower tray **250**.

The support body **271** may include three chamber accommodation parts **272** accommodating the three chamber walls **252d** of the lower tray **250**. The chamber accommodation part **272** may have a hemispherical shape.

The support body **271** may have a lower opening **274** through which the lower ejector **400** passes during the ice separating process. For example, three lower openings **274** may be defined to correspond to the three chamber accommodation parts **272** in the support body **271**.

A reinforcement rib **275** reinforcing strength may be disposed along a circumference of the lower opening **274**.

Also, the adjacent two accommodation part **272** of the three accommodation parts **272** may be connected to each other by a connection rib **273**. The connection rib **273** may reinforce strength of the chamber wells **252d**.

The lower support **270** may further include a first extension wall **285** horizontally extending from an upper end of the support body **271**.

The lower support **270** may further include a second extension wall **286** that is formed to be stepped with respect to the first extension wall **285** on an edge of the first extension wall **285**.

A top surface of the second extension wall **286** may be disposed higher than the first extension wall **285**.

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The first extension part **253** of the lower tray **250** may be seated on a top surface **271a** of the support body **271**, and the second extension part **285** may surround side surface of the first extension part **253** of the lower tray **250**. Here, the second extension wall **286** may contact the side surface of the first extension part **253** of the lower tray **250**.

The lower support **270** may further include a first protrusion groove **287** accommodating the first lower protrusion **257** of the lower tray **250**.

The first protrusion groove **287** may extend in a curved shape. The first protrusion groove **287** may be defined, for example, in a second extension wall **286**.

The lower support **270** may further include a first coupling groove **286a** to which a first coupling member **B2** passing through the first coupling boss **216** of the upper case **210** is coupled.

The first coupling groove **286a** may be provided, for example, in the second extension wall **286**.

The plurality of first coupling grooves **286a** may be disposed to be spaced apart from each other in the direction of the arrow **A** in the second extension wall **286**. A portion of the plurality of first coupling grooves **286a** may be defined between the adjacent two first protrusion grooves **287**.

The lower support **270** may further include a boss through-hole **286b** through which the second coupling boss **217** of the upper case **210** passes.

The boss through-hole **286b** may be provided, for example, in the second extension wall **286**. A sleeve **286c** surrounding the second coupling boss **217** passing through the boss through-hole **286b** may be disposed on the second extension wall **286**. The sleeve **286c** may have a cylindrical shape with an opened lower portion.

The first coupling member **B2** may be coupled to the first coupling groove **286a** after passing through the first coupling boss **216** from an upper side of the lower case **210**.

The second coupling member **B3** may be coupled to the second coupling boss **217** from a lower side of the lower support **270**.

The sleeve **286c** may have a lower end that is disposed at the same height as a lower end of the second coupling boss **217** or disposed at a height lower than that of the lower end of the second coupling boss **217**.

Thus, while the second coupling member **B3** is coupled, the head part of the second coupling member **B3** may contact bottom surfaces of the second coupling boss **217** and the sleeve **286c** or may contact a bottom surface of the sleeve **286c**.

The lower support **270** may further include an outer wall **280** disposed to surround the lower tray body **251** in a state of being spaced outward from the outside of the lower tray body **251**.

The outer wall **280** may, for example, extend downward along an edge of the second extension wall **286**.

The lower support **270** may further include a plurality of hinge bodies **281** and **282** respectively connected to hinge supports **135** and **136** of the upper case **210**.

The plurality of hinge bodies **281** and **282** may be disposed to be spaced apart from each other in a direction of an arrow **A** of FIG. **24**. Each of the hinge bodies **281** and **282** may further include a second hinge hole **281a**.

The shaft connection part **353** of the first link **352** may pass through the second hinge hole **281**. The connection shaft **370** may be connected to the shaft connection part **353**.

A distance between the plurality of hinge bodies **281** and **282** may be less than that between the plurality of hinge

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supports **135** and **136**. Thus, the plurality of hinge bodies **281** and **282** may be disposed between the plurality of hinge supports **135** and **136**.

The lower support **270** may further include a coupling shaft **283** to which the second link **356** is rotatably coupled. The coupling shaft **283** may be disposed on each of both surfaces of the outer wall **280**.

Also, the lower support **270** may further include an elastic member coupling part **284** to which the elastic member **360** is coupled. The elastic member coupling part **284** may define a space in which a portion of the elastic member **360** is accommodated. Since the elastic member **360** is accommodated in the elastic member coupling part **284** to prevent the elastic member **360** from interfering with the surrounding structure.

Also, the elastic member coupling part **284** may include a hook part **284a** on which a lower end of the elastic member **370** is hooked.

<Coupling Structure of Lower Heater>

FIG. **27** is a plan view of the lower support according to an embodiment, FIG. **28** is a perspective view illustrating a state in which a lower heater is coupled to the lower support of FIG. **27**, and FIG. **29** is a view illustrating a state in which the wire connected to the lower heater passes through the upper case in a state in which the lower assembly is coupled to the upper assembly. FIG. **30** is a cross-sectional view showing a state in which the lower heater is installed on the lower support.

Referring to FIGS. **27** to **30**, the ice maker **100** according to this embodiment may further include a lower heater **296** for applying heat to the lower tray **250** during the ice making process.

The lower heater **297** may provide the heat to the lower chamber **252** during the ice making process so that ice within the ice chamber **111** is frozen from an upper side.

Also, since lower heater **296** generates heat in the ice making process, bubbles within the ice chamber **111** may move downward during the ice making process. When the ice is completely made, a remaining portion of the spherical ice except for the lowermost portion of the ice may be transparent. According to this embodiment, the spherical ice that is substantially transparent may be made.

For example, the lower heater **296** may be a wire-type heater.

The lower heater **296** may be located between the lower tray **250** and the lower support **270**.

The lower heater **296** may be installed on the lower support **270**. Also, the lower heater **296** may contact the lower tray **250** to provide heat to the lower chamber **252**.

For example, the lower heater **296** may contact the lower tray body **251**. Also, the lower heater **296** may be disposed to surround the three chamber walls **252d** of the lower tray body **251**.

The lower support **270** may further include a heater coupling part **290** to which the lower heater **296** is coupled. The heater coupling part **290** may include a heater accommodation groove **291** that is recessed downward from the chamber accommodation part **272** of the lower tray body **251**.

Since the heater accommodation groove **291** is recessed, the heater coupling part **290** may include an inner wall **291a** and an outer wall **291b**.

The inner wall **291a** may have, for example, a ring shape, and the outer wall **291b** may be disposed to surround the inner wall **291a**.

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When the lower heater **296** is accommodated in the heater accommodation groove **291**, the lower heater **296** may surround at least a portion of the inner wall **291a**.

The lower opening **274** may be defined in a region defined by the inner wall **291a**. Thus, when the chamber wall **252d** of the lower tray **250** is accommodated in the chamber accommodation part **272**, the chamber wall **252d** may contact a top surface of the inner wall **291a**. The top surface of the inner wall **291a** may be a rounded surface corresponding to the chamber wall **252d** having the hemispherical shape.

The lower heater may have a diameter greater than a recessed depth of the heater accommodation groove **291** so that a portion of the lower heater **296** protrudes to the outside of the heater accommodation groove **291** in the state in which the lower heater **296** is accommodated in the heater accommodation groove **291**.

A separation prevention protrusion **291c** may be provided on one of the outer wall **291b** and the inner wall **291a** to prevent the lower heater **296** accommodated in the heater accommodation groove **291** from being separated from the heater accommodation groove **291**.

In FIG. **26**, the separation prevention protrusions **291c** is provided on the inner wall **291a**.

Since the inner wall **291a** has a diameter less than that of the chamber accommodation part **272**, the lower heater **296** may move along a surface of the chamber accommodation part **272** and then be accommodated in the heater accommodation groove **291** in a process of assembling the lower heater **296**.

That is, the lower heater **296** is accommodated in the heater accommodation groove **291** from an upper side of the outer wall **291a** toward the inner wall **291a**. Thus, the separation prevention protrusion **291c** may be disposed on the inner wall **291a** to prevent the lower heater **296** from interfering with the separation prevention protrusion **291c** while the lower heater **296** is accommodated in the heater accommodation groove **291**.

The separation prevention protrusion **291c** may protrude from an upper end of the inner wall **291a** toward the outer wall **291b**.

A protruding length of the separation prevention protrusion **291c** may be about $\frac{1}{2}$ of a distance between the outer wall **291b** and the inner wall **291a**.

As illustrated in FIG. **28**, in the state in which the lower heater **296** is accommodated in the heater accommodation groove **291**, the lower heater **296** may be divided into a lower round portion **296a** and a lower linear portion **296b**.

The lower round portion **296a** may be a portion disposed along the circumference of the lower chamber **252** and also a portion that is bent to be rounded in a horizontal direction.

The lower linear portion **296b** may be a portion connecting the lower round portions **296a** corresponding to the lower chambers **252** to each other.

The lower round portion **296a** may comprise first lower round portions **296c**, **296d** corresponding to first and third upper chambers **252a**, **252c** of both sides of an outermost section among a plurality of lower chambers **252**.

The first lower round portions **296c**, **296d** may be connected by a pair of lower linear portions **296b**. That is, the pair of lower linear portions **296b** each may be connected to both ends of first lower round portions **296c**, **296d**.

Lengths of the first lower round portions **296c**, **296d** are longer than each of the pair of lower linear portions **296b**.

The pair of lower linear portions **296b** connected to both ends of the first lower round portions **296c**, **296d** may be disposed substantially in parallel.

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A distance (R4) between the pair of lower linear portions **296b** is smaller than double ($2 \times R3$) in a radius of curvature of the first lower round portions **296c**, **296d**.

As the distance (R2) between the pair of lower linear portions **296b** is elongated, lengths of each of the pair of lower linear portions **296b** get long, whereas lengths of the first lower round portions **296c**, **296d** are reduced, and thus a length of the lower heater **296** is reduced when viewing the lower heater **296** as a whole.

When the length of the lower heater **296** is reduced, there is a small amount of heat transmitted to the lower chamber **252** by the lower heater **296**.

In addition, when the distance (R4) of the pair of lower linear portion **296b** is elongated, a distance between the lower linear portion **296b** and the lower chamber **252** is increased, thereby enhancing a time when the heat of the lower linear portion **296b** reaches the lower chamber **252**.

However, according to this embodiment, since the distance (R4) between the pair of lower linear portion **296b** is smaller than double in the radius of curvature in the first lower round portions **296c**, **296d**, an interval between the pair of lower linear portion **296b** and the lower chamber **252** may be reduced to rapidly transfer the heat of the lower linear portion **296b** to the lower chamber **252**.

The distance (R4) between the pair of lower linear portion **296b** may be equal to or larger than a radius of curvature (R3) of the first lower round portions **296c**, **296d**.

As the distance (R4) between the pair of lower linear portions **296d** is reduced, there is a large degree of bending in a boundary between the pair of lower linear portions **296b** and the first lower round portions **296c**, **296d**, thereby providing a lot of concern for the disconnection, and also, heat between two upper chambers that are adjacent to each other may be unnecessary concentrated.

However, according to this embodiment, if the distance (R4) between the pair of lower linear portions **296d** is equal to or larger than the radius of curvature (R3) of the first lower round portions **296c**, **296d**, the above-described problem can be prevented.

The lower round portion **296a** may further comprise a second lower round portion **296e** corresponding to the second upper chamber **252b**.

As an example, a pair of second lower round portions **296e** may be spaced apart from each other. This is because each of the pair of second lower round portions **296e** has to be connected to the first lower round portions **296c**, **296d** by the lower linear part **296b** of both sides.

A length of the second lower round portion **296e** may be shorter than a length of the first lower round portions **296c**, **296d**.

Since the lower round portion **296a** of the lower heater **296** may be separated from the heater accommodation groove **291**, the separation prevention protrusion **291c** may be disposed to contact the lower round portion **296a**.

A through-opening **291d** may be defined in a bottom surface of the heater accommodation groove **291**. When the lower heater **296** is accommodated in the heater accommodation groove **291**, a portion of the lower heater **296** may be disposed in the through-opening **291d**. For example, the through-opening **291d** may be defined in a portion of the lower heater **296** facing the separation prevention protrusion **291c**.

When the lower heater **296** is bent to be horizontally rounded, tension of the lower heater **296** may increase to cause disconnection, and also, the lower heater **296** may be separated from the heater accommodation groove **291**.

However, when the through-opening **291d** is defined in the heater accommodation groove **291** like this embodiment, a portion of the lower heater **296** may be disposed in the through-opening **291d** to reduce the tension of the lower heater **296**, thereby preventing the heater accommodation groove **291** from being separated from the lower heater **296**.

The lower support **270** may include a first guide groove **293** guiding a power input terminal **296c** and a power output terminal of the lower heater **296** accommodated in the heater accommodation groove **291** and a second guide groove **294** extending in a direction crossing the first guide groove **293**.

For example, the first guide groove **293** may extend in a direction of an arrow B in the heater accommodation part **291**.

The second guide groove **294** may extend from an end of the first guide groove **293** in a direction of an arrow A. In this embodiment, the direction of the arrow A may be a direction that is parallel to the extension direction of a rotational central axis C1 of the lower assembly.

Referring to FIG. **28**, the first guide groove **293** may extend from one of the left and right chamber accommodation parts except for the intermediate chamber accommodation part of the three chamber accommodation parts.

For example, in FIG. **28**, the first guide groove **293** extends from the chamber accommodation part, which is disposed at the left side, of the three chamber accommodation parts. That is, a part extending from the first lower round portion **296d** on the left may be accommodated in the first guide groove **293**.

As illustrated in FIG. **28**, in a state in which the power input terminal **296c** and the power output terminal **296d** of the lower heater **296** are disposed in parallel to each other, the lower heater **296** may be accommodated in the first guide groove **293**.

The power input terminal **296c** and the power output terminal **296d** of the lower heater **296** may be connected to one first connector **297a**.

A second connector **297b** to which two wires **298** connected to correspond to the power input terminal **296c** and the power output terminal **296d** are connected may be connected to the first connector **297a**.

In this embodiment, in the state in which the first connector **297a** and the second connector **297b** are connected to each other, the first connector **297a** and the second connector **297b** are accommodated in the second guide groove **294**.

The wire **298** connected to the second connector **297b** is led out from the end of the second guide groove **294** to the outside of the lower support **270** through an lead-out slot **295** defined in the lower support **270**.

According to this embodiment, since the first connector **297a** and the second connector **297b** are accommodated in the second guide groove **294**, the first connector **297a** and the second connector **297b** are not exposed to the outside when the lower assembly **200** is completely assembled.

As described above, the first connector **297a** and the second connector **297b** may not be exposed to the outside to prevent the first connector **297a** and the second connector **297b** from interfering with the surrounding structure while the lower assembly **200** rotates and prevent the first connector **297a** and the second connector **297b** from being separated.

Since the first connector **297a** and the second connector **297b** are accommodated in the second guide groove **294**, one portion of the wire **298** may be disposed in the second guide groove **294**, and the other portion may be disposed outside the lower support **270** by the lead-out slot **295**.

Here, since the second guide groove **294** extends in a direction parallel to the rotational central axis C1 of the lower assembly **200**, one portion of the wire **298** may extend in the direction parallel to the rotational central axis C1.

The other part of the wire **298** may extend from the outside of the lower support **270** in a direction crossing the rotational central axis C1.

According to the arrangement of the wires **298**, tensile force may not merely act on the wires **298**, but torsion force may act on the wires **298** during the rotation of the lower assembly **200**.

When compared that the tensile force acts on the wire **298**, if the torsion acts on the wire **298**, possibility of disconnection of the wire **298** may be very little.

According to this embodiment, while the lower assembly **200** rotates, the lower heater **296** may be maintained at a fixed position, and twisting force may act on the wire **298** to prevent the lower heater **296** from being damaged and disconnected.

The power input terminal **296c** and the power output terminal **296d** of the lower heater **296** are disposed in the first guide groove **293**. Here, since heat is also generated in the power input terminal **296c** and the power output terminal **296d**, heat provided to the left chamber accommodation part to which the first guide groove **293** extends may be greater than that provided to other chamber accommodation parts.

In this case, if intensities of the heat provided to each chamber accommodating part are different, transparency of the made spherical ice after the ice making process and the ice separating process may be changed for each ice.

Thus, a detour accommodation groove **292** may be further provided in the chamber accommodation part (for example, the right chamber accommodation part), which is disposed farthest from the first guide groove **292**, of the three chamber accommodation parts to minimize a difference in transparency for each ice.

For example, the detour accommodation groove **292** may extend outward from the heater accommodation groove **291** and then be bent so as to be disposed in a shape that is connected to the heater accommodation groove **291**.

When a portion **296f** of the lower heater **296** is additionally accommodated in the detour accommodation groove **292**, a contact area between the chamber wall accommodated in the right chamber accommodation part **272** and the lower heater **296** may increase.

Thus, a protrusion **292a** for fixing a position of the lower heater accommodated in the detour accommodation groove **292** may be additionally provided in the right chamber accommodation part **272**.

As an example, a portion **296f** of the first lower round portion **296c** disposed to the right may be disposed in the detour accommodation groove **292**.

Referring to FIG. **29**, in the state in which the lower assembly **200** is coupled to the upper case **120** of the upper assembly **110**, the wire **298** led out to the outside of the lower support **270** may pass through a wire through-slot **138** defined in the upper case **120** to extend upward from the upper case **120**.

A restriction guide **139** for restricting the movement of the wire **298** passing through the wire through-slot **138** may be provided in the wire through-slot **138**. The restriction guide **139** may have a shape that is bent several times, and the wire **298** may be disposed in a region defined by the restriction guide **139**.

FIG. **31** is a cross-sectional view taken along line A-A of FIG. **3**, and FIG. **32** is a view illustrating a state in which ice is completely made in FIG. **30**.

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In FIG. 31, a state in which the upper tray and the lower tray contact each other is illustrated.

Referring to FIG. 31, the upper tray 150 and the lower tray 250 vertically contact each other to complete the ice chamber 111.

The bottom surface 151a of the upper tray body 151 contacts the top surface 251e of the lower tray body 251.

Here, in the state in which the top surface 251e of the lower tray body 251 contacts the bottom surface 151a of the upper tray body 151, elastic force of the elastic member 360 is applied to the lower support 270.

The elastic force of the elastic member 360 may be applied to the lower tray 250 by the lower support 270, and thus, the top surface 251e of the lower tray body 251 may press the bottom surface 151a of the upper tray body 151.

Thus, in the state in which the top surface 251e of the lower tray body 251 contacts the bottom surface 151a of the upper tray body 151, the surfaces may be pressed with respect to each other to improve the adhesion.

As described above, when the adhesion between the top surface 251e of the lower tray body 251 and the bottom surface 151a of the upper tray increases, a gap between the two surface may not occur to prevent ice having a thin band shape along a circumference of the spherical ice from being made after the ice making is completed.

The first extension part 253 of the lower tray 250 is seated on the top surface 271a of the support body 271 of the lower support 270. Also, the second extension wall 286 of the lower support 270 contacts a side surface of the first extension part 253 of the lower tray 250.

The second extension part 254 of the lower tray 250 may be seated on the second extension wall 286 of the lower support 270.

In the state in which the bottom surface 151a of the upper tray body 151 is seated on the top surface 251e of the lower tray body 251, the upper tray body 151 may be accommodated in an inner space of the circumferential wall 260 of the lower tray 250.

Here, the vertical wall 153a of the upper tray body 151 may be disposed to face the first wall 260a of the lower tray 250, and the curved wall 153b of the upper tray body 151 may be disposed to face the second wall 260b of the lower tray 250.

An outer face of the chamber wall 153 of the upper tray body 151 is spaced apart from an inner face of the circumferential wall 260 of the lower tray 250. That is, a space may be defined between the outer face of the chamber wall 153 of the upper tray body 151 and the inner face of the circumferential wall 260 of the lower tray 250.

Water supplied through the water supply part 180 is accommodated in the ice chamber 111. When a relatively large amount of water than a volume of the ice chamber 111 is supplied, water that is not accommodated in the ice chamber 111 may flow into the space between the outer face of the chamber wall 153 of the upper tray body 151 and the inner face of the circumferential wall 260 of the lower tray 250.

Thus, according to this embodiment, even though a relatively large amount of water than the volume of the ice chamber 111 is supplied, the water may be prevented from overflowing from the ice maker 100.

A heater contact part 251a for allowing the contact area with the lower heater 296 to increase may be further provided on the lower tray body 251.

The heater contact portion 251a may protrude from the bottom surface of the lower tray body 251. In one example, the heater contact portion 251a may be formed in a ring

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shape and disposed on the bottom surface of the lower tray body 251. The bottom surface of the heater contact portion 251a may be planar.

The lower tray body 251 may further include a convex portion 251b in which a portion of the lower portion of the lower tray body 251 is convex upward.

A recess 251c may be defined below the convex portion 251b so that the convex portion 251b has substantially the same thickness as the other portion of the lower tray body 251.

In this specification, the “substantially the same” is a concept that includes completely the same shape and a shape that is not similar but there is little difference.

The convex portion 251b may be disposed to vertically face the lower opening 274 of the lower support 270.

The lower opening 274 may be defined just below the lower chamber 252. That is, the lower opening 274 may be defined just below the convex portion 251b.

The convex portion 251b may have a diameter D less than that D2 of the lower opening 274.

When cold air is supplied to the ice chamber 111 in the state in which the water is supplied to the ice chamber 111, the liquid water is phase-changed into solid ice. Here, the water may be expanded while the water is changed in phase. The expansive force of the water may be transmitted to each of the upper tray body 151 and the lower tray body 251.

In case of this embodiment, although other portions of the lower tray body 251 are surrounded by the support body 271, a portion (hereinafter, referred to as a “corresponding portion”) corresponding to the lower opening 274 of the support body 271 is not surrounded.

If the lower tray body 251 has a complete hemispherical shape, when the expansive force of the water is applied to the corresponding portion of the lower tray body 251 corresponding to the lower opening 274, the corresponding portion of the lower tray body 251 is deformed toward the lower opening 274.

In this case, although the water supplied to the ice chamber 111 exists in the spherical shape before the ice is made, the corresponding portion of the lower tray body 251 is deformed after the ice is made. Thus, additional ice having a projection shape may be made from the spherical ice by a space occurring by the deformation of the corresponding portion.

Thus, in this embodiment, the convex portion 251b may be disposed on the lower tray body 251 in consideration of the deformation of the lower tray body 251 so that the ice has the completely spherical shape.

In this embodiment, the water supplied to the ice chamber 111 is not formed into a spherical form before the ice is generated. After the generation of the ice is completed, the convex portion 251b of the lower tray body 251 is deformed toward the lower opening 274, such that the spherical ice may be generated.

In the present embodiment, the diameter D1 of the convex portion 251b is smaller than the diameter D2 of the lower opening 274, such that the convex portion 251b may be deformed and positioned inside the lower opening 274.

In FIG. 31, the line passing vertically through the center of the ice chamber 111 may be referred to as the vertical center line C3. In one example, the vertical center line C3 may pass through the upper opening 154 and lower opening 274.

Further, the line passing through the contact surface of the bottom surface 151a of the upper tray 151 and the top surface 251e of the lower tray 250 in the ice chamber 111

may be defined as a horizontal center line based on the height of the ice chamber 111.

A distance (D3) between two points disposed in opposite sides based on the vertical central line (C3) in the upper round portion 148c of the upper heater 148 may be smaller than a diameter (D7) of the ice chamber 111.

A distance (D3) between two points disposed in the opposite sides based on the vertical central line (C3) in the upper round portion 148c of the upper heater 148 may be greater than a distance (D4) between two points disposed in the opposite sides based on the vertical central line (C3) in the lower round portion 296a of the upper heater 296.

In other words, a radius of curvature (R1) of the upper round portion 148c of the upper heater 148 is greater than a radius of curvature (R3) of the lower round portion 296a of the lower heater 296.

The lower heater 296 has to be disposed close to a lowermost side of the lower tray 250 to freeze the ice latest under the upper chamber 252, and accordingly, bubbles may be gathered at a lowermost side of the lower chamber 252.

Meanwhile, a distance between the upper heater 148 and the horizontal central line of the ice chamber 111 may be less than a distance between the horizontal central line and the upper opening 154. Accordingly, the heat of the upper heater 148 may be rapidly transferred not only to the upper chamber 152 but also to a boundary between the upper tray 150 and the lower tray 250.

Therefore, in this embodiment, a distance (D5) from the horizontal central line to the upper heater 148 is smaller than a distance (D6) from the horizontal central line to the lower heater 296.

A distance from the vertical central line (C3) to at least a portion of the upper heater 148 is longer than a distance from the vertical central line (C3) to at least a portion of the lower heater 296.

The upper heater 148 may be disposed in the same height as the height of a bisector (L1) of bisecting a distance between the upper opening 154 and the horizontal central line or may be higher than the bisector (L1).

As an example, FIG. 31 illustrates that the upper heater 148 is higher than the bisector (L1).

Based on the height of the upper chamber 152, the upper heater 148 may be disposed between the bisector (L1) and the upper opening 154, as an example.

At least a portion of the lower heater 296 may be disposed to vertically overlap the lower ice chamber 252. The lower round portion 296c of the at least lower heater 296 may be disposed to vertically overlap with lower ice chamber 252.

The lower heater 296 may be spaced apart from the vertical central line (C3) of the ice chamber 111.

As an example, the lower round portion 296a of the lower heater 296 may be disposed to surround the lower opening 274. Therefore, an interference between the lower heater 296 and the lower ejector 400 may be prevented in a process that the lower ejector 400 penetrates the lower opening 274.

In an ice making position, at least a portion of the upper heater 296 may be disposed closer to the vertical central line (C3) than the upper heater 148.

Hereinafter, an ice making process by the ice maker according to one embodiment of the present invention will be described.

FIG. 33 is a cross-sectional view taken along line B-B of FIG. 3 in a water supply state, and FIG. 34 is a cross-sectional view taken along line B-B of FIG. 3 in an ice making state.

FIG. 35 is a cross-sectional view taken along line B-B of FIG. 3 in a state in the ice-making completed state, FIG. 36

is a cross-sectional view taken along line B-B of FIG. 3 in an initial state of ice separation, and FIG. 37 is a cross-sectional view taken along line B-B of FIG. 3 in an ice separation completed state.

Referring to FIGS. 33 to 37, first, the lower assembly 200 rotates to a water supply position.

The top surface 251e of the lower tray 250 is spaced apart from the bottom surface 151e of the upper tray 150 at the water supply position of the lower assembly 200.

Although not limited, the bottom surface 151e of the upper tray 150 may be disposed at a height that is equal or similar to a rotational center C2 of the lower assembly 200.

In this embodiment, the direction in which the lower assembly 200 rotates (in a counterclockwise direction in the drawing) is referred to as a forward direction, and the opposite direction (in a clockwise direction) is referred to as a reverse direction.

Although not limited, an angle between the top surface 251e of the lower tray 250 and the bottom surface 151e of the upper tray 150 at the water supply position of the lower assembly 200 may be about 8 degrees.

In this state, the water is guided by the water supply part 190 and supplied to the ice chamber 111.

In this connection, the water is supplied to the ice chamber 111 through one upper opening of the plurality of upper openings 154 of the upper tray 150.

In the state in which the supply of the water is completed, a portion of the supplied water may be fully filled into the lower chamber 252, and the other portion of the supplied water may be fully filled into the space between the upper tray 150 and the lower tray 250.

For example, the upper chamber 151 may have the same volume as that of the space between the upper tray 150 and the lower tray 250. Thus, the water between the upper tray 150 and the lower tray 250 may be fully filled in the upper tray 150. In another example, the volume of the upper chamber 152 may be larger than the volume of the space between the upper tray 150 and the lower tray 250.

In case of this embodiment, a channel for communication between the three lower chambers 252 may be provided in the lower tray 250.

As described above, although the channel for the flow of the water is not provided in the lower tray 250, since the top surface 251e of the lower tray 250 and the bottom surface 151e of the upper tray 150 are spaced apart from each other, the water may flow to the other lower chamber along the top surface 251e of the lower tray 250 when the water is fully filled in a specific lower chamber in the water supply process.

Thus, the water may be fully filled in each of the plurality of lower chambers 252 of the lower tray 250.

In the case of this embodiment, since the channel for the communication between the lower chambers 252 is not provided in the lower tray 250, additional ice having a projection shape around the ice after the ice making process may be prevented being made.

In the state in which the supply of the water is completed, as illustrated in FIG. 34, the lower assembly 200 rotates reversely. When the lower assembly 200 rotates reversely, the top surface 251e of the lower tray 250 is close to the bottom surface 151e of the upper tray 150.

Thus, the water between the top surface 251e of the lower tray 250 and the bottom surface 151e of the upper tray 150 may be divided and distributed into the plurality of upper chambers 152.

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Also, when the top surface **251e** of the lower tray **250** and the bottom surface **151e** of the upper tray **150** are closely attached to each other, the water may be fully filled in the upper chamber **152**.

In the state in which the top surface **251e** of the lower tray **250** and the bottom surface **151e** of the upper tray **150** are closely attached to each other, a position of the lower assembly **200** may be called an ice making position.

In the state in which the lower assembly **200** moves to the ice making position, ice making is started.

Since pressing force of water during ice making is less than the force for deforming the convex portion **251b** of the lower tray **250**, the convex portion **251b** may not be deformed to maintain its original shape.

When the ice making is started, the lower heater **296** is turned on. When the lower heater **296** is turned on, heat of the lower heater **296** is transferred to the lower tray **250**.

Thus, when the ice making is performed in the state where the lower heater **296** is turned on, ice may be made from the upper side in the ice chamber **111**.

That is, water in a portion adjacent to the upper opening **154** in the ice chamber **111** is first frozen. Since ice is made from the upper side in the ice chamber **111**, the bubbles in the ice chamber **111** may move downward.

Since the ice chamber **111** is formed in a sphere shape, the horizontal cross-sectional area may vary based on a height of the ice chamber **111**.

Thus, the output of the lower heater **296** may vary depending on the height at which ice is produced in the ice chamber **111**.

The horizontal cross-sectional area increases as it goes downwardly. Then, the horizontal cross-sectional area becomes maximum at the boundary between the upper tray **150** and the lower tray **250** and decreases as it goes downwardly again.

In the process where ice is generated from a top to a bottom in the ice chamber **111**, the ice comes into contact with the top surface of the convex portion **251b** of the lower tray **250**. In this state, when the ice is continuously made, the block part **251b** may be pressed and deformed as shown in FIG. **34**, and the spherical ice may be made when the ice making is completed.

A control unit (not shown) may determine whether the ice making is completed based on the temperature sensed by the temperature sensor **500**.

The lower heater **296** may be turned off at the ice-making completion or before the ice-making completion.

When the ice-making is completed, the upper heater **148** is first turned on for the ice-removal of the ice. When the upper heater **148** is turned on, the heat of the upper heater **148** is transferred to the upper tray **150**, and thus, the ice may be separated from the surface (the inner face) of the upper tray **150**.

In addition, as the heat of the upper heater **148** is transferred to the boundary between the upper tray **150** and the lower tray **250**, the upper tray **150** and the lower tray **250** can be separated from each other.

After the upper heater **148** has been activated for a set time duration, the upper heater **148** may be turned off and then the drive unit **180** may be operated to rotate the lower assembly **200** in a forward direction.

As illustrated in FIG. **36**, when the lower assembly **200** rotates forward, the lower tray **250** may be spaced apart from the upper tray **150**.

Also, the rotation force of the lower assembly **200** may be transmitted to the upper ejector **300** by the connection unit **350**. Thus, the upper ejector **300** descends by the unit guides

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181 and **182**, and the upper ejecting pin **320** may be inserted into the upper chamber **152** through the upper opening **154**.

In the ice separating process, the ice may be separated from the upper tray **250** before the upper ejecting pin **320** presses the ice. That is, the ice may be separated from the surface of the upper tray **150** by the heat of the upper heater **148**.

In this case, the ice may rotate together with the lower assembly **200** in the state of being supported by the lower tray **250**.

Alternatively, even though the heat of the upper heater **148** is applied to the upper tray **150**, the ice may not be separated from the surface of the upper tray **150**.

Thus, when the lower assembly **200** rotates forward, the ice may be separated from the lower tray **250** in the state in which the ice is closely attached to the upper tray **150**.

In this state, while the lower assembly **200** rotates, the upper ejecting pin **320** passing through the upper opening **154** may press the ice closely attached to the upper tray **150** to separate the ice from the upper tray **150**. The ice separated from the upper tray **150** may be supported again by the lower tray **250**.

When the ice rotates together with the lower assembly **200** in the state in which the ice is supported by the lower tray **250**, even though external force is not applied to the lower tray **250**, the ice may be separated from the lower tray **250** by the self-weight thereof.

While the lower assembly **200** rotates, even though the ice is not separated from the lower tray **250** by the self-weight thereof, when the lower tray **250** is pressed by the lower ejector **400** as shown in FIG. **37**, the ice may be separated from the lower tray **250**.

Particularly, while the lower assembly **200** rotates, the lower tray **250** may contact the lower ejecting pin **420**.

When the lower assembly **200** continuously rotates forward, the lower ejecting pin **420** may press the lower tray **250** to deform the lower tray **250**, and the pressing force of the lower ejecting pin **420** may be transmitted to the ice to separate the ice from the lower tray **250**. The ice separated from the surface of the lower tray **250** may drop downward and be stored in the ice bin **102**.

After the ice is separated from the lower tray **250**, the lower assembly **200** may be rotated in the reverse direction by the drive unit **180**.

When the lower ejecting pin **420** is spaced apart from the lower tray **250** in a process in which the lower assembly **200** is rotated in the reverse direction, the deformed lower tray **250** may be restored to its original form. That is, the deformed convex portion **251b** may be restored to its original form.

In the reverse rotation process of the lower assembly **200**, the rotational force is transmitted to the upper ejector **300** by the connecting unit **350**, such that the upper ejector **300** is raised, and thus, the upper ejecting pin **320** is removed from the upper chamber **152**.

When the lower assembly **200** reaches the water supply position, the drive unit **180** is stopped, and then water supply starts again.

According to the proposed embodiment, since the upper heater is disposed closer to the horizontal central line of the ice chamber than the upper opening, not only may the heat of the upper heater be rapidly transferred to the upper chamber, but also the heat may be rapidly transferred to a boundary between the upper tray and the lower tray.

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When the heat of the upper chamber is transferred to the boundary between the upper tray and the lower tray, the lower tray may be easily separated from the upper tray in the ice separation process.

In addition, according to this embodiment, as the upper heater includes the upper round portion to surround the upper chamber, the heat of the upper heater may be rapidly provided to the upper chamber.

In addition, since the upper round portion of the upper heater is disposed to vertically overlap the ice chamber, the heat of the upper round portion of the upper heater may be rapidly transferred to the ice chamber.

In addition, since the upper heater for ice separation is disposed to surround each of a plurality of upper chambers, the heat may be uniformly transferred to the plurality of upper chambers.

In addition, since the heater coupling part is accommodated in the accommodation part formed in the upper tray and contacts a bottom of the accommodation part in a state that the upper heater is coupled to the heater coupling part of the upper case, the heat of the upper heater can be concentrated in the upper tray.

In addition, the upper heater is accommodated in the accommodation groove in a state that the upper heater is curved in the horizontal direction and the separation prevention protrusion is provided in the heater coupling part, the heater may be stably maintained in a state that the heater is coupling to the heater coupling part.

In addition, in the ice making process, as the upper heater is operated to transfer the heat of the lower heater to a lower side (the lower chamber) of the ice chamber, the ice is frozen from an upper side in the ice chamber, bubbles in water are gathered under the ice chamber. Therefore, there is an advantage that the ice can be made transparent as a whole.

In addition, since the lower heater is disposed in a position spaced apart from the vertical central line of the ice chamber, the lower heater may be prevented from interfering with the lower ejector in the ice separation process.

In addition, according to this embodiment, as the lower heater includes the lower round portion to surround the lower chamber, the heat of the upper heater may be transferred well to the lower chamber.

In addition, since the lower heater is disposed to surround each of circumferences of the plurality of lower chambers, the heat may be uniformly transferred to the plurality of lower chambers.

Hereinafter, another embodiment of the present invention will be described. At this time, the same constituents as the previous embodiment use the same reference numerals.

FIG. 38 is an upper perspective view of an upper support according to another embodiment of the present invention, and FIG. 39 is a lower perspective view of the upper support according to another embodiment of the present invention.

With reference to FIGS. 38 and 39, the upper support 170 of this embodiment may further comprise a wire guiding hook 134 extending from one side to a lower side of the support plate 171 and preventing a flow of a wire 298 that will be described later.

If the flow of the wire 298 is prevented by the wire guiding hook 134, this may prevent the problem that the wire 298 disturbs a rotation of the lower assembly 200, or the wire 298 is disconnected by a rotational operation of the upper assembly 200.

As an example, the support plate 171 may horizontally extend in a direction of a plurality of second lower slots 177, and the wire guiding hook 134 may be installed in one side of an extending part of the support plate 171.

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In addition, the wire guiding hook 134 may be installed not to disturb the rotation of the lower assembly 200 when the upper assembly 110 and the lower assembly 200 are assembled.

The wire guiding hook 134 may comprise a curving part 134b curved one or more times and a support part 134a for supporting the curving part 134b.

Specifically, the curving part 134b may be in a hooked shape curved outside of the support plate 171, and in an example, the curving part 134b may be curved two times after extending from a lower side of the support plate 171 and then be formed back toward the support plate 171.

As an example, the curving part 134b may comprise a first part extending to the lower side of the support plate 171, a second part curved and extending in the horizontal direction from the first part, and a third part curved again in the second part and extending toward the support plate 171.

In addition, the second part may extend from the first part to an opposite direction of the plate opening 172.

In addition, as an interval between the first part and the third part moves away from the support plate 171, the interval may be narrow.

As the third part of the curving part 134b is spaced apart from the support plate 171, the wire 298 may pass through the spaced part, and the third part may have an enough length such that the wire 298 does not protrude through the spaced part.

In addition, the wire 298 passing through the spaced apart may be supported by the second part of the curving 134b.

The support part 134a may be formed to connect the curving part 134b and the support plate 171 while supporting the curving part 134b to improve strength and durability of the curving part 134b.

As an example, the support part 134a may extend vertically from the first part of the curving part 134b to the plate opening 172 and may be connected to the support plate 171.

In detail, the support part 134a may be configured such that a part contacting the support plate 171 is formed more widely, and as the support part 134a extends to the lower side of the support plate 171, it gets narrower.

As another example, the support 134a may be provided in pairs having the same shape, each of which is connected to the first part of the curving part 134b, and may protrude toward the plate opening 172.

Meanwhile, the wire guiding hook 134 may comprise an opening (see 134c of FIG. 44) formed to correspond to a size of the curving part 134b.

The opening (see 134c of FIG. 44) may be formed in the support plate 171 in a position where the wire guiding hook 134 is installed.

For example, the opening (see 134c of FIG. 44) may be formed in a square shape in the support plate 171, and the support 134a may be installed at one side of the opening.

As the wire guiding hook 134 is installed only in one side of the support plate 171, the opening (see 134c of FIG. 44) may serve as preventing a phenomenon of biasing a center of gravity.

FIG. 40 is an upper perspective view of a lower support according to another embodiment of the present invention, and FIG. 41 is a lower perspective view of the lower support according to another embodiment of the present invention. FIG. 42 is a top plan view of the lower support according to another embodiment of the present invention.

FIG. 43 is a perspective view that the lower heater is coupled to the lower support of FIG. 42, FIG. 44 is a view showing a state in which a wire connected to the lower heater penetrates an upper case in a state that a lower

assembly is coupled to an upper assembly, and FIG. 45 is a bottom view showing a state in which a wire connected to the lower heater penetrates an upper case in a state that a lower assembly is coupled to an upper assembly.

With referring to FIGS. 40 to 45, the lower support 270 of this embodiment may comprise a plurality of hinge bodies 281, 282 for connecting each of hinge supports 135, 136 of the upper case 210.

The plurality of hinge bodies 281, 282 may comprise a plurality of hinge body ribs 281b for improving a deformation rate by increasing stiffness.

As an example, the hinge body ribs 281b may be formed to surround a circumference of the hinge body 281 by facing both sides of the hinge body 281.

A hinge body protrusion 281c may be provided in a part where the hinge body ribs 281b contact a top surface of the lower support 270.

In detail, the hinge body ribs 281b may protrude outwards along a circumference from the hinge body 281, and may extend to a bottom end of the lower support 170.

The hinge body protrusion 281c may extend into the lower support 270 from an end of the hinge body rib 281b.

In addition, as the hinge body protrusion 281c is to reduce deformation of the hinge bodies 281, 282, the hinge bodies 281, 282 may be curvilinearly connected without bending with the top surface of the lower support 270.

The lower support 270 may comprise a first guide groove 293 for guiding a power input terminal 296c and a power output terminal 296d of the lower heater 296 accommodated in the heater accommodation groove 291, and a second guide groove 294 extending in a direction of crossing the first guide groove 293.

As an example, the first guide groove 293 may extend from the heater accommodation groove 291 to an arrow B.

The second guide groove 294 may extend from an end of the first guide groove 293 to an arrow A. In this embodiment, the arrow A is a direction alongside an extension direction of a rotational central axis (C1) of the lower assembly 200.

In addition, the second guide groove 294 may comprise slots 295, 299 of which both ends are connected to the outside of the lower support 270.

In detail, a withdrawing slot 299 for withdrawing the wire 268 may be formed in one end of the second guide groove 294.

A central slot 295 may be formed in the other end of the second guide groove 294. The central slot 295 may be formed adjacent to a center of the lower support 270.

The wire 298 is curved after extending toward the central slot 295 in the second guide groove 294, extends towards the withdrawing slot 299, and finally passes through the withdrawing slot 299. Since the wire 298 is disposed in a curved state in the second guide groove 295, at least portion of the curving part may be disposed in the central slot 295 in order to avoid the disconnection in the curving part.

The lower heater 291 accommodated inside or a separation prevention protrusion 293a for preventing the wire 298 from being separated may be provided in at least one of the first guide groove 293 and the second guide groove 294.

A chamber accommodation part (for example, a right chamber accommodation part) disposed farthest from the first guide groove 293 among the three chamber accommodation parts may further comprise a detour accommodation groove 292.

As an example, after the detour accommodation groove 292 is curved by extending outwards from the heater accom-

modation groove 291, the detour accommodation groove 292 may be connected back to the heater accommodation groove 291.

When the lower heater 291 is additionally accommodated in the detour accommodation groove 292, a contact area of a chamber wall accommodated in the chamber accommodation part 272 on the right and the lower heater 296 may be increased.

Therefore, as an example, the chamber accommodation part 272 on the right may further comprise a protrusion 292a for fixing a position of the lower heater accommodated in the detour accommodation groove 292.

In addition, in the chamber accommodation part 272 on the right, a plurality of detour accommodation grooves 292 may be provided, and a penetration opening 291d may be formed to correspond to a protrusion 292b in order to reduce a tension of the lower heater 296 and prevent the lower heater 296 from being separated from the heater accommodation groove 291.

In detail, the chamber accommodation part 272 on the right may comprise the detour accommodation groove 292 on the right based on FIG. 26, and further comprise a detour accommodation groove 292 in a direction of facing the hinge body 281.

In addition, the detour accommodation groove 292 formed in the direction of facing the hinge body 281 may further comprise a penetration opening 291d formed to disconnect the protrusion 292b and the lower heater 296 or prevent the heater accommodation groove 291 from being separated.

As another example, a chamber accommodation part 272 on the left may further comprise a protrusion 292c for fixing a position of the lower heater accommodated in the detour accommodation groove 292. At this time, the detour accommodation groove 292 may be disposed symmetrical to the detour accommodation groove 292 provided in the chamber accommodation part 272 on the right.

With reference to FIGS. 44 and 45, in a state that the lower assembly 200 is coupled to the upper case 120 of the upper assembly 110, the wire 298 withdrawn outside of the lower support 270 through the withdrawing slot 299 formed in one side of the lower support 270 may pass through the wire penetration slot 138 formed in the upper case 120 to extend to the top of the upper case 120.

In detail, the wire 298 penetrating the withdrawing slot 299 is disposed in an upper side of the wire guiding hook 134 to prevent the flow of the wire 298, which allows an interference not to occur in rotating the lower assembly.

A limiting guide 139 for limiting a movement of the wire 298 penetrating the wire penetration slot 138 may be provided in the wire penetration slot 138. The limiting guide 139 may be curved several times, and the wire 298 may be disposed in the wire 298 in an area in which the limiting guide 139 is formed.

By the provided embodiment, the tension of the wire may be reduced by extending a length of the wire connected to the heater, and the wire may be prevented from being disconnected.

In addition, even if the length of the wire is extended by adding the wire guiding hook, the possibility to disconnect the wire by the rotation of the lower assembly may be prevented, and the rotation of the lower assembly may not interfere by the wire.

In addition, the lower assembly may be smoothly rotated by reinforcing a strength of a rotational axis of the lower case.

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What is claimed is:

1. An ice maker comprising:

a first tray defining a plurality of first chambers;

a second tray defining a plurality of second chambers, wherein the plurality of first chambers and the plurality of second chambers are configured to define a plurality of ice chambers based on the first tray and the second tray contacting each other;

a first heater configured to provide heat to the plurality of ice chambers during an ice-making process; and

a second heater configured to provide heat to the plurality of ice chambers during an ice-separation process after the ice-making process,

wherein the ice maker is configured to (i) freeze water in the plurality of ice chambers into ice based on the first and second trays being in a closed state in which the first tray and the second tray are in contact with each other and (ii) separate the ice from the plurality of ice chambers based on the first and second trays being in an open state in which the first tray and the second tray are spaced apart from each other,

wherein the first tray comprises a contact surface that is configured to be in contact with the second tray in the closed state, and

wherein, based on the first and second trays being in the closed state, a first distance between the contact surface and the first heater in a first direction perpendicular to the contact surface is greater than a second distance between the contact surface and the second heater in the first direction.

2. The ice maker of claim 1, wherein the first heater is configured to, based on heating the plurality of ice chambers during the ice-making process, cause bubbles in water received in the plurality of ice chambers to be collected at a portion of each of the plurality of ice chambers.

3. The ice maker of claim 2, wherein the first heater contacts an outer circumferential surface of a chamber wall defining the plurality of ice chambers, and the portion where bubbles are collected corresponds to a portion with which the first heater is in contact.

4. The ice maker of claim 1, further comprising:

an ejector configured to move in the first direction and separate the ice from the plurality of ice chambers.

5. The ice maker of claim 4, further comprising:

a case configured to fix the first tray, and define a tray opening exposing a portion of an outer surface of the first tray,

wherein the ejector is configured to move through the tray opening to separate the ice from the plurality of ice chambers.

6. The ice maker of claim 1, wherein the plurality of ice chambers are defined in an inner surface of a plurality of chamber walls, the plurality of chamber walls being made of a flexible material,

wherein the first heater disposed at the plurality of chamber walls, and

wherein the second heater surrounds the plurality of chamber walls and is spaced apart from the first heater.

7. An ice maker comprising:

a tray body made of a flexible material, the tray body comprising a plurality of chamber walls configured to define a plurality of ice chambers arranged along a first central line extending in a first direction and passing through centers of the plurality of ice chambers;

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a first heater that is disposed at the plurality of chamber walls, the first heater being configured to provide heat to the plurality of ice chambers during an ice-making process; and

a second heater that surrounds the plurality of chamber walls and is spaced apart from the first heater in a second direction orthogonal to the first direction, the second heater being configured to provide heat to the plurality of ice chambers during an ice-separation process after the ice-making process,

wherein a first distance between the first heater and the first central line in the second direction is greater than a second distance between the second heater and the first central line in the second direction, and

wherein the first heater is configured to, based on heating the plurality of ice chambers during the ice-making process, cause bubbles in water received in the plurality of ice chambers to be collected at an end portion of the plurality of ice chambers in the second direction.

8. The ice maker of claim 7, wherein the first heater contacts an outer circumferential surface of each of the plurality of chamber walls, and

wherein the end portion where the bubbles are collected corresponds to a portion with which the first heater is in contact.

9. The ice maker of claim 7, wherein the second heater is spaced apart from an outer circumferential surface of the plurality of chamber walls.

10. The ice maker of claim 7, further comprising an accommodation part disposed in a shape that surrounds the plurality of chamber walls, and

wherein the second heater is in contact with the accommodation part.

11. The ice maker of claim 7, wherein the plurality of ice chambers are defined in an inner surface of the plurality of chamber walls and a first shortest distance between the first heater and the inner surface of the plurality of chamber walls is less than a second shortest distance between the second heater and the inner surface of the plurality of chamber walls.

12. The ice maker of claim 7, further comprising a case that contacts and supports the tray body,

wherein the case comprises a heater coupling part configured to couple the second heater, and

wherein the heater coupling part comprises an outer wall and an inner wall defining a heater accommodation groove for accommodating the second heater, and at least a portion of the inner wall is disposed in a space between the second heater and an outer circumferential surface of the plurality of chamber walls.

13. The ice maker of claim 7, further comprising an ejector configured to separate ice from the plurality of ice chambers and comprises a plurality of ejecting pins, each of the plurality of ejecting pins being configured to move along a second central line extending in the second direction and passing through a center of a corresponding one of the plurality of ice chambers.

14. The ice maker of claim 13, further comprises a case that contacts and supports the tray body, and defines an opening exposing a portion of an outer surface of the plurality of chamber walls, and

wherein the plurality of ejecting pins are configured to move through the opening to separate ice from the plurality of ice chambers.

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15. The ice maker of claim 7, wherein the tray body comprises:

a first tray body configured to define first parts of the plurality of ice chambers; and

a second tray body configured to define second parts of the plurality of ice chambers, the second tray body being configured to:

during the ice-making process, contact the first tray body to thereby close the plurality of ice chambers, and

during the ice-separation process, detach from the first tray body to thereby open the plurality of ice chambers.

16. The ice maker of claim 7, wherein the first heater is configured to operate while at least some water remains in the plurality of ice chambers during the ice-making process, and

wherein the second heater is configured to operate based on completion of the ice-making process.

17. A refrigerator comprising:

a cabinet defining a storage space; and

an ice maker disposed in the storage space, wherein the ice maker comprises:

a first tray defining a plurality of first chambers,

a second tray defining a plurality of second chambers, wherein the plurality of first chambers and the plurality of second chambers are configured to define a plurality of ice chambers based on the first tray and second trays contacting each other,

a first heater disposed at a plurality of chamber walls defining the plurality of ice chambers, the first heater being configured to provide heat to the plurality of ice chambers during an ice-making process, and

a second heater that surrounds the plurality of chamber walls and is spaced apart from the first heater, the second heater being configured to provide heat to the plurality of ice chambers during an ice-separation process after the ice-making process,

wherein a first radial distance from one of centers of the plurality of ice chambers to the first heater is less than a second radial distance from the one of the centers to the second heater,

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wherein the first heater is configured to, based on heating the plurality of ice chambers during the ice-making process, cause bubbles in water received in the plurality of ice chambers to be collected at a portion of each of the plurality of ice chambers,

wherein the ice maker is configured to (i) freeze water in the plurality of ice chambers into ice based on the first and second trays being in a closed state where the first tray and the second tray are in contact with each other, and (ii) separate the ice from the plurality of ice chambers based on the first and second trays being in an open state where the first tray and the second tray are spaced apart from each other,

wherein the first tray comprises a contact surface that is configured to be in contact with the second tray in the closed state, and

wherein, based on the first and second trays being in the closed state, a first distance between the contact surface and the first heater in a first direction perpendicular to the contact surface is greater than a second distance between the contact surface and the second heater in the first direction.

18. The refrigerator of claim 17, wherein the first heater is configured to, based on heating the plurality of ice chambers during the ice-making process, cause the bubbles in water received in the plurality of ice chambers to be collected at a portion of each of the plurality of ice chambers.

19. The refrigerator of claim 18, wherein the first heater contacts an outer circumferential surface of the chamber walls defining the plurality of ice chambers, and

wherein the portion where the bubbles are collected corresponds to a portion with which the first heater is in contact.

20. The refrigerator of claim 17, wherein the plurality of ice chambers are defined in an inner surface of the plurality of chamber walls, the plurality of chamber walls being made of a flexible material,

wherein the first heater disposed at the plurality of chamber walls, and

wherein the second heater surrounds the plurality of chamber walls, and is spaced apart from the first heater.

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